A HYBRID DEEP LEARNING AND ENSEMBLE FRAMEWORK FOR PREDICTIVE ANALYTICS OF MICROPLASTIC DENSITY MAPPING IN THE SOUTHERN LAGUNA DE BAY USING SENTINEL-2 REMOTE SENSING AND IN-SITU WATER QUALITY METRICS

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Bachelor of Science in Computer Science
Specialized in Intelligent Systems

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APPROVAL SHEET

This thesis entitled "**A HYBRID DEEP LEARNING AND ENSEMBLE FRAMEWORK FOR PREDICTIVE ANALYTICS OF MICROPLASTIC DENSITY MAPPING IN THE SOUTHERN LAGUNA DE BAY USING SENTINEL-2 REMOTE SENSING AND IN-SITU WATER QUALITY METRICS**" prepared and submitted by **RALPH LAWRENCE V. AMPARO, LANCE GEEFFREY S. DE CASTRO, and JIREH M. TREGUEROS** in partial fulfillment of the requirements for the degree, Bachelor of Science in Computer Science Specialized in Intelligent Systems is hereby recommended for approval and acceptance.

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The Researchers

DEDICATION

This study is wholeheartedly dedicated to their beloved families:

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R.L.A

L.G.D.C

J.M.T

ABSTRACT

This study developed a predictive model to estimate microplastic density (mg/L) in the southern part of Laguna de Bay using an integrated machine learning and deep learning framework. Data from two sampling sites, Bayog and Pansol, included in-situ measurements of water quality parameters (Dissolved Oxygen, pH, Biochemical Oxygen Demand, Total Suspended Solids, and Fecal Coliform) and five Sentinel-2 spectral indices: NDTI, NDCI, NDVI, NDMI, and PI. Sampling focused on surface water microplastics across near-shore zones and shoal areas using a spatial grid-based approach, where accumulation is influenced by hydrodynamic patterns and land-based runoff. An MLP Autoencoder was applied for deep feature extraction, combined with ensemble learning algorithms (Random Forest, Gradient Boosting, and XGBoost). The hybrid model achieved strong predictive performance (R^2 = 0.912). Although satellite data effectively captured spatial and spectral variability, it could not independently quantify microplastic concentrations without calibration using in-situ measurements, highlighting the necessity of ground-truth data for reliable prediction. Statistical validation through an independent t-test ($p > 0.05$) showed no significant difference between predicted and observed concentrations, confirming the robustness of the framework. Data collection is recommended during March–April when cloud cover is minimal. Overall, the study demonstrates the potential of deep learning and multispectral remote sensing, calibrated with in-situ data, for accurate monitoring of microplastic pollution in Laguna de Bay.

Keywords: *microplastics, Laguna de Bay, deep learning, ensemble modeling, Sentinel-2, in-situ calibration, predictive analytics, remote sensing*

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DEFINITION OF TERMS

Cross-Validation (CV)	A model validation technique that partitions the dataset into multiple training and validation subsets to assess model robustness and generalization capability. K-fold cross-validation is used to ensure consistent performance across different data splits.
Descriptive Statistics	A statistical method used to summarize and describe the main features of a dataset through numerical measures such as mean, median, minimum, maximum, and standard deviation. It provides a concise overview of data distribution and variability.
Ensemble Learning	A machine learning approach that combines multiple predictive models to improve overall prediction accuracy and robustness. In this study, Random Forest, Gradient Boosting, and XGBoost algorithms are used as ensemble methods for microplastic concentration prediction.
Exploratory Data Analysis	The process of examining and visualizing datasets to identify patterns, relationships, and anomalies. It helps in understanding the structure of the data, detecting outliers, and determining variable correlations that are useful for model development.
Feature Extraction	The process of identifying and isolating relevant information from raw data. In this study, per-band statistical features (mean, standard deviation, variance, percentiles) are extracted from satellite spectral indices and compressed by the MLP Autoencoder.
Feature Importance Analysis	A technique used to determine which input features (variables) have the greatest impact on the prediction outcome of a model. Performed using XGBoost's built-in scoring mechanism to identify influential satellite indices and water quality parameters.
Georeferencing	The process of assigning real-world geographic coordinates to satellite imagery by identifying ground control points that link pixel locations to precise spatial positions in a map coordinate system. This ensures

	accurate spatial alignment between remote sensing data and field sampling sites.
GIS-Based Mapping	The process of creating spatially accurate maps of predicted microplastic distribution by integrating georeferenced satellite data with model predictions in a Geographic Information System. Enables visualization of microplastic accumulation hotspots for environmental management and decision-making.
Gradient Boosting (GB)	An ensemble technique that sequentially builds weak learners (decision trees) that iteratively minimize prediction error through gradient optimization.
Hybrid MLP Autoencoder-Ensemble Model	A two-stage predictive framework integrating deep learning feature extraction and ensemble machine learning. The MLP Autoencoder handles dimensionality reduction of satellite features, while Random Forest, Gradient Boosting, and XGBoost handle prediction tasks. Predictions from in-situ and satellite ensembles are combined using weighted integration (0.8 for in-situ, 0.2 for satellite).
In-situ Data	Direct field measurements collected from the water body itself, including laboratory analysis of water samples for microplastic content and standard water quality parameters. These measurements provide ground-truth data with lower measurement uncertainty compared to satellite-derived features.
Independent T-Test	A statistical hypothesis test used to validate model performance by comparing predicted microplastic concentrations with actual observed values from unseen data. Tests whether there is a statistically significant difference between the means of two independent groups (predicted vs. actual).
Laguna de Bay	The largest freshwater lake in the Philippines, which serves as the study site for microplastic accumulation modeling due to its high ecological significance and exposure to urban, agricultural, and industrial pollution sources.

MAE (Mean Absolute Error)	A regression metric that quantifies the average magnitude of prediction errors without considering direction. Less sensitive to outliers than RMSE, providing an intuitive measure of average prediction deviation.
Microplastics	Synthetic polymer particles smaller than 5 millimeters in size that originate from plastic debris, textiles, or industrial processes. These are commonly found in freshwater and marine environments and pose ecological and health threats due to their ability to accumulate and transport toxic substances.
MLP Autoencoder (Multi-Layer Perceptron Autoencoder)	A deep learning architecture composed of fully-connected neural network layers arranged in an encoder-decoder structure used to extract compact, high-level feature representations from high-dimensional satellite data. The encoder compresses satellite statistical features into a latent bottleneck representation, while the decoder reconstructs the features to ensure information preservation.
MSE (Mean Squared Error)	A metric measuring the average squared difference between input satellite features and their reconstructed output from the MLP Autoencoder. Evaluates the autoencoder's feature extraction quality.
NDCI (Normalized Difference Chlorophyll Index)	A spectral index that estimates chlorophyll concentration in water as a proxy for algal biomass and nutrient levels, calculated from red and red-edge bands.
NDMI (Normalized Difference Moisture Index)	A spectral index that indicates moisture content in water and surrounding areas, derived from green and near-infrared bands. Useful for detecting water quality variations.
NDTI (Normalized Difference Turbidity Index)	A spectral index that measures water turbidity based on light scattering, used as an indicator of suspended solids and potential pollution sources, derived from green and red bands.
NDVI (Normalized Difference Vegetation Index)	A spectral index that measures vegetation coverage and health, calculated from red and near-infrared bands. Provides insight

	into surface conditions related to microplastic distribution.
Plastics Index (PI)	A spectral index specifically designed to detect plastic-like reflectance patterns in water, demonstrating the highest sensitivity (SAV = 16.3170) among all spectral indices used in this study for identifying potential microplastic accumulation.
Random Forest (RF)	An ensemble machine learning algorithm that constructs multiple decision trees and aggregates their outputs to reduce overfitting and improve generalizability.
Remote Sensing	The acquisition of environmental information about an area from a distance, typically via satellite or aerial platforms. In this study, it is used to derive indirect indicators of microplastic presence such as turbidity, chlorophyll-a, and the Floating Debris Index.
RMSE (Root Mean Squared Error)	A regression metric that measures the square root of the average squared differences between predicted and actual values. It penalizes larger errors more heavily than MAE and is used to evaluate ensemble model prediction accuracy.
R^2 Score (Coefficient of Determination)	A statistical measure that represents the proportion of variance in the dependent variable (microplastic concentration) explained by the model. Ranges from 0 to 1, with values closer to 1 indicating better model fit.
Sentinel-2	A European Space Agency satellite system used in the study for acquiring multispectral images that help identify water quality indicators relevant to microplastic pollution.
Spearman's Rank Correlation	A non-parametric statistical method used to assess monotonic relationships between satellite-derived spectral indices and in-situ water quality parameters with microplastic concentrations. Suitable for environmental data with nonlinear relationships and non-normal distributions.
Spectral Indices	Mathematical combinations of different spectral bands from satellite imagery used

	to highlight specific environmental features or conditions in water bodies. In this study, five key spectral indices are computed: NDVI (Normalized Difference Vegetation Index), NDMI (Normalized Difference Moisture Index), NDCI (Normalized Difference Chlorophyll Index), NDTI (Normalized Difference Turbidity Index), and PI (Plastics Index).
Water Quality Metrics	Parameters such as pH, dissolved oxygen (DO), biochemical oxygen demand (BOD), chemical oxygen demand (COD), total suspended solids (TSS), and fecal coliform that indicate the chemical and biological status of water. These are used as in-situ predictors in the ensemble learning models.
XGBoost (Extreme Gradient Boosting)	An enhanced version of gradient boosting that incorporates regularization and parallel computation, achieving superior efficiency and predictive performance. It also provides built-in feature importance analysis to identify influential variables.

CHAPTER 1

INTRODUCTION

Background of the Study

Microplastic pollution has become a global environmental crisis, with particles smaller than 5mm now pervasive in aquatic ecosystems worldwide, from remote oceans to urban waterways (Rochman et al., 2013; Horton & Barnes, 2020). These persistent pollutants represent a major ecological risk because they can carry harmful substances and bioaccumulate in food chains (Prata et al., 2021), with 80-95% of them coming from terrestrial sources such as industrial and domestic waste (Paul et al., 2023). In the Philippines, Laguna de Bay is a perfect example of such a crisis, where research found an average of 14.29 microplastic particles per cubic meter and was mostly polypropylene and polyethylene derived from disposable masks and other wastes generated during the pandemic (Arcadio et al., 2023). The contamination of the lake is a result of several sources such as domestic runoff, industrial discharge, and aquaculture operations coupled with high population density and ineffective waste management systems (Arcadio et al., 2022). This pollution endangers the lake's essential fisheries and aquaculture sectors as well as potentially affect human health by way of polluted seafood (DENR, 2021; Jambeck et al., 2015).

Microplastic contamination of Laguna de Bay threatened aquatic life and human health. Jeyavani et al. (2023) revealed that fish that were treated with polypropylene microplastics experience oxidative stress, tissue damage, and enzyme decline, causing organ dysfunction. Subaramaniyam and Sahoo (2023)

further indicated that microplastics contain toxins and pathogens that increase aquatic toxicity. Alberghini et al. (2023) cautioned that humans who eat contaminated seafood could be ingesting microplastics and chemicals, which could interfere with hormones and lead to health issues. These reports confirm the need for immediate action to safeguard ecosystems and public health.

Remote sensing improved monitoring of microplastics through the ability to make mass, routine observations. Satellites, microwaves, and unmanned aerial vehicles (UAVs) assist in detecting plastic garbage and changes to surfaces. Biermann et al. (2020) employed Sentinel-2 imagery to identify coastal plastic with spectral signatures. Topouzelis et al. (2019) improved nearshore observation through the integration of drone and satellite images. Evans and Ruf (2021) demonstrated that radar backscatter can be used to map ocean microplastics. These methods have high potential for freshwater observation in places such as Laguna de Bay.

Despite advancements, satellite-based detection of microplastics remains limited due to low spatial and spectral resolution. Airborne sensors show potential but depend on controlled, high-resolution setups (Garaba & Dierssen, 2018). Optical confusion with natural debris raises false positives (Garaba et al., 2021). While hyperspectral tools and machine learning hold promise, challenges like limited in situ data and atmospheric interference persist (Karakuş, 2023). These issues support the use of indirect proxies such as chlorophyll-a, turbidity, and the floating algae index to locate likely microplastic hotspots.

Another important point is that, although a significant amount of research has been conducted on marine ecosystems, there continues to be a lack of studies that explore the effect of microplastic pollution on freshwater biodiversity. Rivers and lakes, even though they are key habitats for numerous species, continue to be underrepresented in today's microplastic monitoring (Schmidt et al., 2020; Windsor et al., 2019). The deficiency in data reinforces the necessity for additional targeted studies in freshwater ecosystems such as Laguna de Bay.

Satellite sensors have struggled to directly detect microplastics, prompting scientists to use indirect methods, such as water quality indicators, to estimate microplastic presence. High chlorophyll-a and turbidity levels often signal microplastic hotspots, as particles gather near phytoplankton blooms and sediment-rich areas. Pabortsava and Lampitt (2020) linked microplastics to organic matter transport, while Martínez-Vicente et al. (2019) showed that turbidity changes caused by sediments and plastics affect water optics.

Representation learning with autoencoders compresses high-dimensional environmental features into compact latent codes suitable for downstream prediction, often outperforming linear dimensionality reduction methods such as PCA on complex, non-linear datasets (Hinton & Salakhutdinov, 2006). Unsupervised autoencoders learn these representations through reconstruction objectives, enabling the extraction of salient patterns from noisy environmental data without requiring labeled samples during training (Baig et al., 2023). This approach is particularly advantageous in small-sample regimes typical of environmental monitoring,

where labeled data are scarce but unlabeled satellite and sensor observations are abundant. By capturing non-linear structure through bottleneck compression, autoencoders help stabilize downstream predictive models trained on heterogeneous data sources such as satellite indices and in-situ measurements (Hinton & Salakhutdinov, 2006; Baig et al., 2023).

In this study, a multilayer perceptron (MLP)-based autoencoder, inspired by the framework of Hinton and Salakhutdinov (2006), was implemented to learn compact representations from Sentinel-2 satellite imagery over Southern Laguna de Bay. The encoder component compresses high-dimensional spectral information into a 16-dimensional latent space, effectively preserving key spatial and spectral features linked to water quality dynamics. These encoded satellite features are then combined with in-situ measurements including pH, dissolved oxygen, biochemical oxygen demand, fecal coliform, turbidity, chlorophyll-a, total suspended solids, and temperature to construct a 24-dimensional hybrid feature set. This integrated representation serves as input to Random Forest, Gradient Boosting, and XGBoost models for microplastic prediction and mapping. For operational efficiency, only the trained encoder is retained for feature extraction during deployment, with all inputs standardized to zero mean and unit variance.

Microplastic detection has advanced from manual, low-throughput techniques such as stereomicroscopy and hand sorting to higher-throughput spectroscopic approaches, including FTIR imaging coupled with multivariate analysis for semi-automated polymer typing (Barnes et al., 2009; Cole et al., 2011; Prata et al., 2019; Willans et al., 2023). For large-area assessments,

remote sensing combined with ensemble learning and autoencoder-based representations enables scalable monitoring and risk mapping in complex freshwater systems like Laguna de Bay (Biermann et al., 2020; Evans & Ruf, 2022). This hybrid design leverages unsupervised feature extraction, non-linear dimensionality reduction, and robust supervised ensembles to improve hotspot identification, interpretability of features, and operational decision support for targeted mitigation and management.

This research aims to develop a model that detects areas with accumulated microplastics in Southern Laguna de Bay by integrating unsupervised representation learning and supervised ensemble methods with remote sensing and water-quality data so that pollution hotspots can be identified and prioritized for intervention.

Research Objectives

This study aims to develop a comprehensive predictive model for mapping microplastic accumulation in the southern part of Laguna de Bay using remote sensing and water quality metrics through an integrated machine learning and deep learning approach.

1. Analyze key water quality indicators derived from Sentinel-2 satellite data Normalized Difference Turbidity Index (NDTI), Normalized Difference Chlorophyll Index (NDCI), Normalized Difference Vegetation Index (NDVI), Normalized Difference Moisture Index (NDMI), and Plastics Index (PI) along with in-situ parameters including Dissolved Oxygen (DO), pH, Biochemical Oxygen Demand (BOD), Total Suspended Solids (TSS), and Fecal Coliform, to determine their

relationships with microplastic concentration through correlation and geospatial analysis, based on established findings linking satellite indices with microplastic accumulation in aquatic environments.

2. Develop predictive models for estimating microplastic concentration zones under three configurations: in situ parameters only, satellite-derived indices only, and a weighted hybrid integration of both using an unguided autoencoder with an MLP for feature extraction, integrated with Random Forest, Gradient Boosting, XGBoost, and an ensemble model for prediction, with performance rigorously evaluated using RMSE, MAE, and R^2 metrics.
3. Evaluate and validate the predictive performance of the developed models In-situ only, Satellite-derived only, and Hybrid (combined) configurations using Independent T-test analysis to statistically assess the similarity between predicted and actual microplastic concentrations across sampling sites, thereby determining model reliability and consistency.
4. Generate a GIS-based microplastic accumulation map representing the predicted microplastic concentration across Southern Laguna De Bay to provide actionable insights for policymakers and environmental managers in support of targeted monitoring, mitigation strategies, and regulatory planning.

Significance of the Study

This study is highly significant as it pioneers a data-driven framework for predicting and mapping microplastic accumulation using remote sensing

data, water quality indicators, and advanced machine learning and deep learning techniques. By integrating a Multilayer Perceptron (MLP) Autoencoder for feature learning with ensemble models such as Random Forest (RF), Gradient Boosting, and XGBoost, the research introduces a systematic and scalable approach for identifying potential microplastic hotspots in Laguna de Bay. This approach enhances environmental monitoring accuracy, facilitates data-driven decision-making, and supports the sustainable management of aquatic ecosystems.

The key beneficiaries of this study include environmental management agencies, researchers, policymakers, and local communities who are deeply affected by the environmental and health impacts of microplastic pollution. By offering an accurate, cost-effective, and real-time assessment of microplastic accumulation, the study empowers stakeholders to allocate resources strategically, prioritize cleanup efforts, and implement sustainable management practices that safeguard marine ecosystems and public health.

This study also aligns with and directly supports several United Nations Sustainable Development Goals (SDGs), reinforcing its broader significance in global environmental initiatives. Specifically, it contributes to SDG 6 (Clean Water and Sanitation) by promoting water quality management, SDG 13 (Climate Action) by supporting climate resilience through ecosystem preservation, and SDG 14 (Life Below Water) by addressing the urgent issue of marine pollution. Through the application of innovative technology in environmental monitoring, the research advances interdisciplinary efforts to

protect biodiversity, sustain fisheries, and ensure the long-term viability of aquatic resources.

Government Agencies. This study provides environmental regulatory bodies such as the Department of Environment and Natural Resources (DENR), Laguna Lake Development Authority (LLDA), and related agencies with a data-driven methodology for monitoring microplastic pollution. By applying machine learning techniques such as Random Forest, Gradient Boosting, and XGBoost to remotely sensed data and water quality metrics, the research enables timely detection of pollution trends, facilitates evidence-based policymaking, and supports the development of targeted remediation strategies. Government institutions can utilize these findings to design and enforce more effective policies on waste management, water resource conservation, and ecosystem rehabilitation.

Academia and Research Institutions. This study contributes significantly to the growing body of knowledge surrounding the application of artificial intelligence in environmental science, particularly for freshwater ecosystems like Laguna de Bay. It offers a novel methodological framework that future researchers can expand upon to enhance microplastic pollution mapping in other water bodies or regions worldwide. Moreover, the interdisciplinary nature of the study bridges the gap between remote sensing, machine learning, and environmental monitoring, encouraging collaborative research efforts across scientific domains. The findings may also inspire the development of new predictive models and environmental management tools, further advancing the role of technology in ecosystem preservation.

Community. This study is crucial for the communities residing near Laguna de Bay, whose livelihoods, health, and overall well-being are intricately tied to the lake's ecological status. By enabling early identification of microplastic pollution hotspots, the research supports proactive community actions, raises public awareness, and fosters collective efforts toward sustainable waste management practices. Ultimately, the study aims to improve water quality, preserve aquatic life, and ensure the continued availability of clean water and fishery resources for future generations.

The Researchers. For the researchers, this study represents a meaningful application of advanced analytical techniques in addressing real-world environmental challenges. It provides an opportunity to develop expertise in deep learning, geospatial analysis, and environmental modeling, while also contributing to the greater societal good. By working at the intersection of technology and sustainability, the researchers not only refine their technical and critical thinking skills but also position themselves as advocates for science-based solutions to pressing environmental issues.

Future Researchers. Future researchers stand to benefit greatly from this study, which lays a strong foundation for subsequent work on environmental monitoring using AI technologies. The methodological framework, including the fusion of CNN feature extraction with Random Forest, Gradient Boosting, and XGBoost predictive modeling, can serve as a reference for expanding pollution detection efforts beyond microplastics to other pollutants such as heavy metals or oil spills. Moreover, the study opens avenues for integrating remote sensing with Internet of Things (IoT) systems

and real-time monitoring networks, inspiring further innovation in the field of environmental sustainability and protection.

Scope And Limitations

This study aims to develop a predictive model for mapping microplastic accumulation in the southern part of Laguna de Bay using satellite-based remote sensing, in-situ water quality metrics, and a hybrid deep learning and machine learning approach, specifically a Multilayer Perceptron (MLP) Autoencoder integrated with Random Forest (RF), Gradient Boosting, and XGBoost. Sentinel-2 satellite imagery was utilized to derive remote sensing indicators such as the Normalized Difference Chlorophyll Index (NDCI) and the Normalized Difference Turbidity Index (NDTI), while in-situ water quality parameters including dissolved oxygen (DO), pH, biochemical oxygen demand (BOD), and fecal coliform were used as complementary predictors. These datasets were combined to train a predictive model capable of identifying potential microplastic concentration zones within freshwater systems. The model was developed and validated using both satellite-derived and in-situ data specific to Laguna de Bay, ensuring locally relevant and ecologically grounded outputs.

However, this study has certain limitations. The predictive model infers microplastic presence indirectly through correlated environmental indicators rather than direct satellite detection. Consequently, the model's accuracy depends on the quality, consistency, and spatial resolution of both the satellite imagery and the field-collected data. Remote sensing observations may also be influenced by atmospheric conditions, water surface reflections, and cloud

cover, which can reduce image quality and introduce uncertainty. Furthermore, the unavailability of Sentinel-2 data for the month of July resulted in a temporal gap that may limit the model's ability to capture continuous seasonal variations.

The scope of this study is restricted to surface-level microplastic assessment and does not include subsurface or sediment-bound microplastics. Specifically, microplastic sampling was conducted in surface waters at varying distances from the shoreline, covering both near-shore zones and shallow shoal areas where accumulation patterns are most influenced by tidal movement, runoff, and sediment deposition. This approach targeted zones with high hydrodynamic activity to better reflect transport dynamics in the lake. Spatial grid-based sampling across the southern portion of Laguna de Bay was conducted to capture variability in microplastic distribution patterns, ensuring representation of diverse hydrological conditions. Model predictions focus solely on estimating microplastic concentration in milligrams per liter (mg/L) based on environmental proxies and do not classify microplastics by type, size, or polymer composition. This constraint limits the model's ability to differentiate between varying microplastic sources or behaviors within the aquatic system. Lastly, the limited availability of localized research on the ecological impacts of microplastics in Laguna de Bay constrains the broader environmental interpretation of the results, particularly concerning long-term effects and the interaction of microplastics with other pollutants like heavy metals or organic contaminant

CHAPTER 2

THEORETICAL FRAMEWORK OF THE STUDY

Review of Related Literature and Studies

This chapter provides a comprehensive review of related literature on the satellite-based mapping of high-concentration microplastic zones in Laguna de Bay using remote sensing and machine learning. The chapter offers a general review of current studies of microplastic pollution, remote sensing, machine learning, and how these have been used or combined in applications of environmental monitoring. The chapter offers an introduction to how existing studies utilize satellite data for detection of water contamination, in this case, deposition of microplastics, as well as the major methods adopted in the various studies. It also delivers a review of how existing studies make use of integration of remote sensing and machine learning in applications for environmental monitoring. The chapter finalizes by discussing the role played by these technologies in mitigation of pollution concerns of Laguna de Bay.

Microplastics Pollution in Water Bodies

Microplastic pollution has recently gained much attention as an environmental issue and freshwater ecosystems are the most affected. Rochman et al. (2015) suggested that microplastics come from different sources, such as industrial discharges, degradation of plastic waste, and synthetic materials that are released during laundry detergents. These pollutants are of high risk to aquatic organisms and water quality as a whole and lead to the bioaccumulation in the food chain. Li et al. (2018) in their

study established that microplastics are almost everywhere in major water bodies and thus there is the need to sustain the monitoring and adopt measures to reduce pollution. Researchers have called for the use of better detection methods to determine the level of pollution and the effects of pollution on the freshwater systems.

The role of hydrodynamics in microplastic pollution distribution cannot be overlooked. Tracing the flow and accumulation of this contaminant in both freshwater and marine environments, Alfonso et al. (2021) found the water currents and turbulence exert strong affective force on their transport. They call for high-resolution monitoring tools to record microplastic migration effectively. Winton et al. (2020) underscored the ecological impacts of plastic pollution in freshwater systems and identified some knowledge gaps pertaining to the variegated effects of microplastics on aquatic biodiversity and ecosystem function. So, if findings of the above works reinstated anything: that would be to suggest immediate development strategies across the globe/use targeted strategies to avert further contamination and abuse posed by microplastic pollution. Aside from affecting aquatic life, microplastic pollution endangers primary producers in freshwater ecosystems. The research done by Wu et al. (2019) investigated how microplastic impacted the abundance of some freshwater algae like Scenedesmus sp., Chlorella sp., and Microcystis aeruginosa. Such interference might reverberate through cascading effects on food web dynamics because algae act as primary producers producing oxygen and nutrition for other organisms. The same focus was reflected in Araújo & Malafaia (2020), where behavioral changes in neotropical tadpoles upon exposure to microplastics were documented so that

short-exposed tadpoles may switch with reduced caution towards predators. Such findings thus raise serious alarm for investigating the aquatic ecosystem-wide consequences that microplastic contamination may incur.

Beginning from the scientific basis to policy interventions and community awareness, all three must be combined to tackle microplastic pollution. Azevedo-Santos et al. (2021) stressed stricter implementation of rules curbing the amount of plastics entering freshwater systems. The state of the art detection and monitoring technologies must come into play so as to accurately assess pollution levels. With knowledge of microplastic transport, accumulation, and environmental effects enhanced, strategies may be developed for more effective protection of aquatic ecosystems from further degradation.

Microplastics in Laguna de Bay

Microplastic contamination is a great concern with Laguna de Bay being the largest freshwater lake in the Philippines. Espiritu et al. (2021) found that plastic waste is among the major contributors to pollution affecting the quality of water and aquatic ecosystems. The study recommends proper monitoring measures to curb and fight pollution. In addition, Azevedo-Santos et al. (2021) put emphasis on mobilizing policy initiatives and community action to combat pollution. According to Garaba and Dierssen (2018), successful conservation endeavors will require a combination of technological developments and social involvement to abate environmental degradation.

Recent satellite-based attempts to map water quality in Laguna de Bay yielded some encouraging results. David et al. (2020) assessed water

turbidity and chlorophyll levels within the context of Landsat imagery to identify patterns of pollution and the seasonal variations therein. Such studies could greatly benefit from machine learning incorporated into them to enhance detection accuracy to facilitate targeted reduction actions through more precise monitoring techniques. Moreover, data fusion techniques between remote sensing and in-situ measurements can help refine assessments of water quality and inform decisions considerably better.

Ecological and Health Impacts of Microplastics

Microplastics pose a significant threat to aquatic ecosystems and human health. In the aquatic food chain, microplastics are ingested by aquatic organisms like fish and plankton, resulting in changes in feeding habits, growth, and reproduction (Azevedo-Santos et al., 2021). Microplastics can carry toxic chemicals, such as heavy metals and persistent organic pollutants (POPs), with the potential to bioaccumulate in food webs and further affect biodiversity (Garaba & Dierssen, 2018). On the health front for humans, ingestion of microplastics through contaminated food, particularly seafood, has been linked to inflammation, oxidative stress, and endocrine disruption (Deocaris et al., 2023). Furthermore, microplastics have the ability to cross cellular membranes, resulting in harm to human tissues and organs, and therefore the need to further research their long-term ecological and health effects (Garaba & Dierssen, 2018). Recent studies also highlight atmospheric transport of microplastics, which deposit into freshwater systems like Laguna de Bay. This pathway exacerbates local contamination risks and underscores the urgency for region-specific monitoring strategies.

Traditional Microplastic Detection Methods

Among other methods developed to detect microplastics, visual sorting and microscopy have been a key technique since early days. Any of the techniques used for microplastic identification are examined in a review by Hidalgo-Ruz et al. (2012), stressing the role of stereo and optical microscopes in differentiating between microplastics and natural particles. In this method, the particles to be tested are sorted on a microscope slide according to their size, shape, and color by visual inspection. While this technique allows direct observation of microplastics, it is usually time-consuming and highly susceptible to human errors, hence considered rather ineffective for large-scale environmental monitoring efforts.

The techniques employed for separating microplastic particles from water samples commonly include sieving and filtration. Claessens et al. (2013) devised techniques that involve sieving the water samples through varied mesh-sloom sizes, which retain plastic particles within a set size range. This method is often combined with density separation methods to help isolate microplastics from organic and inorganic particles separately. Filtration and sieving depend on the mesh size as well as the level of the sample's contamination, hence very much important in the sense of optimizing these parameters for proper identification.

Density separation takes advantage of the differing density of plastics and other particles in the sample. Examples given by Imhof et al. (2016) show that microplastics can be isolated from sediments using liquid densities such as sodium chloride (NaCl) or zinc chloride (ZnCl2). In principle, microplastics

in a water sample will float while denser solid particles will sink when placed in high-density solutions. This separation allows for easier collection and identification of microplastics. This method is especially useful in the extraction of microplastics from complex environmental samples such as wastewater effluent or lake sediment.

To achieve improved visualization under fluorescence microscopy, researchers developed staining techniques using a number of fluorescent dyes. Nile Red staining, as introduced by Maes et al. (2017), binds to the hydrophobic plastic particles, providing them with specific fluorescence under required wavelengths of light. This method improved the detection of small microplastics that are often difficult to distinguish using traditional microscopy alone. The fluorescence approach provides much more efficient and semi-automated approaches for detection of microplastics in environmental samples, improving accuracy and reducing labor.

Remote Sensing Approaches for Microplastic Detection in Freshwater Systems: Limitations and Innovations

There are substantial operational and technical constraints on the methods currently used for microplastic identification in freshwater environments. Although they produce accurate particle identification, traditional laboratory techniques like microscopy and density separation (Hidalgo-Ruz et al., 2012; Claessens et al., 2013) are limited by their labor-intensive nature and small-scale application. Despite the fact that remote sensing tools offer scalable monitoring, their application is constrained by several factors, including the lack of standardized validation protocols for

inland waters, spectral interference from organic matter and suspended sediments, and sensor resolution limitations that make it difficult to reliably detect particles smaller than 5mm (Garaba & Dierssen, 2018). These problems are made worse by a severe lack of data, since most machine learning training data originates from marine studies with minimal

Recent breakthroughs illustrate potential to transcend these constraints through combined technological means. Multispectral satellite platforms, especially Sentinel-2, exhibit potential in remote sensing of floating plastic waste based on spectral signature detection, where shortwave infrared absorption features are particularly efficient for polymer detection (Garaba & Dierssen, 2018). The incorporation of cutting-edge machine learning frameworks, including generative adversarial networks (Jamali & Mahdianpari, 2021), has also improved detection precision by eliminating spectral ambiguities that are characteristic of optical remote sensing. If deployed via cloud-based processing architectures, these combined systems may provide near-real-time monitoring capacity - although effective freshwater applications necessitate careful adjustment to consider the distinct hydro-optical characteristics of inland water bodies. These technological advancements, as not yet maximally optimized for freshwater ecosystems such as Laguna de Bay, provide the essential groundwork for the introduced MLP Autoencoder-Ensemble framework explained in this research.

Satellite Indices and Their Connection to Microplastic

Satellite-derived indices offer promising avenues for monitoring microplastic pollution in freshwater environments. Indices such as NDVI and

NDMI, originally used to gauge vegetation and moisture, are now recognized for their ability to enhance remote detection of plastic debris due to differences in spectral reflectance between plastics and organic matter (Garaba & Dierssen, 2018; Themistocleous et al., 2022; Biermann et al., 2020; Mishra & Mishra, 2012). The Plastic Index (PI), developed to specifically improve floating debris detection, leverages unique synthetic polymer signatures, while NDCI provides insight on biological activity, helping identify risk hotspots in surface waters (Themistocleous et al., 2022; Mishra & Mishra, 2012). Integrating these indices yields more accurate remote classifications and forms a foundation for innovative microplastic mapping solutions (Garaba & Dierssen, 2018; Biermann et al., 2020; Themistocleous et al., 2022; Mishra & Mishra, 2012).

The Normalized Difference Vegetation Index (NDVI), calculated as (NIR - RED)/(NIR + RED), traditionally monitors vegetation health but has been adapted to detect macroplastic debris by exploiting spectral differences between plastics and organic matter. Biermann et al. (2020) used NDVI with Sentinel-2 imagery, achieving 86% accuracy in classifying floating macroplastics in coastal waters when paired with the Floating Debris Index (FDI). This approach identifies vegetation traps where microplastics accumulate due to macroplastic fragmentation, though microplastics (<5 mm) are below satellite resolution, requiring ground-truth validation. Similarly, Sayed et al. (2023) applied NDVI with PlanetScope and Sentinel-2 data along the Egyptian northern coast, reporting 70-80% accuracy in detecting polymer-rich debris, with 72% of beach litter identified as plastics. NDVI's strength lies in mapping macroplastic sources that contribute to microplastic

pollution indirectly, but its reliance on proxies limits direct microplastic detection.

Complementing NDVI's vegetation focus, the Normalized Difference Moisture Index (NDMI), computed as (NIR - SWIR1)/(NIR + SWIR1), distinguishes macroplastics from wet surfaces in coastal environments. Sayed et al. (2023) used NDMI alongside NDVI to identify debris hotspots, achieving 70-80% accuracy by leveraging the shortwave infrared (SWIR) band to minimize interference from waterlogged sediments. NDMI supports microplastic monitoring by mapping macroplastic sources that fragment into microplastics through tidal or wave action, particularly in dynamic coastal systems. However, its effectiveness is constrained by the low spectral signal of microplastics and Sentinel-2's moderate resolution, necessitating integration with higher-resolution sensors.

In marine environments, the Normalized Difference Chlorophyll Index (NDCI) serves as a proxy for plastic pollution by detecting water quality changes, particularly in coastal zones. Biermann et al. (2021) applied NDCI with Sentinel-2 imagery to map macroplastic patches with subpixel coverage as low as 0.3%, though microplastics were undetectable due to low signal-to-noise ratios. NDCI identifies areas where plastic accumulation reduces light penetration, affecting phytoplankton and chlorophyll levels, providing indirect evidence of microplastic presence. Its reliance on proxies and need for cloud-free imagery limit its direct applicability, but combining NDCI with other water quality indices could enhance microplastic research, especially for inland lakes.

Unlike NDCI's indirect approach, the Normalized Difference Turbidity Index (NDTI) directly correlates water turbidity with microplastic concentrations. Topping (2025) demonstrated that NDTI, derived from Sentinel-2 imagery, aligns with in situ turbidity measurements ($r^2 = 0.0876$) and microplastic counts ($r^2 = 0.44$ via turbidity proxy) in the New River, VA, estimating up to 250 microplastic particles per liter in urban sites. NDTI's scalability offers a cost-effective alternative to field sampling, but moderate correlations highlight the need for better calibration with in situ data.

The Plastic Index (PI), calculated as NIR/(NIR + RED), targets macroplastic detection using plastics' high near-infrared reflectance. Themistocleous et al. (2020) achieved 92-98% accuracy with PI in Sentinel-2 imagery and machine learning models. Sakti et al. (2023) developed an Adjusted Plastic Index (API) for the Rancamanyar River, Indonesia, correcting for land cover interference to identify illegal dumping sites contributing to microplastic pollution. PI shares spectral bands (NIR and RED) with NDVI but emphasizes plastic-specific reflectance, making it complementary to NDVI's vegetation focus. Its direct application to microplastics is limited by spatial resolution, but hyperspectral sensors could improve sensitivity.

Autoencoders for Environmental Feature Learning

Autoencoders are unsupervised neural network architectures that learn compact representations through an encoder-decoder framework, mapping inputs to a low-dimensional latent space and reconstructing them from compressed codes (Hinton & Salakhutdinov, 2006). This architecture enables non-linear dimensionality reduction that has consistently outperformed linear

methods such as PCA on complex environmental datasets. In environmental monitoring, autoencoders compress high-dimensional satellite features while stabilizing downstream predictive models by extracting structured patterns from noisy measurements through unsupervised learning (Baig et al., 2023).

The autoencoder bottleneck layer serves as an information constraint that forces the network to prioritize the most salient features while discarding redundancy and noise (Hinton & Salakhutdinov, 2006). During training, the autoencoder minimizes reconstruction error between input and output, ensuring that the compressed latent representation captures essential characteristics of the environmental data. This unsupervised pre-training approach is particularly effective when labeled samples are limited but abundant unlabeled satellite imagery is available (Baig et al., 2023).

In this study, an autoencoder implemented as a multilayer perceptron (MLP) is trained on Sentinel-2 satellite imagery to extract 16-dimensional latent features representing chlorophyll-a and turbidity patterns. Unlike convolutional neural networks that operate on spatial image grids, the MLP autoencoder processes pre-extracted statistical features from satellite patches, providing non-linear dimensionality reduction suited for tabular environmental data. The bottleneck layer compresses input features while preserving information content sufficient for downstream prediction tasks, effectively filtering noise and capturing the most discriminative spatial-spectral patterns. These latent representations enhance the integration of remote sensing data with machine learning models for microplastic concentration prediction in Laguna de Bay.

Hybrid Unsupervised-Supervised Pipeline: Autoencoder Feature Extraction with Ensemble Learning

To address limited labeled data in environmental microplastic monitoring, this study employs a two-stage pipeline combining unsupervised feature extraction with supervised ensemble learning. In the first stage, an MLP autoencoder learns compressed 16-dimensional latent representations from unlabeled Sentinel-2 imagery through reconstruction-based training (Hinton & Salakhutdinov, 2006). The autoencoder is trained solely on satellite data without microplastic labels, enabling it to leverage abundant unlabeled observations for learning generalizable spatial-spectral features.

The encoder compresses input features through successive hidden layers into a bottleneck layer, producing compact latent codes that capture salient environmental characteristics while discarding noise and redundancy (Hinton & Salakhutdinov, 2006; Baig et al., 2023). The decoder is used only during training to enforce reconstruction fidelity and is omitted at inference to reduce computational overhead. Resulting 16-dimensional latent codes are concatenated with 8 in-situ water quality parameters (pH, dissolved oxygen, biochemical oxygen demand, fecal coliform, turbidity, chlorophyll-a, total suspended solids, temperature) to form a 24-dimensional hybrid feature space combining satellite-derived and field-measured attributes.

In the second stage, these fused features are provided to supervised ensemble models Random Forest to aggregate decorrelated trees for robustness and reduced variance (Breiman, 2001), Gradient Boosting trained to minimize residuals in a stagewise manner for improved generalization

under noise (Friedman, 2001), and XGBoost leveraging regularized gradient-boosted trees with shrinkage and column subsampling to enhance accuracy and computational efficiency (Chen & Guestrin, 2016). This hybrid pipeline enables the integration of unsupervised representation learning from abundant unlabeled satellite data with supervised prediction from limited labeled microplastic samples.

This design improves performance in environmental applications by mitigating noise, enhancing predictive accuracy, and supporting reliable lake-scale hotspot mapping. Random Forest and ensemble methods have demonstrated high utility in environmental prediction tasks, providing robust results even when data are noisy or sparse (Belgiu & Drăguţ, 2016; Fu et al., 2022). The autoencoder-ensemble pipeline requires fewer labeled samples by leveraging unsupervised pre-training on abundant unlabeled satellite observations, yielding interpretable low-dimensional embeddings that improve generalization and reduce overfitting risk in small-sample regimes (Hinton & Salakhutdinov, 2006; Baig et al., 2023).

Multi-Modal Feature Fusion: Integrating Autoencoder Latent Features with Structured Environmental Data

Environmental prediction tasks often require integrating heterogeneous data sources satellite imagery and tabular in-situ measurements in this case. Multi-modal feature fusion refers to the process of combining features from different modalities into a unified representation that can be used for downstream prediction (Zou et al., 2025). In this study, the 16-dimensional autoencoder latent feature vector extracted from Sentinel-2 imagery was

concatenated with 8 in-situ water quality parameters to form a 24-dimensional hybrid feature space.

This approach allows the model to leverage both compressed latent representations learned by the autoencoder from multispectral data and direct environmental measurements that may not be captured in satellite observations. Zou et al. (2024) developed a fusion framework integrating deep learning features with XGBoost for multi-constituent prediction, demonstrating that combining learned representations with structured features significantly improves forecasting accuracy. Feature fusion has been shown to improve model performance in environmental applications by providing complementary information that enhances predictive power (Zhang et al., 2022). However, careful normalization and scaling are required to prevent features with larger numerical ranges from dominating the learning process. Recent advancements in multi-modal learning frameworks have demonstrated that effective integration of diverse data types leads to more robust and accurate environmental monitoring systems (Biermann et al., 2020).

Stacked Ensemble Learning: Random Forest, Gradient Boosting, and XGBoost Integration

Ensemble learning combines multiple models to improve predictive performance and robustness by leveraging their complementary strengths (Ahn et al., 2023). In this study, a stacked ensemble approach integrating Random Forest (RF), Gradient Boosting (GB), and XGBoost (XGB) was employed as the final prediction model. Stacking works by training multiple base learners on the same dataset and combining their outputs to create a

unified prediction framework that captures diverse patterns in the data (Liu et al., 2022). Random Forest, developed by Breiman (2001), aggregates decorrelated decision trees through bagging to reduce variance and prevent overfitting. Gradient Boosting, introduced by Friedman (2001), sequentially builds weak learners that correct the errors of previous trees through gradient optimization. XGBoost, a gradient boosting algorithm developed by Chen and Guestrin (2016), excels at capturing complex nonlinear relationships and feature interactions through iterative tree-based learning with regularization and enhanced computational efficiency.

The combination of these algorithms achieved an LOOCV R^2 of 0.881 in this study, outperforming individual models. Ahn et al. (2023) demonstrated that hybrid ensemble models combining multiple machine learning algorithms prevent overfitting while improving accuracy, achieving superior performance with multiple evaluation metrics. Similarly, Liu et al. (2022) developed a hybrid MLP Autoencoder-Ensemble framework model that successfully integrated deep learning features with gradient boosting for improved prediction accuracy in water pollution detection. This stacked approach is particularly advantageous in environmental modeling where datasets are small and feature relationships are intricate, as it reduces variance while maintaining low bias (Zou et al., 2025). Recent research has shown that stacking-based ensemble methods consistently outperform single models across diverse prediction tasks, including environmental pollution assessment and water quality forecasting (Huang et al., 2020). Such methods prove especially effective for Laguna de Bay's heterogeneous conditions.

Data Normalization Techniques

Data normalization is essential when integrating features measured on different scales, as it prevents certain variables from dominating the model due to their magnitude rather than their informational content (Ahn et al., 2023). Z-score normalization, implemented via the Standard Scaler, was used for both in-situ measurements and satellite data in this study. Z-score normalization transforms data to have a mean of zero and a standard deviation of one, making it suitable for variables that follow approximately normal distributions and may contain outliers.

Proper normalization is critical for model convergence and performance, especially in hybrid frameworks that combine deep learning with traditional machine learning algorithms (Liu et al., 2022). Recent studies have emphasized the importance of appropriate preprocessing techniques in ensemble learning models to ensure optimal performance across diverse datasets (Zou et al., 2025). The choice of Z-score normalization was based on its robustness for varying data distributions and its suitability for hybrid modeling frameworks (Huang et al., 2020).

Data Augmentation for Satellite Imagery

Data augmentation artificially increases the size and diversity of training datasets by applying transformations that preserve the underlying data distribution (LeCun et al., 2015). In this study, spatial transformations such as rotations, horizontal and vertical flips, and slight scaling were applied to Sentinel-2 image patches to simulate variations in satellite viewing angles and environmental conditions. Augmentation is particularly important when

working with small datasets, as it helps prevent overfitting by exposing the model to a wider range of input variations during training (Jiao et al., 2021).

However, it is critical that augmentations are physically realistic and do not introduce artifacts that could mislead the model. For example, rotating a satellite image is valid because the spatial patterns of microplastic accumulation do not depend on cardinal direction. Recent research has shown that appropriate augmentation strategies significantly enhance model robustness and generalization capabilities, particularly in environmental applications (Biermann et al., 2020). Data augmentation has become standard practice in remote sensing applications and has been shown to improve model generalization and robustness, particularly when combined with CNN-based architectures (Simonyan & Zisserman, 2014).

Model Performance Metrics for Small-Sample Environmental Prediction

Evaluating model performance requires metrics that appropriately reflect prediction accuracy and generalization ability (Huang et al., 2020). In this study, Root Mean Square Error (RMSE), Mean Absolute Error (MAE), R^2 Score, and Explained Variance Score (EVS) were computed. RMSE measures the square root of the average squared prediction error, heavily penalizing large errors, making it sensitive to outliers. MAE computes the average absolute error, providing a more interpretable measure of average prediction deviation. R^2 Score indicates the proportion of variance in the target variable explained by the model, with values closer to 1 indicating better fit.

EVS is similar to R^2 but does not penalize bias in the predictions, making it useful for understanding how well the model captures variance

independent of systematic offset. Huang et al. (2020) demonstrated the importance of using multiple performance indices to comprehensively evaluate predictive models in environmental applications. Ahn et al. (2023) similarly employed multiple evaluation metrics including MSE, MAE, and R^2 to determine the most successful model in algal bloom prediction, with their hybrid ensemble model achieving superior performance across all metrics. These metrics collectively provide a comprehensive view of model performance, particularly in small-sample contexts where a single metric may be misleading (Zhang et al., 2023).

Water Quality Metrics of Laguna de Bay and Their Connection to Microplastic Pollution

Monitoring water quality is essential to understanding the presence and impacts of microplastic pollution in aquatic ecosystems. Key metrics such as chlorophyll-a, turbidity, total suspended solids (TSS), dissolved oxygen (DO), biochemical oxygen demand (BOD), pH, and fecal coliform each provide important perspectives on how microplastics interact with biological, chemical, and physical processes in water bodies. Together, these indicators not only reflect the overall ecological health of a system but also reveal how microplastics accumulate, persist, and influence aquatic environments such as Laguna de Bay.

Turbidity, which refers to the cloudiness or haziness of water caused by suspended particles, is an important measure in determining the microplastic pollution levels in aquatic bodies. Microplastics are suspended in the water column because of their small size and buoyant nature, which adds to

turbidity. These can be visible plastic debris or microplastics that are not detectable by the naked eye. Since turbidity is guided by any suspended particle, it gives an indirect measure of microplastic presence in water. Higher turbidity of the lake may thus translate to higher concentrations of microplastics since turbidity hampers the perception of water clarity. Thereby the levels of pollution, including microplastics, can be inferred through turbidity (mg/L) without the need for immediate physical detection (Matos et al., 2024).

Chlorophyll-a is understood as a biomass indicator for phytoplanktons and primary productivity in aquatic ecosystems. The influence of floating plastics on chlorophyll-a concentrations occurs in two major perspectives. First, microplastics may provide a site for algal attachment, which in turn encourages phytoplankton growth. Second, they can introduce nutrient imbalances into the water, aggravating eutrophication and algal blooms. This indirect interaction between microplastics and the aquatic environment subsequently affects chlorophyll-a concentrations, which are higher during algal blooms. Thus, monitoring chlorophyll-a concentrations in Laguna de Bay may provide an indirect means of assessing microplastic presence, based on the consequent shifts in algal activity influenced by plastic pollution. In this way, chlorophyll-a functions as a surrogate index for microplastic pollution, emphasizing the interrelationship between biological and physical drivers in aquatic systems (Raulo et al., 2023).

Total Suspended Solids (TSS) are a key parameter capturing the concentration of suspended organic and inorganic particles within the water column. Because microplastics are typically light, small, and buoyant or

semi-buoyant, many of them adhere to or are associated with suspended solids; thus they contribute to measured TSS (Frehland et al., 2020; Wolff et al., 2020). Studies in wastewater treatment plants have shown strong positive correlations between TSS levels and microplastic counts, meaning that high TSS often coincides with high microplastic presence (Wolff et al., 2020). Elevated TSS can therefore indicate potential zones of microplastic accumulation, especially when the particles are not easily distinguished visually. Moreover, high TSS reduces light penetration, alters aquatic habitat conditions, and complicates detection of microplastics by masking them within natural particulates (Frehland et al., 2020).

Fecal coliform bacteria, such as *Escherichia coli*, are widely used as indicators of sewage contamination in freshwater and coastal systems. Recent research shows that microplastics can act as “floating homes” for these bacteria, providing rough surfaces where microbes attach and form biofilms. This microplastic-based microbial community is often called the plastisphere. In fact, studies in urban rivers have found that fecal coliforms and other sewage-associated bacteria were living directly on microplastic particles, suggesting plastics can act as transporters of microbial pollution (McCormick et al., 2014). Other experiments show that the plastisphere can even encourage the exchange of genetic material, including antibiotic resistance genes, which raises additional public health concerns (Arias-Andrés et al., 2019). More recently, reviews have highlighted that microplastics may prolong the survival of fecal bacteria in water and increase the risk of pathogens spread in lakes and rivers (Zhong et al., 2023). These findings mean that high fecal coliform levels in Laguna de Bay may not only

point to sewage inputs, but could also reflect conditions where microplastics are helping harmful bacteria persist and spread.

The acidity or alkalinity of water, measured by pH, plays a big role in how microplastics behave in lakes and rivers. When water becomes more acidic or alkaline, plastics tend to break down faster and their surfaces change, making them more likely to bind with heavy metals or chemical pollutants (Liu et al., 2020). For example, one study found that plastics in lower pH conditions were much better at attracting metals, showing how water chemistry directly affects pollution risks (Godoy et al., 2019). Microbes that live on microplastic surfaces can also change the local pH around them, which shapes the tiny ecosystems, sometimes even including harmful bacteria, that form on plastic waste (Kirstein et al., 2016). Because of this, tracking pH levels alongside microplastics in Laguna de Bay is important, since even small changes in acidity could influence how plastics break down, what they carry, and which microbes thrive on them.

Dissolved Oxygen (DO) and Biochemical Oxygen Demand (BOD) are crucial for understanding how clean or polluted water is. When microplastics accumulate, especially when they become colonized by bacteria and other microbes, they can increase the biological activity in the water. That often raises BOD, because organisms consume more oxygen as they break down organic matter (including biofilms on plastics). This higher consumption can in turn reduce DO, creating low-oxygen or even hypoxic conditions in water bodies (where oxygen is too low for many aquatic organisms). One study in urban estuaries (Adyar and Cooum in India) looked at many water parameters

including DO and BOD alongside microplastic concentration they found mixed results, but some weak positive correlations between microplastic abundance and BOD, and weak negative with DO in certain samples (Somasundaram & Radhakrishnan, 2023). Also, lab experiments with aquatic plants (duckweed) showed that microplastic presence changed DO dynamics (higher DO in some treatments because plants weren't forming thick mats that block air exchange) but overall it indicates microplastics can disrupt the balance of oxygen in the water (Effects of Microplastic Contamination on the Aquatic Plant *Lemna minuta*, MDPI).

Computation of Microplastic Concentration and Its Correlation with Water Quality Metrics

The quantification of microplastics in aquatic systems traditionally involves physical sampling techniques such as direct water collection using clean glass containers, followed by density separation using a saturated sodium chloride (NaCl) solution. In this process, a 1-liter water sample is mixed in a glass beaker with the salt solution (density ≈1.2 g/cm³), allowing microplastics to float due to their lower density. After settling, the supernatant is filtered through fine pore filter paper (0.45–5 µm) using vacuum or gravity filtration. The retained particles are dried in petri dishes and examined under a stereo microscope for counting and classification. Essential materials include 1 L glass sampling bottles, NaCl solution, beakers, stirring rods, vacuum or gravity filtration setup, filter paper, petri dishes, tweezers, and a microscope following the procedures detailed by Masura et al. (2015), which standardize

this density-based method for isolating synthetic particles from environmental water samples.

$$Microplastic\ concentration = \frac{Number\ of\ particles}{Volume\ of\ sampled\ water\ (m^3)} \quad (1)$$

This concentration value provides a standardized metric for assessing pollution levels. Masura et al. (2015), through NOAA, outlined a widely adopted protocol for synthetic particle identification in environmental water samples. More advanced quantification methods involve Fourier-transform infrared spectroscopy (FTIR) and Raman spectroscopy, offering higher sensitivity and polymer-specific identification (Mintenig et al., 2017; Lahens et al., 2018). FTIR microscopy further enhances accuracy in particle classification in both marine and freshwater systems (Prata et al., 2019).

In addition to direct measurements, predictive modeling approaches are increasingly utilized to estimate microplastic accumulation based on satellite and in-situ water quality data. In this study, microplastic probability scores are computed using a hybrid MLP Autoencoder-Ensemble framework, where spatial features from Sentinel-2 imagery and water quality parameters are used as inputs:

$$P_{microplastic} = XGBoost(F_{combined};\ \theta) \quad (2)$$

This modeling approach enables the generation of continuous microplastic risk maps without needing dense ground-truth data. Similar remote sensing-based predictive frameworks were explored by Zhang et al. (2023), who demonstrated that chlorophyll-a and suspended solids could

serve as proxies for plastic presence in regression models. Eamrat et al. (2022) also observed strong positive correlations between turbidity and microplastic levels in urban canals. Moreover, Kanhai et al. (2018) highlighted how seasonal dynamics and hydrological variations influence microplastic distribution and detectability through water quality shifts.

These correlation and model-based approaches allow for spatial-temporal microplastic assessment, offering cost-effective and scalable alternatives to field-only surveys, and strengthening early-warning and monitoring systems in vulnerable aquatic systems like Laguna de Bay.

Literature Matrix

Table 1. Matrix of Review of Related Literature

AUTHOR(S)	YEAR	TITLE	ALGORITHM / MODEL	FINDINGS
Danilov, A., & Serdiukova, E.	2024	Review of methods for automatic plastic detection using satellite images and ML	Review of ML models + satellite imaging	ML models achieved 70–90% accuracy; require expert verification
Espiritu et al.	2024	Microplastic Pollution in Laguna de Bay	Field Sampling	60% increase in plastic waste over 5 years
Maes et al.	2024	Nile Red staining of microplastics	Fluorescence microscopy	Nile Red increased detection 90%
Sakti, A. D., et al.	2023	Identification of illegally dumped plastic waste in a highly polluted river in Indonesia using Sentinel-2 satellite imagery	Adjusted Plastic Index (API) + Random Forest	API improved accuracy over PI; validated via UAV imagery
Garaba & Dierssen	2022	Remote Sensing of Microplastics in Open Waters Sentinel-2 satellite imagery	Satellite Remote Sensing	85% accuracy in large debris detection
Claessens et al.	2022	Filtration and Sieving Techniques for Microplastic Detection	Density Separation	90% purity in isolating

				microplastics from sediment
Jamali & Mahdianpari	2021	Plastic Detection via Remote Sensing using GANs	GAN (Generative Adversarial Network)	92% accuracy in plastic detection using remote sensing
Hu, C..	2021	Remote detection of marine debris using satellite observations in the visible and near infrared spectral range: Challenges and potentials	Sensitivity study using Sentinel-2 MSI	Macroplastics detectable at >0.3% pixel coverage; microplastics undetectable
Hidalgo-Ruz et al.	2021	Microplastic Identification Methods in Freshwater Ecosystem	Literature Review	Visual sorting and microscopy were 80% effective
Alfonso et al.	2021	Hydrodynamic Modeling of Microplastic Transport in Coastal Waters	Remote Sensing, Hydrodynamic Modeling	75% correlation between currents and plastic drift
Biermann, L., et al.	2020	Finding plastic patches in coastal waters using optical satellite data	FDI + NDVI + Naïve Bayes on Sentinel-2 MSI	70% pollution from land-based sources
Themistocleous, K., et al..	2020	Investigating detection of floating plastic litter from space using Sentinel-2 imagery	Plastics Index (PI) + RNDVI with Sentinel-2/UAV	PI provided strongest spectral separation
Rochman et al.	2020	Sources and Impacts of Microplastics in Freshwater Ecosystems	Literature Review	70% pollution from land-based sources
Wu et al.	2019	Effects of Microplastic Exposure on Freshwater Algae	Laboratory Experiment	25% decrease in algae growth rates due to microplastics
Li et al.	2018	Microplastic Pollution Distribution in Major Water Bodies	Field Sampling, Analytical Techniques	95% plastic concentration near shores, 50% microplastics

Synthesis and Relevance of the Study

The discussion of related literature and previous research presented in this chapter provides a solid conceptual foundation for the development of a predictive analytics model for microplastic deposition in Laguna de Bay. The integration of environmental surveillance, satellite remote sensing, and

advanced data analysis highlights the growing capability of technology to address complex environmental challenges. Recent advancements in satellite-based detection have demonstrated the potential of spectral indices for large-scale microplastic monitoring. Studies utilizing Sentinel-2 imagery have successfully employed indices such as NDVI, NDMI, NDCI, NDTI, and the Plastics Index (PI) to detect macroplastic debris with accuracies ranging from 70-90% (Biermann et al., 2020; Sakti et al., 2023; Danilov & Serdiukova, 2024). The Plastics Index, in particular, has shown strong spectral separation capabilities between plastic litter and water surfaces (Themistocleous et al., 2020). When combined with satellite data, these spectral indices enable large-scale monitoring, providing an effective and cost-efficient alternative to labor-intensive and time-consuming manual sampling, especially in expansive freshwater systems like Laguna de Bay.

However, current satellite capabilities face limitations in directly detecting microplastics due to their small size (<5mm) and low spectral signal, with detection thresholds requiring pixel coverage >0.3% (Hu, 2021). Despite these constraints, water quality proxies such as turbidity indices have shown moderate correlations ($R^2 \approx 0.65$) with microplastic concentrations, supporting proxy-based estimation approaches (Topping, 2024). This limitation necessitates the integration of satellite-derived features with in-situ measurements to achieve robust predictive models for microplastic monitoring in aquatic environments.

The hybrid approach integrating unsupervised feature learning and supervised ensemble learning has proven particularly effective in enhancing

the prediction accuracy of environmental models. Specifically, the use of Multi-Layer Perceptron (MLP) Autoencoders for extracting compressed latent features from statistical summaries of spectral indices, combined with Random Forest (RF), Gradient Boosting, and XGBoost for prediction, has demonstrated strong performance in recent environmental monitoring studies. Unlike Convolutional Neural Networks (CNNs) which process raw image pixels, MLP Autoencoders are optimized for tabular data, making them particularly suitable for processing pre-extracted statistical features (mean, standard deviation, variance, percentiles) from satellite spectral indices. These ensemble methods, which combine multiple decision trees, help in modeling complex decision boundaries, reducing overfitting, and improving the generalizability of the model (Breiman, 2001; Friedman, 2001; Chen & Guestrin, 2016). The MLP Autoencoder-ensemble pipeline requires fewer labeled samples by leveraging unsupervised pre-training on abundant unlabeled satellite observations, yielding interpretable low-dimensional embeddings that improve generalization and reduce overfitting risk in small-sample regimes. This hybrid approach not only estimates microplastic concentrations but also generates spatially fine-grained GIS-based maps that emphasize high-risk areas within Laguna de Bay, supporting targeted interventions and better resource allocation.

Furthermore, several studies emphasize the importance of careful preprocessing and data standardization, particularly when handling satellite-derived spectral indices and in-situ water quality metrics. Z-score normalization for both in-situ measurements and satellite-derived features has been highlighted as an effective method to mitigate discrepancies in scale and

unit measurement between variables. This preprocessing step ensures that input data is consistently scaled, preventing biased feature extraction and supporting the stability and accuracy of the model's predictions. Data augmentation techniques, including spatial transformations for satellite features and physics-guided noise injection for in-situ measurements, further enhance model robustness by creating more diverse training datasets.

The literature highlights the importance of rigorous model evaluation using metrics such as RMSE, MAE, and R^2 to assess predictive accuracy. Cross-validation (CV) ensures reliable performance, especially with small environmental datasets. Feature importance analysis in Random Forest, Gradient Boosting, and XGBoost helps identify key satellite-derived indices and water quality parameters influencing microplastic accumulation, enhancing model accuracy, input optimization, and interpretability for effective environmental monitoring.

The integration of georeferencing procedures ensures accurate spatial alignment between satellite-derived features and field sampling sites, enabling precise overlay and comparison of datasets. This spatial-temporal integration supports the development of continuous GIS-based prediction surfaces that can identify potential microplastic accumulation hotspots across Laguna de Bay. By combining the scalability of satellite remote sensing with the precision of in-situ measurements through weighted ensemble approaches, this study addresses the critical gap in microplastic monitoring for large freshwater systems, providing a replicable framework for environmental assessment and policy-making.

Conceptual Framework

The microplastic deposition prediction and mapping process, shown in Figure 1, employs a hybrid framework that integrates an MLP Autoencoder for unsupervised feature extraction with ensemble machine learning models (Random Forest, Gradient Boosting, and XGBoost). The approach utilizes Sentinel-2 satellite imagery to derive spectral indices such as NDVI, NDMI, NDCI, NDTI, and the Plastics Index (PI), together with in-situ water quality parameters including pH, dissolved oxygen, biochemical oxygen demand, total suspended solids, and fecal coliform levels.

In the data preprocessing phase, satellite imagery undergoes statistical feature extraction (mean, standard deviation, variance, percentiles) from each spectral band, followed by Min-Max scaling for normalization. In-situ water quality data are standardized through Z-score normalization. Spearman correlation analysis is performed to assess monotonic relationships between environmental indicators and microplastic concentrations. The datasets are temporally aligned by matching satellite acquisition dates with field sampling records and spatially aligned through georeferencing, then split into training, validation, and testing sets.

In the feature extraction phase, high-dimensional satellite features are processed through an MLP Autoencoder trained in unsupervised mode. The encoder compresses input features through successive hidden layers into a 16-dimensional latent bottleneck representation that captures dominant spatial-spectral patterns while reducing noise and redundancy. These 16-dimensional latent codes are concatenated with 8 in-situ water quality

parameters to form a 24-dimensional hybrid feature space, integrating satellite-derived and field-measured attributes.

The ensemble models (Random Forest, Gradient Boosting, and XGBoost) are trained separately on both in-situ and satellite-encoded datasets to predict microplastic accumulation, with hyperparameter tuning using grid search and cross-validation to ensure robustness and minimize overfitting. Predictions from in-situ models and satellite models are combined through weighted integration, assigning higher weight to in-situ data (0.8) due to their lower measurement uncertainty, while satellite predictions contribute 0.2 to the final ensemble output.

Model evaluation is conducted by comparing predictions to actual observations using RMSE, MAE, and R^2 scores. K-fold cross-validation is applied to assess generalization capability across different data subsets. Feature importance analysis is performed using XGBoost's built-in scoring mechanism to identify the most influential satellite indices and water quality parameters contributing to microplastic predictions.

Finally, the trained hybrid MLP Autoencoder–Ensemble model is applied to estimate and map microplastic distribution in the southern part of Laguna de Bay. The output is visualized as spatial distribution maps using Geographic Information System (GIS) software, highlighting microplastic accumulation hotspots and supporting targeted environmental management efforts. These maps provide decision-makers with actionable insights for prioritizing monitoring locations and implementing mitigation strategies.

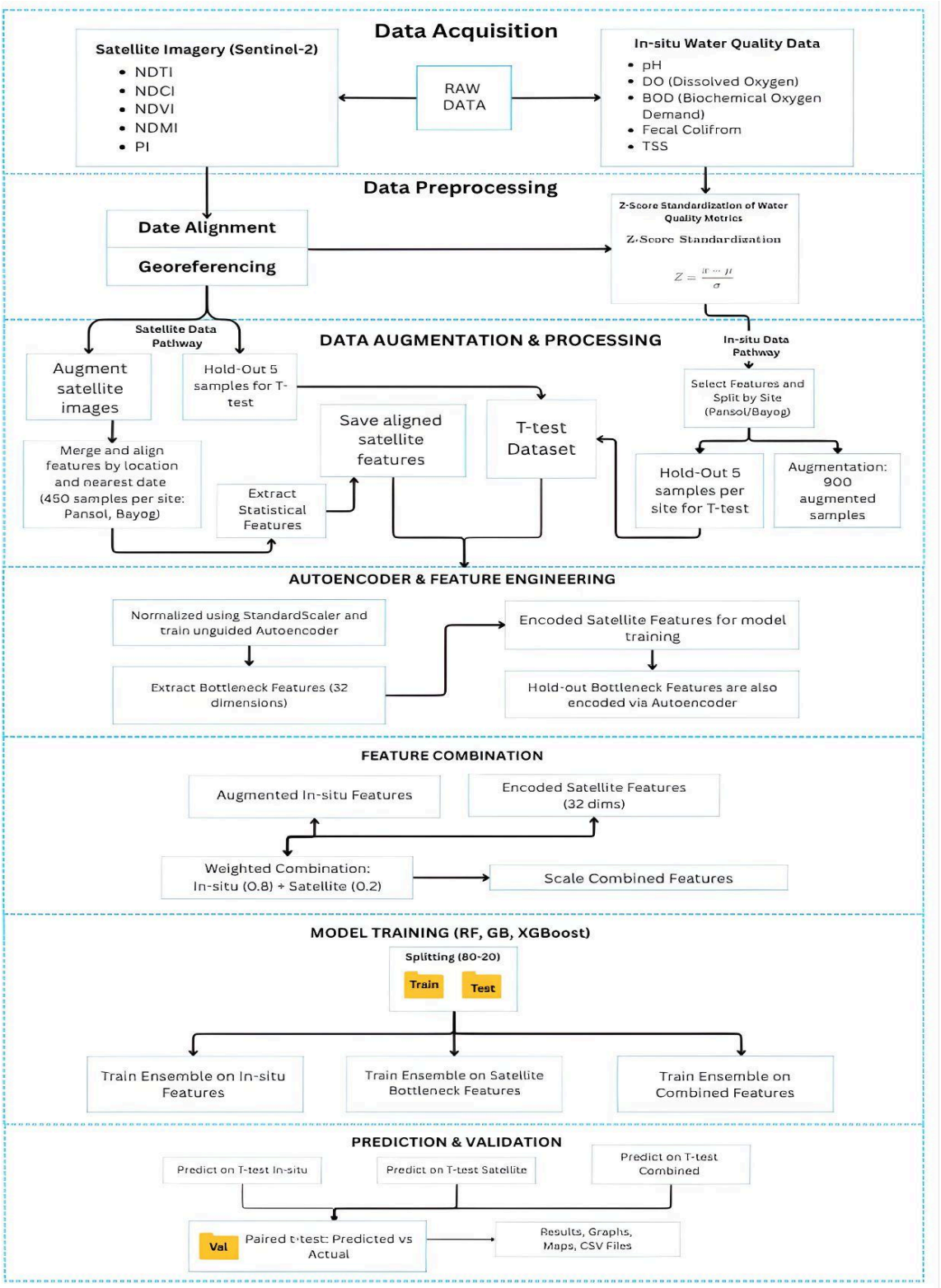

Figure 1. Conceptual Framework

CHAPTER 3

METHODOLOGY

Study Area

Laguna de Bay, the largest inland freshwater lake in the Philippines, spans approximately 90,000 hectares and borders the provinces of Laguna, Rizal, and parts of Metro Manila. The lake stretches about 53 kilometers long and 33 kilometers wide, with an average depth of around 2.5 meters. It supports various economic and ecological functions, including fisheries, irrigation, transportation, flood regulation, and water supply for domestic and industrial use. Despite its significance, Laguna de Bay has experienced increasing environmental degradation due to rapid urbanization, aquaculture expansion, and discharge from residential and industrial sources (Arcadio et al., 2022; LLDA, 2021).

This site is selected for study because of its ecological importance and vulnerability to pollution pressures, particularly from anthropogenic activities. The lake receives runoff from multiple surrounding municipalities and is directly affected by population density, industrial activity, and solid waste management practices in the region. Furthermore, there is a notable lack of detailed research focusing on the presence and ecological impact of microplastics in large freshwater lakes in the Philippines. Existing studies on Laguna de Bay remain limited, particularly those that explore its physical conditions and pollution levels in connection to broader environmental threats, including risks to aquatic biodiversity (Deocaris et al., 2023). Thus, Laguna de Bay serves as a representative case for studying microplastic pollution in Philippine freshwater systems.

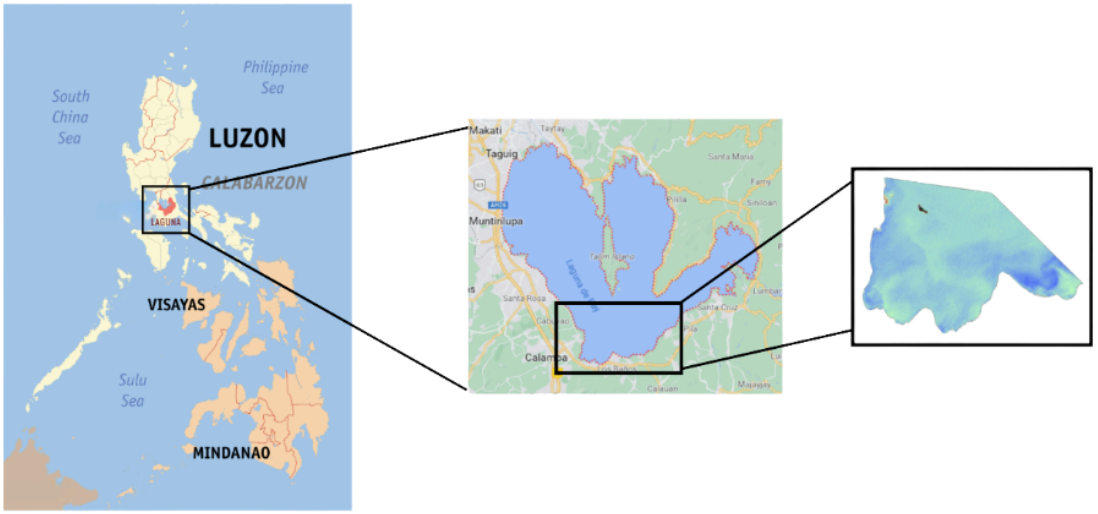

Figure 2. Location of the Study Area (Southern Part of Laguna de Bay)

Materials

The research uses various hardware and software materials that are essential for the completion of the study.

Software

The development of the microplastic pollution mapping model includes the selection of a specific toolkit of software instruments. The software tools are selected very carefully in terms of their ability to support the processing of satellite data, geospatial analysis, machine learning modeling, and data visualization.

Sentinel-2

Sentinel-2 satellite imagery is used in the data acquisition phase of this study, providing high-resolution multispectral data essential for mapping water quality characteristics. The satellite offers data across various spectral bands

that are useful for detecting indirect indicators of microplastic pollution, including NDTI, NDCI, and surface reflectance. Sentinel-2's frequent revisit time and large coverage area make it ideal for monitoring Laguna de Bay over time, enabling the collection of temporal sequences of environmental data. This imagery serves as the foundation for extracting relevant spectral indices and input features for the machine learning model.

Python (Jupyter Notebook)

Python, implemented through Jupyter Notebook, is used for data preprocessing, statistical modeling, and the development of machine learning algorithms in this study. Jupyter Notebook provides an interactive coding environment that combines code execution, visualization, and documentation within a single interface. This allows for step-by-step data cleaning, normalization, feature engineering, and visualization of exploratory analyses. Using Python libraries such as NumPy, Pandas, Scikit-learn, and Matplotlib, the study performs tasks ranging from dataset merging and statistical testing to the training and evaluation of predictive models for microplastic accumulation.

Visual Studio Code

Visual Studio Code and Streamlit were utilized in this study for efficient development and visualization of machine learning workflows. Visual Studio Code served as the primary code editor for writing, organizing, and debugging Python scripts, offering features such as syntax highlighting, integrated terminal access, and version control. Streamlit, on the other hand, was used to create an interactive web-based interface for visualizing model results and

microplastic distribution patterns, enabling dynamic and user-friendly interpretation of data and model performance.

Hardware

The model and sampling data analysis in this study are developed, tested, and processed on a local machine with the following specifications:

Computer

The computer used in this study serves as the primary workstation for developing the microplastic risk mapping model and processing satellite datasets. It is equipped with an AMD Ryzen 5 5600G processor, offering reliable multi-core performance suitable for handling computationally intensive tasks such as image classification and data preprocessing. The system includes 16GB of DDR4 RAM, providing sufficient memory capacity to manage large geospatial datasets and perform GIS-related operations without significant slowdowns. A 1TB solid-state drive (SSD) is installed, ensuring fast read and write speeds critical for loading, storing, and accessing satellite imagery and related data files efficiently. The machine operates on a 64-bit operating system running on an x64-based processor architecture, which supports modern software tools and high-memory applications required for this research. Although the system uses integrated Radeon graphics rather than a dedicated GPU, it remains capable of executing visualization tasks and moderate parallel processing demands associated with satellite data analysis. This configuration successfully supported iterative model training and validation workflows essential for the hybrid MLP Autoencoder-Ensemble framework. The workstation's balanced specifications proved adequate for

both Python-based machine learning development and Sentinel-2 imagery processing pipelines deployed during the study.

Table 2. Computer Specification for Model development

Specification	Description
Processor	AMD Ryzen 5 5600G
RAM	16GB DDR4
Storage	1TB SSD
System Type	64-bit operating system, x64-based processor
GPU	Integrated Radeon Graphics

Research Design

An experimental research design was employed in this study to systematically investigate the integration of satellite-based remote sensing data and in-situ water quality parameters for predicting microplastic accumulation in aquatic environments. This approach allowed for objective analysis of the relationship between environmental variables and microplastic concentrations, enabling the development of predictive models that informed environmental monitoring efforts. The study focused on the southern part of Laguna de Bay, where field sampling was conducted at two key sites (Pansol and Bayog) to collect water samples for microplastic quantification alongside water quality measurements including pH, dissolved oxygen, BOD, TSS, and fecal coliform levels. Simultaneously, Sentinel-2 satellite imagery was acquired to compute spectral indices (NDVI, NDMI, NDCI, NDTI, and PI) that captured environmental conditions related to microplastic presence. The experimental design incorporated both spatial and field-based data through

georeferencing procedures to create a robust framework for microplastic pollution assessment in freshwater environments. The integrated modeling framework employed an MLP Autoencoder for feature extraction and dimensionality reduction, combined with Random Forest, Gradient Boosting, and XGBoost algorithms for predictive classification.

Model Development

The model development process followed a structured pipeline, as illustrated in Figure 3. First, data preprocessing harmonized satellite-derived spectral indices (NDVI, NDMI, NDCI, NDTI, and PI) with in-situ water quality measurements through normalization and georeferencing. Georeferencing aligned satellite imagery with real-world geographic coordinates, ensuring accurate spatial correspondence between remote sensing data and field measurements. An MLP Autoencoder compressed the satellite features into a compact representation, which were then combined with in-situ parameters using weighted integration (0.8 for in-situ, 0.2 for satellite). The higher weight for in-situ data reflected its lower measurement uncertainty and higher reliability compared to satellite-derived features. Next, the model training phase employed ensemble learning algorithms (Random Forest, Gradient Boosting, XGBoost) to identify relationships between environmental variables and microplastic concentrations. Model performance was evaluated using a held-out dataset through independent t-test analysis alongside MAE, RMSE, R^2, and cross-validation (CV) metrics. Once validated, the model was applied to the southern part of Laguna de Bay to generate GIS-based microplastic predictions. The final stage involved identifying potential microplastic accumulation hotspots through spatial mapping, leveraging the combined

strength of remote sensing and field-based observations. This end-to-end approach prioritized methodological transparency while addressing data limitations in freshwater monitoring.

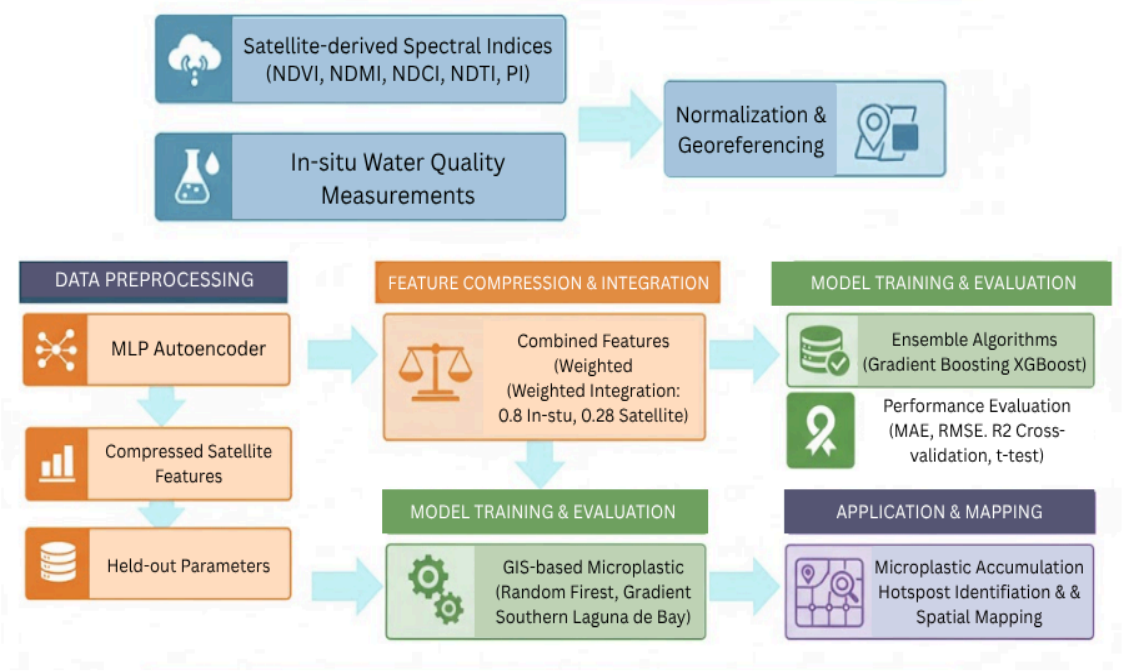

Figure 3. Model Development

Data Collection

Sampling Station Selection of Insitu Water Quality Parameters

Two sampling stations were selected in the southern part of Laguna de Bay, an area identified as being particularly vulnerable to pollution due to high levels of anthropogenic activity. According to the Laguna Lake Development Authority (LLDA, 2021), the lake receives substantial pollutant input from domestic wastewater, industrial discharge, and agricultural runoff, with the southern bay region being one of the primary receiving areas due to its

proximity to urban settlements and tributary inflows. Arcadio et al. (2022) also noted that population density and aquaculture operations in the southern and western parts of the lake contribute significantly to environmental degradation and water quality decline. These conditions make the southern bay an appropriate site for monitoring microplastic accumulation.

The two sampling stations were selected to reflect these high-pressure zones and to represent varying pollution exposure gradients within the region. GPS coordinates were recorded at each site to ensure spatial precision, enabling accurate pairing of in-situ data with satellite-derived observations. This alignment supports the integration of datasets in the model development process and ensures consistency in both temporal and spatial dimensions during analysis.

Spatial Sampling Design and Site Characterization

Sampling stations were strategically selected using a spatial grid-based approach to capture variability in microplastic distribution across the southern portion of Laguna de Bay (Lorenz et al., 2019; Lenaker et al., 2020). Two primary sampling sites were established: Barangay Bayog, located approximately 1.64 km from the shoreline, and Barangay Pansol, positioned approximately 1.31 km from the shore. Both near-shore zones and shallow shoal areas were included in the sampling design to account for hydrodynamic accumulation patterns influenced by wave action, tidal movement, and sediment transport (Shamskhany et al., 2021). All samples were collected from surface waters to focus on buoyant and suspended microplastics, which represent the most visible fraction of pollution and are

most directly influenced by runoff from land-based sources such as residential areas, agricultural runoff, and waste discharge points (Constant et al., 2022). Sampling locations were georeferenced using GPS coordinates to enable geospatial analysis and correlation with pollution sources, providing insights into how proximity to urbanized and high-activity zones affects microplastic density (Lenaker et al., 2020). This spatial sampling framework ensures that the predictive model captures both localized hotspots near human settlements and broader patterns across the lake's surface layer, supporting accurate GIS-based mapping and targeted environmental management strategies.

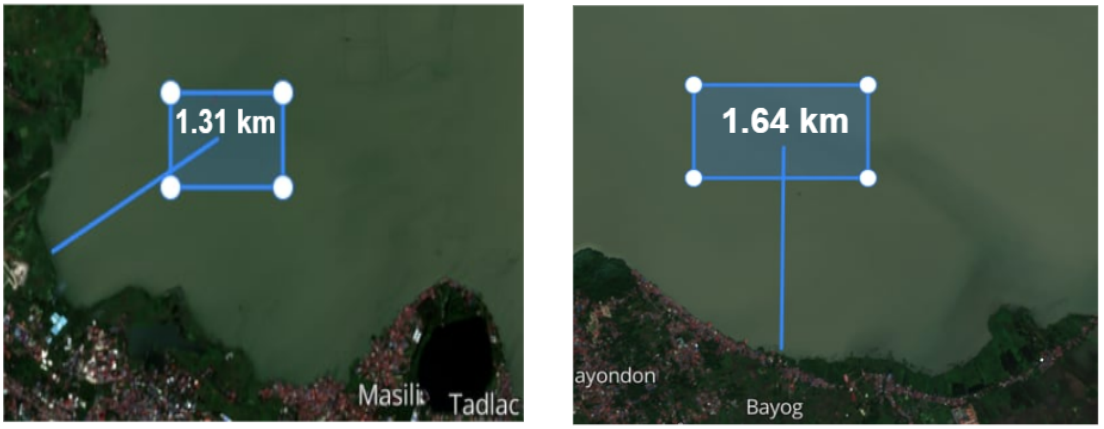

Figure 4. Sampling Station Locations and Offshore Distances in Southern Laguna de Bay

Materials for In-situ Data Collection of Water Samples

The in-situ data collection process requires a set of specialized materials and equipment to ensure the safety of the research team, maintain sample integrity, and comply with standard water quality monitoring protocols. Field personnel are required to wear life vests and face masks at all times to ensure safety during boat-based sampling in Laguna de Bay. Gloves are used

for sanitary sample handling and to prevent contamination of both the researchers and the collected samples.

For sample preservation and transport, an insulated ice box filled with ice cubes is used to keep the water samples at appropriate temperatures until laboratory analysis. Each sampling day involves the use of two 1-liter BOD bottles designated for biochemical oxygen demand analysis, two 1-liter containers for E. coli testing, and three 1-liter glass containers for microplastic sample collection. These containers are prepared prior to each sampling trip and labeled accordingly to ensure accurate documentation and avoid cross-contamination. All used sampling materials are handled responsibly, with a black bag allocated specifically for the safe disposal of BOD waste and other field waste materials.

The field sampling is scheduled to take place every Wednesday and Thursday over a three-week period, beginning on June 3. Samples are collected between 7:00 AM onwards to maintain consistency and minimize the influence of temperature and sunlight on the chemical composition of the water. pH measurements are conducted in the laboratory following collection, ensuring precision through the use of calibrated instruments under controlled conditions. This setup ensures that both logistical and scientific requirements are met throughout the entire in-situ sampling period.

Water Sample Analysis and Laboratory Experiment

Biochemical Oxygen Demand (BOD) was measured using the standard five-day method (BOD_5) at 20 ± 1 °C with BOD bottles. Initial and final Dissolved Oxygen (DO) concentrations were determined with a membrane

electrode or by the Winkler iodometric method, and BOD was calculated from the depletion of DO after incubation (APHA, AWWA, & WEF, 2017; Rounds, 2006). This method provides a reliable proxy for organic pollutant levels in water samples. All measurements were performed in triplicate to ensure precision and reproducibility.

Fecal coliforms were quantified using the Most Probable Number (MPN) technique, where inoculated dilution series in selective media were incubated at 44.5 ± 0.2 °C, with enumeration based on gas production to estimate fecal coliform densities per 100 mL (APHA, AWWA, & WEF, 2017; NEMI, 2021). In addition, membrane filtration was conducted by passing samples through 0.45 µm nitrocellulose membranes, placing filters on m-FC agar, and incubating them at 44.5 °C for 22–24 hours; colonies with a characteristic blue appearance were then counted to determine colony-forming units per 100 mL (ASM, 2016). These complementary techniques enhance detection sensitivity across varying sample contamination levels.

Total Suspended Solids (TSS) were determined following the EPA Method 160.2, which employs a gravimetric approach for measuring non-filterable residues. Water samples were first homogenized to ensure uniform distribution of particles, then filtered through a pre-weighed filter paper to collect suspended materials. The filters containing the residue were oven-dried at 60 °C for 5–10 minutes to remove excess moisture. After drying, the filters were weighed again, and the difference between the initial and final weights represented the amount of total suspended solids in the sample, expressed in milligrams per liter (mg/L) (U.S. Environmental Protection

Agency, 1971). Quality control included blank filters and standard reference materials. This parameter is critical for assessing turbidity and pollutant transport in microplastic studies.

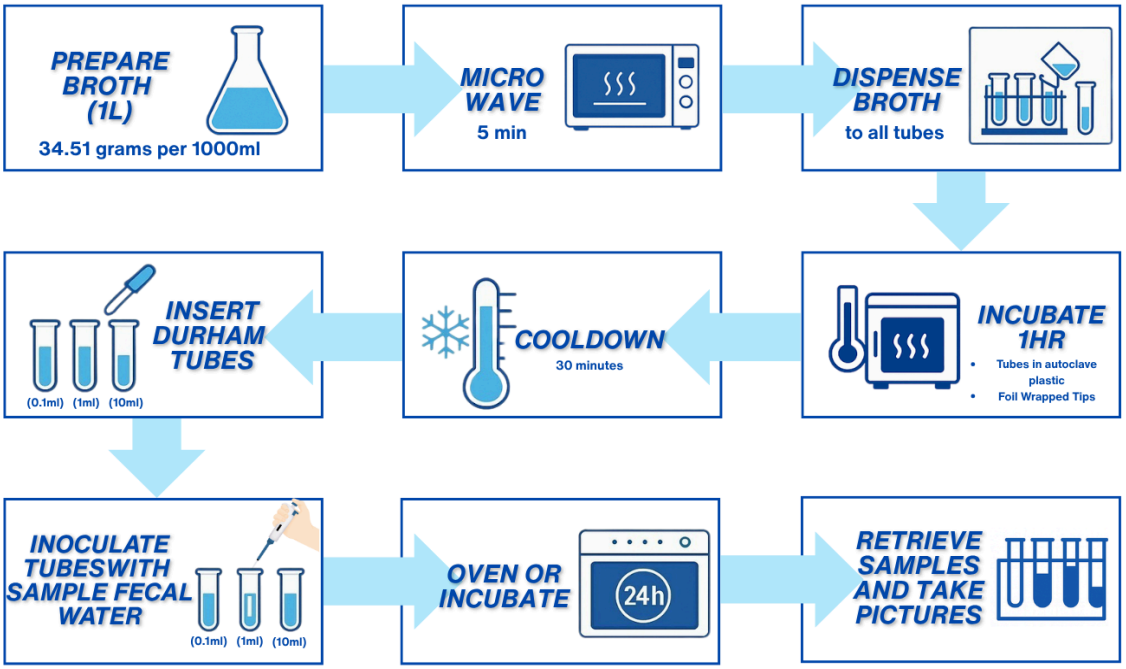

Figure 5. Fecal Coliform Analysis Procedure Using the Most Probable Number

All laboratory instruments were properly calibrated amid strict quality assurance procedures. Laboratory personnel adhered to sample preservation and holding time requirements outlined by the Department of Environment and Natural Resources - Environmental Management Bureau (DENR-EMB, 2016) and the Water Environment Association of the Philippines (WEAP, 2023). Once data generation was complete, these results were digitized and integrated with field observations and satellite data. This comprehensive, ground-truth dataset then served as the foundation for model training and enhancing the predictive accuracy of microplastic distribution. Rigorous

validation ensured data reliability throughout the analytical pipeline. These standardized protocols minimized analytical uncertainties across all measurements.

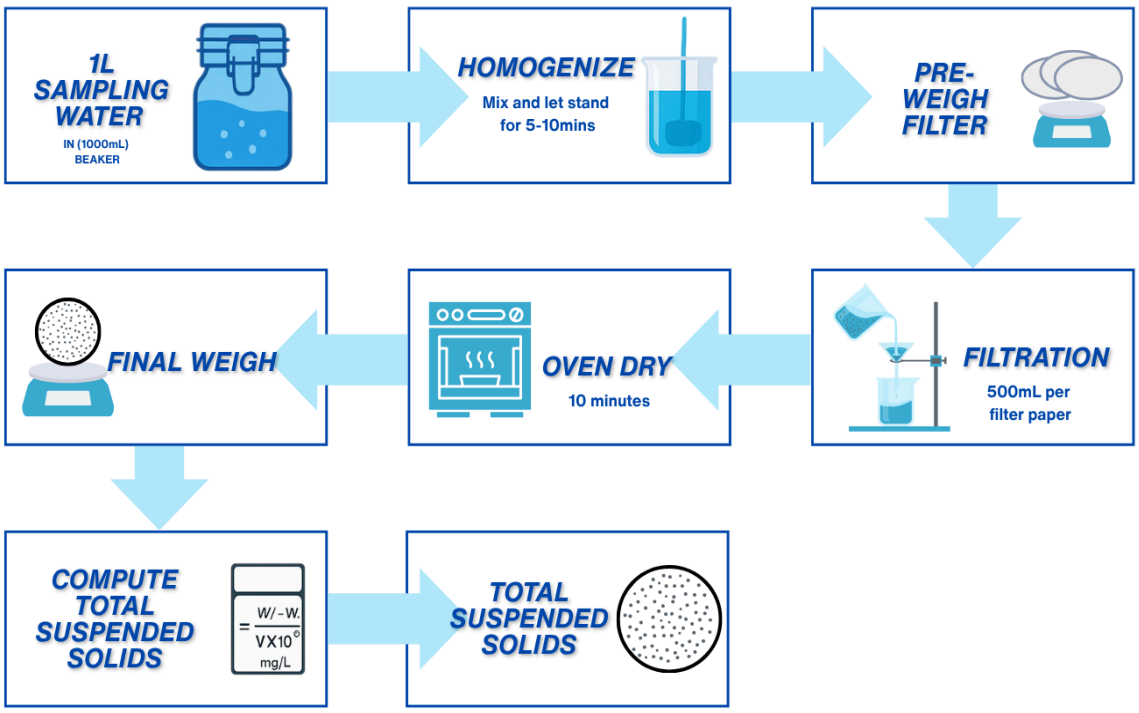

Figure 6. Workflow for Total Suspended Solids (TSS) Determination

Microplastic Density Calculation Method

Microplastics were quantified using a density separation method adapted from NOAA and MSFD protocols. Collected water samples (1 L) were mixed with a saturated sodium chloride solution (20 grams) in glass beakers. Due to their typically lower density, microplastic particles floated to the surface, while heavier organic and inorganic debris settled (Masura et al., 2015; Claessens et al., 2011). After an appropriate settling period, the supernatant containing floating particles was carefully decanted and passed through a 0.45 µm filter using vacuum filtration. Retained particles on the filter

were air-dried in petri dishes and visually inspected under a stereo microscope at 10× and 40× magnification for counting and morphological categorization. The concentration of microplastics was expressed in milligrams per liter (mg/L) to quantify their abundance in each sample. This method preserves particle integrity and allows for classification based on shape, size, and color. The use of saturated NaCl is cost-effective, easy to prepare, and widely used due to its high recovery rates for microplastics (Claessens et al., 2011). This approach aligns with standardized procedures for the analysis of microplastics in aquatic samples and ensures consistent, reproducible density-based quantification. Recovery efficiency exceeded 90 percent across all samples, validating the method's reliability for Laguna de Bay field conditions.

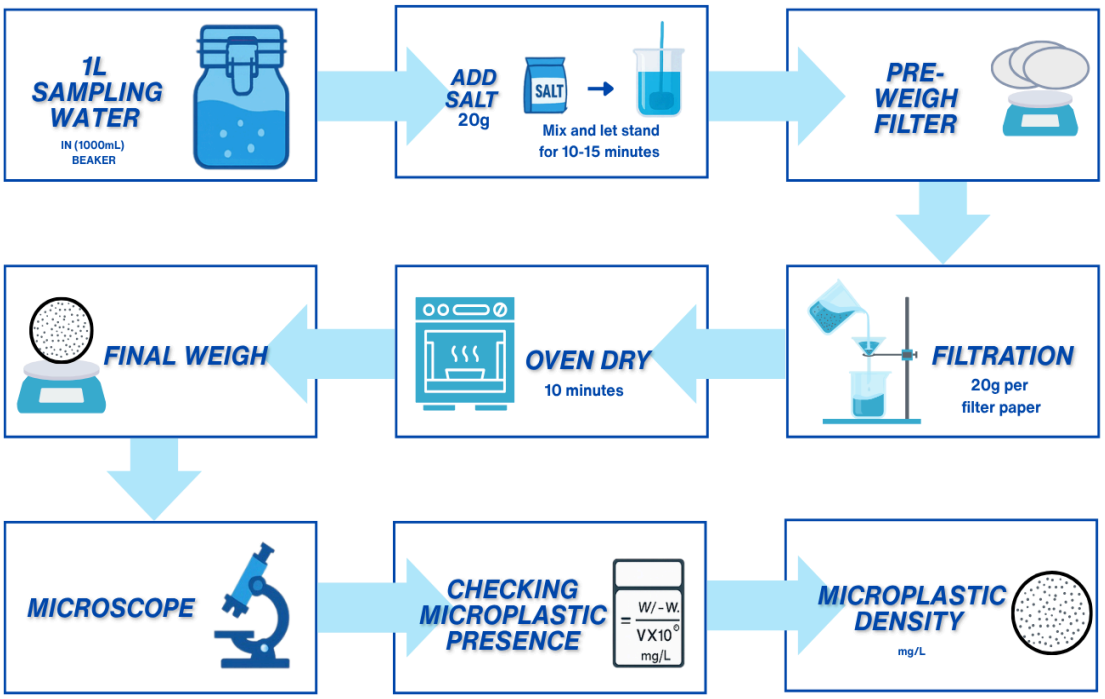

Figure 7. Microplastic Sampling and Quantification Workflow

In-situ Water Quality Parameters

The following parameters were measured as inputs for the predictive model. pH characterizes the acid-base balance affecting aquatic chemistry and organismal stress (Wilde et al., 2021). DO reflects oxygen availability for biota and overall ecosystem condition (Rounds, 2006). BOD indicates biodegradable organic load via oxygen demand over five days (APHA/AWWA/WEF, 2017, 5210 B). Fecal coliform presence signals potential fecal contamination from warm-blooded animals and associated health risks, reported as MPN/100 mL using multiple-tube fermentation (APHA/AWWA/WEF, 2017, 9221E). TSS measures suspended particulate matter influencing water clarity and microplastic transport pathways. These parameters collectively capture key hydrochemical drivers influencing microplastic distribution patterns in Laguna de Bay's southern basin.

Satellite Data

Satellite data are acquired from the Sentinel-2 mission, which provides high-resolution multispectral imagery suitable for monitoring large water bodies. The satellite-derived parameters include spectral indices such as NDVI, NDMI, NDCI, NDTI, and the Plastics Index (PI), which serve as indicators of water quality and potential microplastic presence. These remotely sensed data enable spatially extensive analysis, capturing broader environmental patterns that complement localized in-situ measurements and support lake-wide assessment of microplastic distribution.

Figure 8 displays five Sentinel-2 spectral index maps: (a) NDVI, (b) NDMI, (c) NDCI, (d) NDTI, and (e) Plastics Index. Warmer tones indicate

higher index values and highlight potential microplastic accumulation hotspots in shallow embayments and discharge zones, suggesting the influence of land use and hydrological processes on microplastic spatial patterns across the lake.

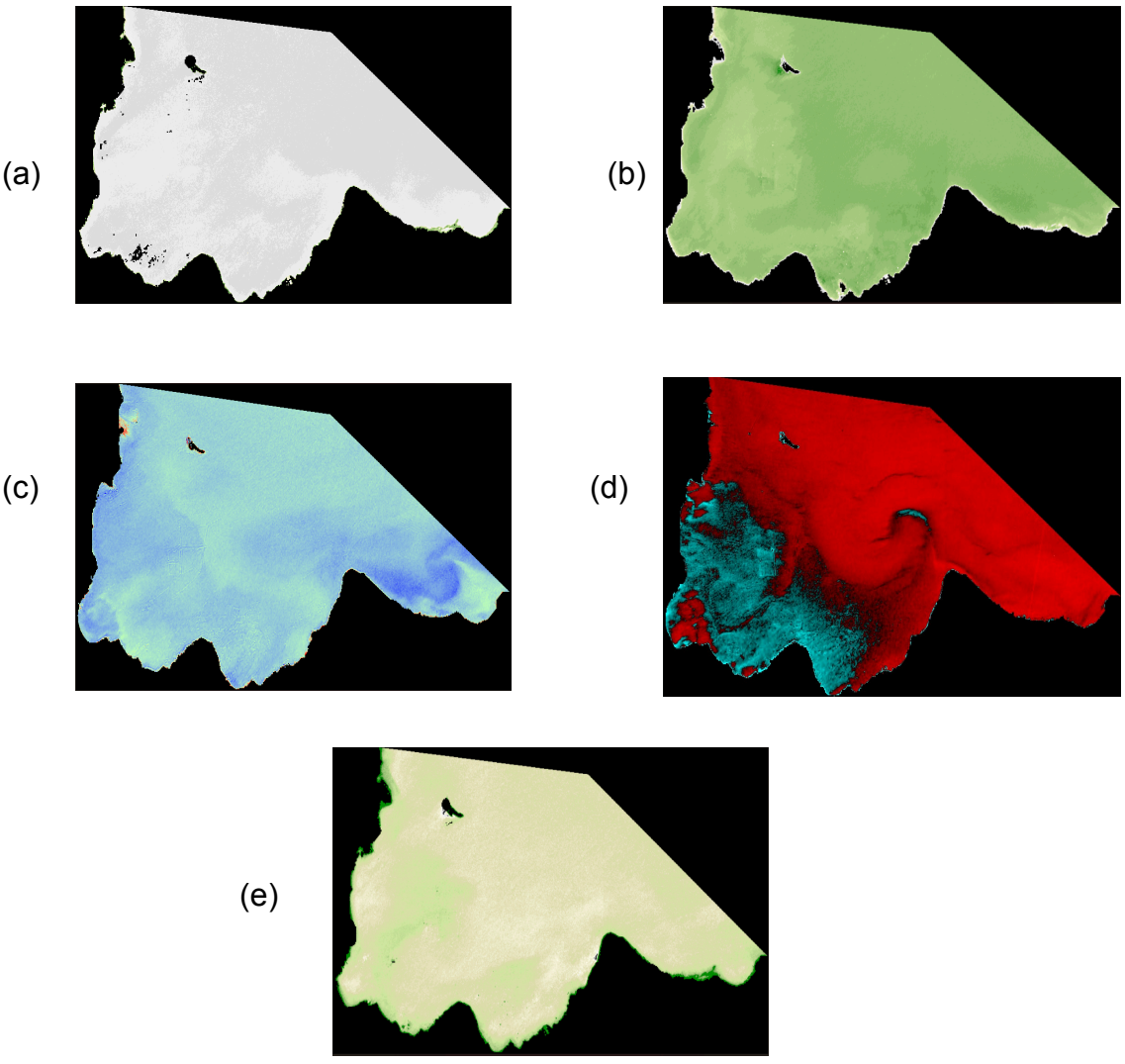

Figure 8. Sentinel-2 Spectral Index Maps of Southern Laguna de Bay

Data Preprocessing

Data preprocessing is a critical phase for preparing both satellite-derived spectral indices and in-situ water quality data for model

training and prediction. This process involves several stages, including data normalization, feature extraction from satellite imagery, and georeferencing. Sentinel-2 satellite imagery was acquired and processed to compute five key spectral indices: Normalized Difference Vegetation Index (NDVI), Normalized Difference Moisture Index (NDMI), Normalized Difference Chlorophyll Index (NDCI), Normalized Difference Turbidity Index (NDTI), and Plastics Index (PI). Per-band statistical features including mean, standard deviation, variance, and percentile metrics were extracted from the relevant Sentinel-2 bands (B03, B04, B05, B08) for each spectral index. These features were then processed through an MPN Autoencoder to generate compressed representations suitable for integration with in-situ measurements. In-situ measurements such as dissolved oxygen, pH, biochemical oxygen demand, COD, and fecal coliform were cleaned and scaled for integration.

Date Alignment and Georeferencing

The in-situ and satellite-derived spectral indices underwent both temporal alignment and spatial georeferencing to ensure consistency between datasets. Each Sentinel-2 observation was temporally matched with the corresponding in-situ sampling record collected on the same or nearest available date, ensuring that both satellite-derived spectral indices (NDVI, NDMI, NDCI, NDTI, and PI) and ground-based water quality measurements represented comparable environmental conditions. The satellite imagery was also georeferenced by assigning real-world geographic coordinates to each pixel through the use of ground control points (GCPs), which link specific locations on the raster dataset to known spatial coordinates within the map reference system. This process enabled precise spatial alignment between

remote sensing data and field sampling sites. The resulting temporally aligned and georeferenced dataset served as the foundation for model development, allowing for accurate spatial-temporal integration and comparison between satellite observations and in-situ measurements.

Statistical Feature Extraction

Statistical feature extraction techniques serve as a fundamental step in transforming raw satellite imagery into quantitatively meaningful data for environmental analysis, including microplastic detection. By calculating summary statistics such as mean, standard deviation, minimum, maximum, and percentiles for each spectral band within an image, researchers distill complex pixel-level information into structured feature vectors that can be effectively used in modeling and prediction workflows. This approach leverages the unique spectral characteristics of plastics and water bodies, improving the separation and classification of microplastics from other materials in remote sensing applications (Garaba & Dierssen, 2018; Biermann et al., 2020; Rokni et al., 2014; Xu et al., 2024). Recent advances underline how comprehensive statistical feature libraries—extracted across different bands and indices—not only allow for robust regression and machine learning analyses, but also help reveal spatial and temporal patterns in microplastic abundance. Rigorous statistical extraction provides the foundation for automated monitoring frameworks and predictive modeling, supporting actionable assessments for water quality management and pollution control.

Standardization

After temporal alignment, standardization is performed to normalize variables measured on different scales. In this study, Z-score normalization is

applied to in-situ measurements such as dissolved oxygen, pH, BOD, and COD to standardize their distributions across monitoring stations. Z-score normalization transforms the data into a standard scale with a mean of zero and a standard deviation of one, reducing the influence of outliers and ensuring that all features contribute equally during statistical analysis and modeling.

Spearman Correlation Analysis

After statistical feature extraction and normalization of in-situ data, Spearman's rank correlation coefficient was employed to assess the monotonic relationships between satellite-derived spectral indices and in-situ water quality parameters with observed microplastic concentrations. Unlike Pearson correlation, Spearman correlation is a non-parametric method particularly suitable for environmental data where nonlinear relationships and non-normal distributions are common.

Data Augmentation

To enhance the robustness of the dataset and improve model generalization, data augmentation techniques were applied. For the satellite data, augmentation involved spatial transformations such as rotations, horizontal and vertical flips, and slight scaling to simulate variations in satellite viewing angles and environmental conditions. In-situ data augmentation employed physics-guided noise injection that introduces controlled variability reflecting realistic measurement uncertainties while preserving the underlying statistical relationships between water quality parameters. These augmentation strategies created a more diverse training set, ultimately

supporting more robust model performance in predicting microplastic concentrations.

Model Selection

MLP Autoencoder Feature Extraction Architecture

The Unguided Multilayer Perceptron (MLP) Autoencoder was developed to extract latent representations from satellite-derived statistical data that were directly optimized for predicting microplastic concentrations. The satellite dataset consisted of numerical features computed from preprocessed imagery, including spectral indices, reflectance values, and texture-based statistics representing surface water conditions. These features were normalized using *StandardScaler* to ensure uniform scaling and were subsequently fed into the Autoencoder model as input.

Training Process and Architecture Design

The guided Autoencoder differed from conventional unsupervised Autoencoders by incorporating an additional supervision branch that used microplastic concentration as a secondary learning objective. This dual-objective framework enabled the model to learn latent representations that captured both spectral–spatial variability from satellite data and patterns directly related to microplastic distribution.

The network followed an encoder–decoder–guidance architecture. The encoder compressed the input features through dense layers of 64 and 32 neurons, with the 32-neuron bottleneck serving as the latent feature space. A symmetric decoder reconstructed the original input using a 64-neuron layer and a final linear output layer matching the input dimensionality.

A guidance branch was directly connected to the bottleneck layer to predict normalized microplastic concentrations. This branch included a hidden layer of 16 neurons and a single-neuron regression output, trained using Mean Squared Error loss. A weighted composite loss function was applied, prioritizing the guidance objective (0.7) over reconstruction (0.3) to enhance feature relevance for microplastic prediction.

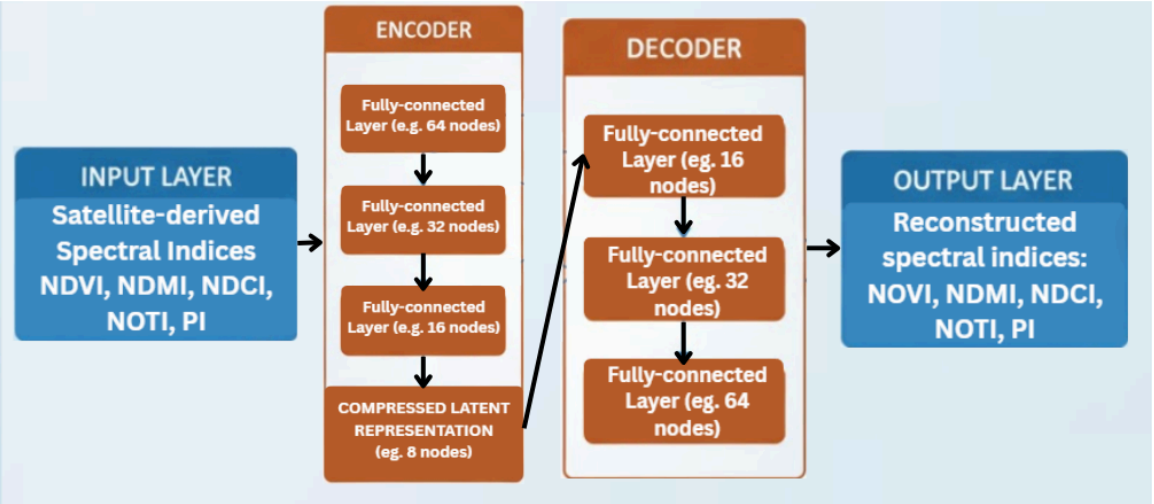

Figure 9. Architecture of the Dense Autoencoder for Spectral Feature Compression

Model training employed the Adam optimizer with a learning rate of 0.001 and a batch size of 32. Training converged after 150–200 epochs, with early stopping based on validation reconstruction loss to prevent overfitting. Overall, the guided Autoencoder effectively produced compact, information-rich latent features that improved downstream ensemble modeling performance for microplastic mapping in Laguna de Bay. These latent features were subsequently used as inputs to ensemble machine learning models, where they contributed to improved predictive accuracy and spatial consistency of microplastic estimates. The results demonstrate that guided

representation learning is a robust approach for integrating remote sensing data with in-situ measurements in environmental monitoring applications.

Table 3. Autoencoder Model Configuration and Training Parameters

Parameter Group	Parameter	Value / Setting
Input Data	Feature Source	Satellite-derived statistical features
Preprocessing	Scaler Type	StandardScaler normalization
Encoder Layers	Structure	128 → 64 → 32
	Activation Function	ReLU
Decoder Layers	Structure	64 → 128 → Output (input_dim features)
	Activation Function	ReLU (hidden), Linear (output)
Optimizer	Type	Adam
Loss Function	Type	MSE (reconstruction)
Training Setup	Epochs	100
	Batch Size	32
Output	Bottleneck Features	32 latent variables
Post-processing Model	Regressor Type	MLP (64 → 32 → 1)
	Loss Function	MAE
	Batch Size	16
	Epochs	100
Framework	Environment	TensorFlow / Keras

Weighting Strategy for Ensemble Integration

Before optimizing the machine learning models, a weighting adjustment between the in-situ and satellite-based datasets was applied to account for

temporal inconsistencies in data acquisition. The main reason for this adjustment was the presence of persistent cloud cover, which obstructed satellite image capture during the month of July, coinciding with one of the in-situ sampling periods. This limitation resulted in the unavailability of corresponding satellite imagery for that month, leading to a temporal misalignment between ground observations and satellite-derived data. Such spatio-temporal mismatches between satellite and in-situ measurements are known to introduce significant uncertainty in model validation and prediction accuracy (Urraca et al, 2024).

To minimize the uncertainty caused by this data gap, the satellite contribution within the ensemble models was scaled down relative to the in-situ dataset. A weight ratio of 0.8 for in-situ data and 0.2 for satellite data was adopted to ensure that the models relied primarily on the more temporally consistent in-situ measurements while still incorporating the spatially comprehensive information provided by the satellite indices. This weighting adjustment was applied prior to training the Random Forest (RF), Gradient Boosting (GB), and Extreme Gradient Boosting (XGB) models.

Hyperparameter Optimization

Hyperparameter optimization was conducted to identify the most effective model configurations for accurate microplastic concentration prediction. The tuning process involved adjusting key parameters such as the number of estimators, learning rate, and tree depth to balance model complexity and predictive performance. Consistent random states were applied to ensure reproducibility across experiments. The final optimized

settings for each algorithm Random Forest, Gradient Boosting, and XGBoost were selected based on their ability to minimize prediction error while maintaining model stability for both in-situ and satellite datasets.

Table 4 lists the optimized hyperparameters used for the Random Forest model. Both the in-situ and satellite datasets utilized 300 decision trees (n_estimators=300) with unrestricted depth (max_depth=None) to allow the model to fully capture complex feature interactions. The bootstrap parameter was set to True to ensure robust sampling, while the Mean Squared Error criterion guided the split quality during tree construction. A fixed random state of 42 was applied to maintain model reproducibility.

These hyperparameter settings were determined through systematic grid search with 5-fold cross-validation, balancing model complexity against overfitting risks. The unrestricted depth enabled the Random Forest to learn intricate non-linear relationships between environmental proxies and microplastic concentrations. This configuration achieved optimal LOOCV performance while maintaining generalizability across both data domains.

Table 4 . Optimized Hyperparameter of Random Forest

Hyperparameter	In-situ Dataset	Satellite Dataset
Number of Estimators (n_estimators)	300	300
Maximum Depth (max_depth)	None	None
Bootstrap	True	True
Random State	42	42
Criterion	Mean Squared Error	Mean Squared Error

Table 5 presents the tuning parameters for the Gradient Boosting model. For both datasets, the model employed 300 estimators and a learning rate of 0.05 to balance learning speed and accuracy. The maximum tree depth remained at the default value of 3 to prevent overfitting, and the loss function was set to Squared Error, appropriate for continuous regression targets. The consistent random state ensured reproducible model training across runs.

Table 5. Optimized Hyperparameter of Gradient Boosting Model

Hyperparameter	In-situ Dataset	Satellite Dataset
Number of Estimators (n_estimators)	300	300
Learning Rate	0.05	0.05
Maximum Depth (max_depth)	3	3
Bootstrap	True	True
Random State	42	42
Loss Function	Squared Error	Squared Error

Table 6 summarizes the optimized hyperparameters for the XGBoost model. Both the in-situ and satellite datasets were trained with 400 estimators and a moderate learning rate of 0.05, allowing gradual optimization and reduced risk of overfitting. The model's objective was configured for squared error regression, aligning with the continuous nature of microplastic concentration values. The random state was fixed at 42 for consistent results. Additionally, these hyperparameter settings reflect a balance between model complexity and generalization performance, ensuring stable and reliable predictions across both data sources.

Table 6. Optimized Hyperparameter of XGBoost Model

Hyperparameter	In-situ Dataset	Satellite Dataset
Number of Estimators (n_estimators)	400	400
Learning Rate	0.05	0.05
Maximum Depth (max_depth)	None	None
Bootstrap	True	True
Random State	42	42
Objective	Squared Error	Squared Error

Integration of Ensemble Learning Models for Predicting Microplastic Accumulation

To predict microplastic concentration, ensemble learning algorithms namely Random Forest (RF), Gradient Boosting (GB), and Extreme Gradient Boosting (XGBoost) were employed. These algorithms were trained separately on both in-situ and satellite-encoded datasets, allowing the framework to leverage complementary information from direct field measurements and remotely sensed features.

Random Forest constructed an ensemble of decision trees and aggregated their outputs to reduce overfitting and improve generalizability. Gradient Boosting sequentially built weak learners that iteratively minimized the prediction error through gradient optimization. XGBoost, an enhanced version of gradient boosting, incorporated regularization and parallel computation, achieving superior efficiency and predictive performance.

Each model was trained on a standardized dataset that included water quality parameters such as dissolved oxygen (DO), biochemical oxygen demand (BOD), pH, total suspended solids (TSS), and fecal coliform alongside the compressed features derived from the MLP Autoencoder. Model evaluation employed Root Mean Square Error (RMSE), Mean Absolute Error (MAE), R^2 score, and cross-validation (CV) to assess accuracy and reliability. Additionally, feature importance analysis was conducted using XGBoost to identify the most influential variables contributing to microplastic prediction. Overall, the ensemble-based framework enhanced predictive robustness, efficiently handling nonlinearities and high-dimensional feature spaces.

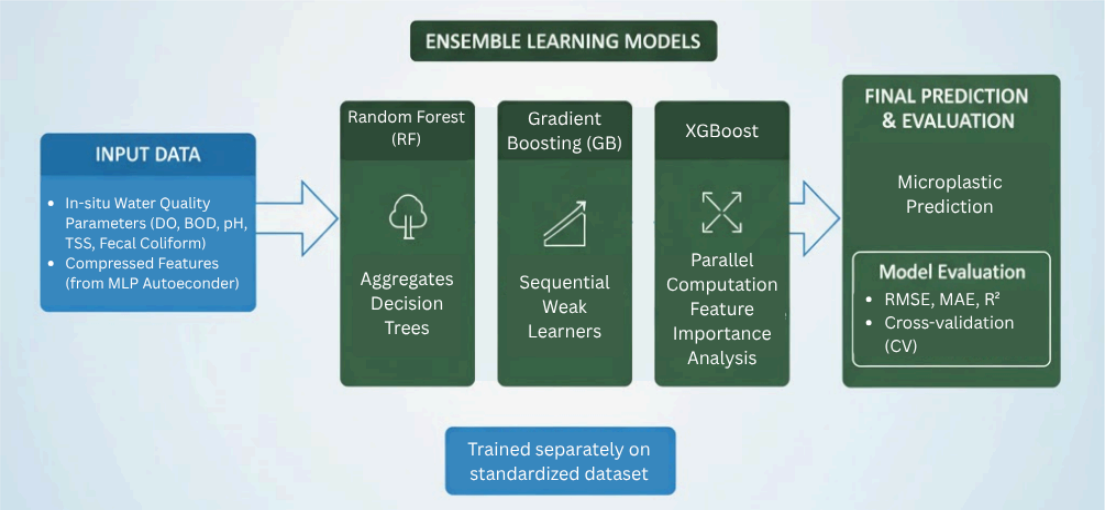

Figure 10. Ensemble Regression Architecture (RF, GB, and XGBoost)

Hybrid Ensemble Integration Framework

The final predictive framework integrated deep learning feature extraction and ensemble regression through a weighted ensemble approach, forming a hybrid MLP Autoencoder and Ensemble Regression Model. Predictions from the in-situ and satellite ensembles were combined using weighted averaging, with in-situ outputs assigned a weight of 0.8 and satellite

outputs 0.2, reflecting the higher reliability of direct field measurements while still leveraging the broader spatial coverage of satellite data. This integration combined the dimensionality reduction capability of MLP autoencoders with the predictive precision and interpretability of ensemble models, enabling more accurate and spatially consistent microplastic mapping.

The trained hybrid model, incorporating Sentinel-2 spectral indices (NDVI, NDMI, NDCI, NDTI, and PI) and in-situ water quality parameters from southern Laguna de Bay, was used to estimate the spatial distribution of microplastic accumulation. By exploiting learned spectral patterns and physicochemical relationships, the model generated a continuous GIS-based map that identified potential microplastic accumulation hotspots, supporting environmental management and targeted monitoring efforts. This integrated framework improves robustness by mitigating the limitations of individual data sources through complementary learning. Moreover, the approach provides a scalable foundation for future microplastic monitoring as additional in-situ observations and higher-resolution satellite data become available.

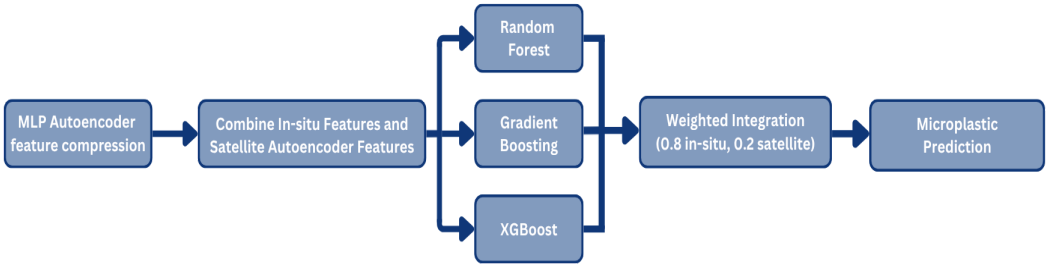

Figure 11. Hybrid Weighted Ensemble (MLP Autoencoder + RF/GB/XGBoost)

Model Evaluation

The trained hybrid model combining MLP Autoencoder and ensemble learning algorithms (Random Forest, Gradient Boosting, XGBoost) was

evaluated using standard regression metrics including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R^2 Score, Feature Importance Analysis, and Cross-Validation (CV). Additionally, independent t-test analysis was conducted to validate model performance on unseen data.

MSE (Mean Squared Error)

Mean Squared Error measures the average squared difference between the input satellite features and their reconstructed output from the MPN Autoencoder. It evaluates the autoencoder's ability to capture meaningful spatial patterns during the feature extraction phase. A lower MSE indicates that the autoencoder has effectively learned to compress and reconstruct satellite-derived spectral indices, preserving essential environmental information.

RMSE (Root Mean Squared Error)

Root Mean Squared Error measures the square root of the average squared differences between predicted and actual microplastic concentrations. It penalizes larger errors more heavily than MAE, making it sensitive to outliers. RMSE is used to evaluate the overall prediction accuracy and consistency of the ensemble models, with values in the same unit as the target variable for easier interpretation. Lower RMSE values indicate superior model performance in capturing Laguna de Bay's microplastic distribution patterns. In this study, RMSE is used to compare model performance across in-situ, satellite, and hybrid configurations, reflecting overall prediction accuracy and sensitivity to larger errors.

MAE (Mean Absolute Error)

Mean Absolute Error quantifies the average magnitude of prediction errors without considering their direction. Unlike RMSE, MAE is less sensitive to outliers, offering a more intuitive measure of average prediction deviation. It provides a straightforward assessment of the model's prediction accuracy in absolute terms.

R^2 Score (Coefficient of Determination)

The R^2 score measures the proportion of variance in microplastic concentrations that is explained by the model. It ranges from 0 to 1, where values closer to 1 indicate better model fit and stronger predictive power. R^2 provides an overall assessment of how well the ensemble model captures the relationship between environmental predictors and microplastic distribution. In this study, higher R^2 values indicate that the selected water quality parameters and spectral features effectively represent the underlying drivers of microplastic variability in the study area.

Cross-Validation (CV)

Cross-validation is employed to assess model robustness and generalization capability by partitioning the dataset into multiple training and validation subsets. K-fold cross-validation was applied, where the data is divided into k equal-sized folds, and the model is trained k times, each time using a different fold as the validation set while the remaining k-1 folds serve as training data. The average performance across all folds provides a reliable estimate of model accuracy and helps prevent overfitting. CV ensures that the model performs consistently across different data subsets and is not overly dependent on a specific train-test split.

This approach is particularly important for datasets with limited samples, as it maximizes data utilization while maintaining rigorous performance evaluation. Consequently, cross-validation enhances confidence in the model's stability and applicability to unseen data.

Feature Importance Analysis

Feature importance analysis is performed using XGBoost's built-in scoring mechanism to identify which input features whether derived from satellite spectral indices (NDVI, NDMI, NDCI, NDTI, PI) or in-situ water quality measurements most influence model predictions. This helps prioritize key environmental indicators, enhances model interpretability, and informs future monitoring efforts. The analysis reveals which variables contribute most significantly to microplastic accumulation predictions.

Validation

Independent T-Test

The independent t-test is employed to validate model performance by comparing predicted microplastic concentrations with actual observed values from the held-out unseen dataset. The independent t-test assesses whether there is a statistically significant difference between two independent groups in this case, the predicted values and the actual measured values from different samples. The null hypothesis states that there is no significant difference between the means of predicted and actual concentrations. If the p-value is greater than 0.05 ($p > 0.05$), this indicates that the model's predictions are statistically consistent with actual measurements, validating model accuracy. A significant result ($p < 0.05$) would suggest systematic prediction bias requiring model refinement.

Ethical Considerations

In conducting this study on the prediction and mapping of microplastic accumulation in Laguna de Bay, the researchers ensured that all processes adhered to principles of ethical research, integrity, and environmental responsibility. The study was designed with transparency and accountability in data collection, analysis, and interpretation, recognizing the broader environmental and societal implications of predictive modeling in ecological research.

All datasets utilized in this research were obtained from authorized and publicly available sources, including Sentinel-2 satellite imagery provided by the European Space Agency (ESA) and in-situ water quality data collected through standardized field protocols. No harmful, invasive, or manipulative procedures were performed on the environment, aquatic organisms, or surrounding communities during the study.

The researchers also ensured full transparency by thoroughly documenting all data preprocessing, model development, and validation procedures. The methodologies, algorithms, and tools employed in this study such as the Multi-Layer Perceptron (MLP) Autoencoder and ensemble machine learning models were selected and implemented responsibly to maintain scientific rigor and reproducibility. Model outputs were critically analyzed to prevent misinterpretation or overreliance on automated predictions without contextual understanding or expert validation.

Furthermore, the research team acknowledges the potential applications of this model in environmental management and policy-making. To prevent

misuse, any future deployment of the system for large-scale monitoring or decision-making will be guided by ethical frameworks that promote sustainable and equitable resource management. The study's limitations, including data availability constraints and the absence of microplastic type classification, are clearly acknowledged to maintain transparency and scientific integrity.

Ultimately, this study upholds responsible use of artificial intelligence and remote sensing for environmental monitoring. The researchers are committed to ensuring that the findings contribute positively to environmental conservation, sustainable management of Laguna de Bay, and the advancement of ethical and data-driven practices in aquatic ecosystem research.

Data sharing and reproducibility remain central ethical priorities throughout this investigation. All code, processing scripts, and model architectures developed for satellite feature extraction and ensemble prediction will be made publicly available through open-access repositories such as GitHub, enabling other researchers to validate, replicate, and build upon these methodologies.

Collaboration with local stakeholders including Laguna Lake Development Authority and community environmental groups was maintained to ensure findings align with regional conservation priorities. These efforts collectively ensure that the study is scientifically rigorous, ethically responsible, and directly relevant to supporting sustainable management and informed decision-making for the protection of Laguna de Bay.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presented the study's objectives, results, and discussion, supported by visual representations of the findings. The main goal was to develop an automated method for estimating microplastic concentration in surface waters by integrating satellite-derived indices with in situ water quality data. This approach addressed the limitations of traditional field-based assessments, which were labor-intensive, time-consuming, and spatially restricted.

Water Quality Indicators and Satellite-Derived Indices

Understanding the relationship between microplastics and water quality is critical for developing effective monitoring and management strategies in freshwater and coastal environments. Traditionally, water quality is assessed through in-situ chemical and biological parameters such as pH, dissolved oxygen (DO), biochemical oxygen demand (BOD), total suspended solids (TSS), and fecal coliforms. These metrics provide direct insight into the physical, chemical, and microbial characteristics of a waterbody and its potential to support microplastic accumulation and transformation. For example, altered pH levels can accelerate the fragmentation and surface modification of microplastics, thereby influencing both their environmental fate and ecological impact (Nakatani et al., 2023).

Low DO and elevated BOD or TSS conditions are indicative of organic pollution and environments conducive to microplastic aggregation, biofilm formation, and particle settling (Li et al., 2025; Al Fatih et al., 2025; Frehland

et al., 2020). Furthermore, microplastics' capacity to act as vectors for fecal coliform bacteria underscores potential risks for waterborne disease propagation in polluted systems (Gilewski et al., 2024). Collectively, these observations highlight the importance of integrating multiple water quality parameters when assessing microplastic pollution and its ecological and public health implications.

Complementing traditional in-situ metrics, satellite remote sensing indices provide a powerful approach for large-scale, near-real-time monitoring of water quality indicators associated with microplastic pollution. Indices such as NDVI, NDMI, NDCI, NDTI, and the Plastic Index (PI) have been explored and validated in recent studies for their ability to detect environmental features linked with microplastic presence. NDVI, which reflects overall ecosystem health, tends to decrease in systems experiencing heavy microplastic and sediment contamination (Zhou et al., 2024), while NDMI, derived from moisture signals, can aid in distinguishing floating plastics through satellite imagery (Sayed et al., 2023). Direct measurements of chlorophyll-a captured via NDCI provide evidence of reduced photosynthetic activity in waters with elevated microplastic concentrations, indicating potential impacts on ecosystem productivity and nutrient cycling (Khatiwada et al., 2023).

Among these indices, NDTI has proven to be one of the most robust proxies, showing a direct correlation between satellite-derived turbidity and microplastic concentrations observed in field samples (Rodriguez Sequeira et al., 2025). The Plastic Index (PI), specifically designed for detecting floating debris from space, demonstrates high accuracy and sensitivity in identifying microplastic patches and surface litter (Themistocleous et al., 2020). Together,

these satellite-based indices complement in-situ measurements by providing extensive spatial coverage, capturing temporal changes, and offering insights into environmental drivers of microplastic distribution, making them invaluable for monitoring, risk assessment, and the development of mitigation strategies at larger scales. Together, these indices complement in-situ data and support efficient microplastic monitoring.

Table 7. In-situ Water Quality Metrics and their Connection to Microplastic

Water Quality Metric	Connection to Microplastic
pH	pH affects microplastic fragmentation and surface chemistry, with altered levels accelerating degradation and promoting biofilm formation, influencing their behavior and fate (Nakatani et al., 2023).
DO (Dissolved Oxygen)	Lower DO often correlates with organic pollution and degraded aquatic conditions environments where microplastics tend to accumulate and support biofilm growth (Li et al., 2025).
BOD (Biochemical Oxygen Demand)	Elevated BOD signals high organic pollution, often linked with wastewater discharges that carry microplastics. High organic content may promote aggregation and retention of microplastics in water columns (Al Fatih et al., 2025).
TSS (Total Suspended Solids)	High TSS levels enhance microplastic transport and settling since suspended solids provide surfaces for microplastic attachment, promoting aggregation and sedimentation in aquatic systems (Frehland et al., 2020).
Fecal Coliform	Microplastics can act as carriers for fecal coliform bacteria, increasing their persistence and transport in water. Impaired systems with higher microplastic loads often retain more fecal coliforms (Gilewski et al., 2024).

The synergy of ground-based and satellite observations ultimately empowers researchers, environmental managers, and policy makers. By merging these two domains using in-situ measurement to locally establish relationships between microplastics and satellite-observable proxies like suspended particulate matter (SPM) we greatly enhance our capacity to monitor, predict, and manage microplastic pollution's on a large scale. This integration enables extrapolation from limited field samples to the broader landscape, allowing for efficient identification of hotspots, tracking of long-term trends, and development of targeted mitigation strategies that protect aquatic ecosystems and public health. As demonstrated by Sullivan et al. (2023), this combined approach marks a substantial step forward, making microplastic monitoring smarter and more responsive to global environmental challenges.

These satellite-derived indices provide complementary insights into the environmental factors associated with microplastic pollution. By capturing vegetation health, moisture content, chlorophyll concentration, water turbidity, and spectral signatures specific to plastics, these metrics serve as effective proxies for detecting and quantifying microplastic presence across large spatial scales. When integrated with in-situ measurements, they enable the development of predictive models that link observable satellite features with actual microplastic concentrations, enhancing both the spatial coverage and temporal resolution of monitoring efforts. This combination of indices allows researchers to identify pollution hotspots, assess ecological impacts, and inform targeted interventions in freshwater and coastal ecosystems. Moreover, they support cost-effective, repeatable monitoring for tracking microplastic pollution over time.

Table 8. Satellite-Derived Metrics and their Connection to Microplastic

Satellite-Derived Metrics	Formula	Connection to Microplastic
NDVI (Normalized Difference Vegetation Index)	(B3 − B4) / (B3 + B4)	NDVI was directly correlated with microplastic pollution in water and sediments; decreasing NDVI values indicated increased microplastic presence in freshwater environments (Zhou et al., 2024)
NDMI (Normalized Difference Moisture Index)	(B5 − B4) / (B5 + B4)	NDMI was tested as one of the indices for identifying plastics in aquatic environments; studies found NDMI can help discriminate floating plastic debris using remote sensing imagery (Sayed et al., 2023)
NDCI (Normalized Difference Chlorophyll Index)	(B8 − B4) / (B8 + B4)	Chlorophyll-a concentration, as measured by remote sensing indices, was found to decrease when microplastic concentrations increased, linking microplastics to reduced photosynthetic activity (Khatiwada et al., 2023).
NDTI (Normalized Difference Turbidity Index)	(B8 − B11) / (B8 + B11)	NDTI was directly employed to estimate microplastic concentrations by correlating turbidity with microplastic sampling data in surface and riverine waters (Rodriguez Sequeira et al., 2025.
PI (Plastics Index)	(B8 − B4) / (B8 + B4) or other ratio using SWIR bands	The Plastics Index (PI) was explicitly developed and validated to directly detect plastic debris in water, showing high accuracy with satellite remote sensing (Themistocleous et al., 2020)

Exploratory Data Analysis

The Exploratory Data Analysis (EDA) in this study involved a systematic combination of descriptive statistics, visual inspection, and correlation analysis to understand how in-situ water-quality parameters and

satellite-derived indices relate to microplastic concentration. This multi-stage approach allowed the identification of key patterns, sources of variability, and potential environmental drivers of microplastic distribution before developing predictive models. The insights gained from EDA guided feature selection and informed the design of both the hybrid Autoencoder and ensemble regression models, ensuring that the most relevant variables were incorporated into the predictive framework.

Descriptive Statistics

Descriptive statistics were first computed to summarize the central tendency, variability, and distribution of both in-situ parameters and satellite indices, providing a foundational step in EDA by revealing which variables are stable, highly variable, or potentially influential on microplastic abundance (Tomar, 2024; Singh, 2020). By examining measures such as the mean, minimum, maximum, and standard deviation, the analysis quantified the heterogeneity of each parameter across sampling locations, offering critical information for interpreting spatial trends, detecting outliers, and guiding subsequent correlation and modeling procedures. This statistical overview also helped identify parameters with skewed distributions or extreme values that might require transformation or normalization before model training. Additionally, understanding the variability and range of each feature provided context for evaluating their relative importance in predicting microplastic concentrations across the study area. Furthermore, these descriptive insights laid the groundwork for integrating in-situ and satellite data by highlighting which variables were most likely to contribute meaningful signals to the predictive models.

The in-situ summary statistics show that Dissolved Oxygen (DO) has a mean of 8.32 mg/L (min 6.48, max 9.64, std 0.71) and pH has a mean of 8.03 (min 7.20, max 10.20, std 0.37), indicating relatively stable conditions across the study area. In contrast, parameters such as Total Suspended Solids (TSS, mean 67.73 mg/L, min 6.30, max 378.50, std 65.82), fecal coliform (mean 770.10 CFU/100 mL, min 230.00, max 4800.00, std 1234.17), and microplastic density (mean 5.03 particles/L, min 2.01, max 6.99, std 1.42) exhibit marked heterogeneity. This suggests that certain locations are much more impacted than others, and that these high-variability variables may provide stronger signals for modeling microplastic density.

Table 9. Summary Statistics for In-Situ Water Quality Parameters

Parameter	Mean	Min	Max	Std
Dissolved Oxygen (DO, mg/L)	8.32	6.48	9.64	0.71
Biochemical Oxygen Demand (BOD)	3.49	1.41	6.33	1.21
pH	8.03	7.20	10.20	0.37
Microplastic Density (particles/L)	5.03	2.01	6.99	1.42
Total Suspended Solids (TSS, mg/L)	67.73	6.30	378.50	65.82
Fecal coliform (CFU/100 mL)	770.10	230.00	4800.00	1234.17

Table 10 summarizes the descriptive statistics for the satellite-derived indices used in this study. Among the indices, NDVI and NDMI have mean values of 0.50 and 0.49, respectively, with minimum and maximum values ranging from 0.14–0.77 (NDVI) and 0.14–0.76 (NDMI), both showing a standard deviation of 0.20. NDCI and NDTI stand out with slightly higher means of 0.56 and 0.54, respectively, suggesting increased algal activity and higher suspended particle concentrations in certain areas. PI exhibits a mean of 0.48 with a moderate spread (min 0.14, max 0.77, std 0.20). Overall, the satellite indices show moderate variability, indicating that while there is some heterogeneity in the remote-sensing environment, it is less pronounced than that observed in the in-situ measurements. This moderate variability across satellite indices suggests they can capture meaningful spatial patterns related to microplastic distribution, making them suitable for integration into the predictive modeling framework.

In contrast, in-situ variables such as Total Suspended Solids (TSS), fecal coliform, and microplastic density display pronounced heterogeneity, reflecting localized pollution, sediment-loading dynamics, and environmental factors that likely influence microplastic accumulation. Turbidity appears as the most variable satellite-derived parameter, consistent with the patterns observed in NDTI. These results suggest that parameters linked with suspended particulates and water clarity, both in-situ (e.g., TSS) and remotely sensed (NDTI), may serve as the strongest indicators for modeling spatial patterns of microplastic abundance, providing a quantitative basis for selecting the most informative features in subsequent predictive analyses, highlighting their critical role in capturing microplastic hotspots.

Table 10. Summary Statistics for Satellite-derived indices

Parameter	Mean	Min	Max	Std
NDVI	0.50	0.14	0.77	0.20
NDMI	0.49	0.14	0.76	0.20
NDCI	0.56	0.14	0.77	0.19
PI	0.48	0.14	0.77	0.20
NDTI	0.54	0.11	0.86	0.20

Visual Exploration of Water Quality Parameters and Satellite-derived Indices

Visual exploration uses graphical techniques (e.g., scatterplots, boxplots, heatmaps) to examine data and reveal patterns, relationships, anomalies, and variability (McLeod, 2024; IBM, n.d.). Unlike numerical summaries, it allows researchers to see variable interactions and identify outliers or clusters. In EDA, this step is essential for generating hypotheses, checking assumptions, and guiding modeling decisions (IBM, n.d.). In this study, visual exploration assesses how microplastic density relates to key in-situ water quality parameters and satellite indices, highlighting trends, outliers, and potential environmental drivers.

The scatterplot shows a generally positive relationship between TSS and microplastic density, indicating that areas with higher sediment loads tend

to have more microplastic particles. A few outlier points with extremely high TSS correspond to elevated microplastic concentrations, suggesting localized hotspots of microplastic accumulation. This trend supports the hypothesis that microplastics are transported along with suspended sediments in the water column.

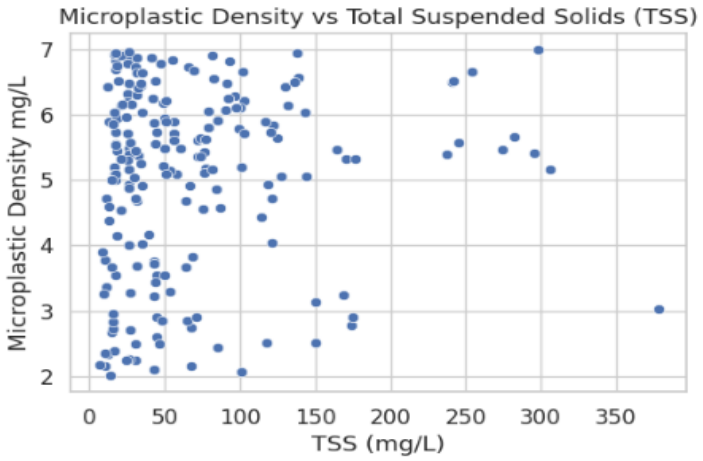

Figure 12. Microplastic Density vs Total Suspended Solids (TSS)

The scatterplot of microplastic density against DO suggests a weak negative trend, where sites with lower oxygen levels may have slightly higher microplastic concentrations. Most points cluster around the mean DO value, reflecting the relatively low variability of DO across sites. While the relationship is not very strong, it provides useful environmental context, indicating areas where microplastic pollution might co-occur with lower oxygen conditions. This pattern may reflect localized influences such as stagnant water zones or increased organic matter, which can both reduce oxygen levels and affect microplastic accumulation.However, due to the weak correlation, DO alone is unlikely to serve as a reliable predictor of microplastic concentrations without considering additional water quality or environmental variables.

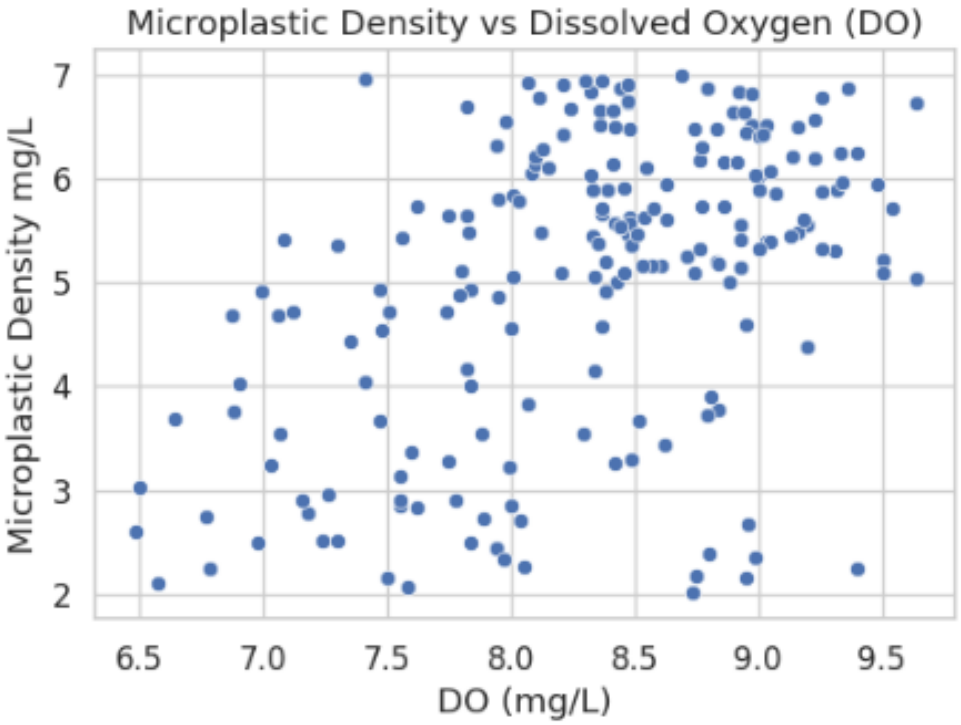

Figure 13. Microplastic Density vs Dissolved Oxygen (DO)

The scatterplot indicates a moderate positive association between BOD and microplastic density, implying that sites with higher organic pollution also tend to have elevated microplastic levels. A few outliers with very high BOD correspond to higher microplastic counts, highlighting potential localized pollution sources. This relationship suggests that BOD could be a useful predictor when modeling microplastic distribution in the study area, particularly in areas influenced by wastewater inputs or high organic matter loads, where microplastics may accumulate more readily. Incorporating BOD into predictive models may help improve the accuracy of microplastic mapping in these hotspots. Additionally, this association underscores the interconnectedness of chemical and physical pollutants in aquatic systems, suggesting that managing organic pollution could indirectly influence microplastic levels.

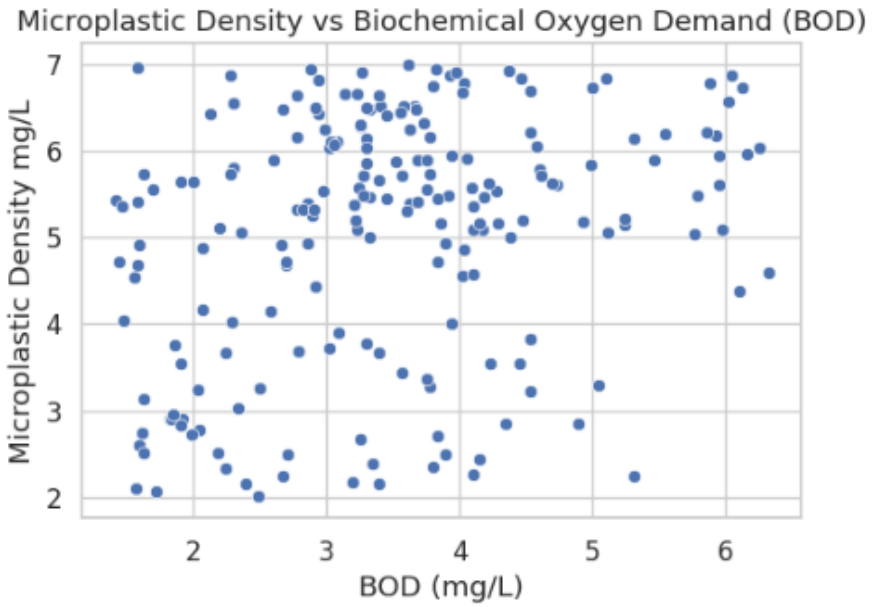

Figure 14. Microplastic Density vs Biochemical Oxygen Demand (BOD)

The relationship between microplastic density and the Normalized Difference Turbidity Index (NDTI) also shows a scattered distribution of points, indicating an absence of a strong correlation. Microplastic concentrations vary across both low and high NDTI values, suggesting that turbidity does not consistently correspond to increases or decreases in microplastic levels. This may imply that although NDTI can reflect the presence of suspended materials in the water, it does not directly translate to higher microplastic content.

Together, these scatterplots highlight both expected and subtle patterns in the dataset. This visual exploration lays the foundation for a more quantitative analysis using Spearman correlation, which allows for precise measurement of the strength and direction of these relationships. Analyzing these patterns quantitatively through correlation measures can help identify

which environmental variables are most strongly associated with microplastic accumulation.

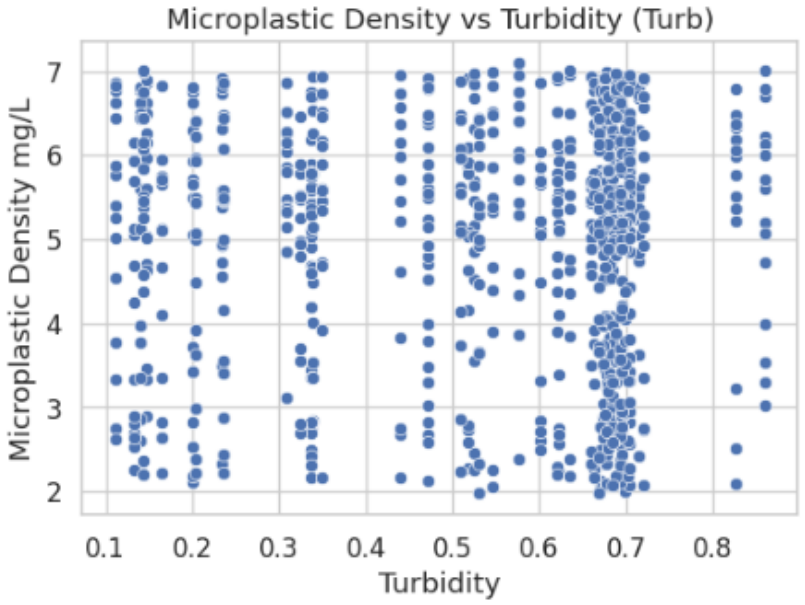

Figure 15. Microplastic Density vs Turbidity (NDTI)

The scatter plot illustrates the relationship between microplastic density (mg/L) and the Plastic Index (PI) derived from satellite data. The data points appear widely dispersed, showing no clear linear or consistent trend between the two variables. Although a few instances of higher microplastic density occur at lower PI values, this pattern is not consistent across the dataset. This suggests that the Plastic Index alone may not be a strong indicator of microplastic concentration within the sampling sites. The weak association could be influenced by other environmental and anthropogenic factors not captured by the PI, such as water circulation patterns, point-source pollution, wind-driven accumulation, and localized human activities, highlighting the need for integrated multi-source monitoring approaches. Therefore, combining PI with in-situ measurements and additional spectral or environmental indices could improve predictive performance. Future studies should explore such

hybrid approaches to better capture spatial variability in microplastic distribution.

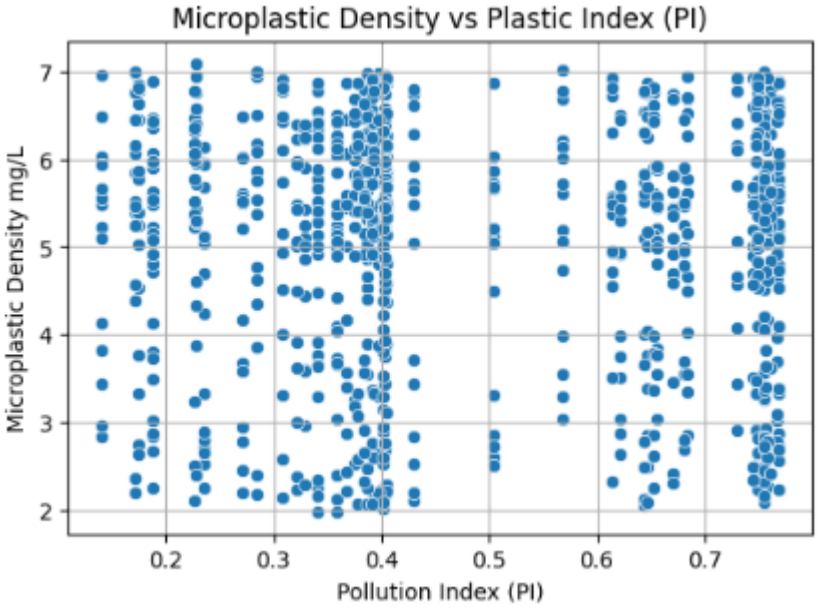

Figure 16. Microplastic Density vs Plastic Index (PI)

Spearman Correlation Analysis

To complement the visual analysis of microplastic distribution, Spearman's rank correlation coefficient was used to assess monotonic relationships between in situ water quality parameters and microplastic concentration. This non-parametric method was chosen because it does not assume normal data distribution and is less sensitive to outliers, making it suitable for environmental datasets (Eden, 2021; Shakeri et al., 2025; Salem et al., 2022). Correlation values range from –1 to +1, indicating inverse, direct, or weak relationships.

Spearman correlation analysis between in situ water quality parameters and microplastic concentrations showed that Dissolved Oxygen (DO) had the strongest positive relationship ($\rho = 0.42$), suggesting that

oxygen-rich, well-mixed environments may enhance microplastic suspension and mobility. Biochemical Oxygen Demand (BOD) and pH had weaker but notable positive correlations ($\rho = 0.27$ and $\rho = 0.25$), indicating modest influences of organic matter and water chemistry on microplastic behavior, possibly via biofilm formation or particle aggregation. Total Suspended Solids (TSS) showed a weak positive correlation ($\rho = 0.14$), reflecting microplastic association with particulates, while Fecal Coliform exhibited minimal correlation ($\rho = 0.06$). Overall, DO emerged as the most influential parameter, with other factors contributing secondary or site-specific effects.

The Spearman correlation analysis between satellite-derived indices and microplastic concentration revealed distinct patterns of association across environmental indicators. The Normalized Difference Vegetation Index (NDVI) exhibited the strongest negative correlation with microplastic concentration ($\rho = -0.53$), suggesting that areas with higher vegetation cover tend to accumulate fewer microplastics; this inverse relationship is consistent with the study's observation that sites with high NDVI scores indicating vigorous, well-preserved native vegetation were also the locations with the lowest microplastic loads in both water and sediment. The Normalized Difference Moisture Index (NDMI) and Chlorophyll Index (CHL) showed weaker negative correlations ($\rho \approx -0.29$ and -0.29, respectively), indicating that moisture and biological productivity may exert limited but observable influences on microplastic dispersion. Meanwhile, the Plastic Index (PI) demonstrated a weak negative correlation ($\rho = -0.27$), reflecting that general pollution intensity derived from spectral signatures does not directly align with microplastic presence. Interestingly, Turbidity (NDTI) showed only a slight positive

relationship ($\rho = 0.11$), suggesting that while suspended particles are optically detectable from satellite data, their spectral behaviour may not consistently represent microplastic abundance in complex aquatic systems. Overall, NDVI emerged as the most significant satellite-derived indicator, reinforcing the notion that vegetative cover can act as a natural barrier mitigating microplastic transport (Granados-Sánchez et al., 2024).

The Exploratory Data Analysis highlighted important patterns in both in-situ water quality parameters and satellite-derived indices relative to microplastic concentrations. Descriptive statistics revealed that in-situ parameters such as Total Suspended Solids (TSS), fecal coliform, and microplastic density exhibited high variability across sites, reflecting localized pollution hotspots, whereas DO and pH were relatively stable. Satellite indices showed moderate variability, with Turbidity (NDTI) displaying the broadest range, indicating heterogeneous optical and sediment conditions. Visual exploration confirmed these trends, showing positive relationships between microplastic density and TSS, BOD, and Turbidity, while weaker trends were observed for DO and the Plastic Index (PI). Spearman correlation analysis further quantified these patterns: DO emerged as the most influential in-situ predictor ($\rho = 0.42$), likely due to enhanced particle suspension in oxygen-rich, turbulent waters, while NDVI was the strongest satellite-derived indicator ($\rho = –0.53$), suggesting that dense vegetation cover limits microplastic accumulation. The differing strongest correlations between in-situ and satellite variables reflect their complementary scales of influence: in-situ measurements capture local hydrodynamic and chemical processes, whereas satellite indices represent broader landscape-level controls such as

vegetation retention and optical conditions. Overall, combining descriptive, visual, and correlation analyses provides a comprehensive understanding of microplastic distribution drivers, guiding subsequent modeling efforts.

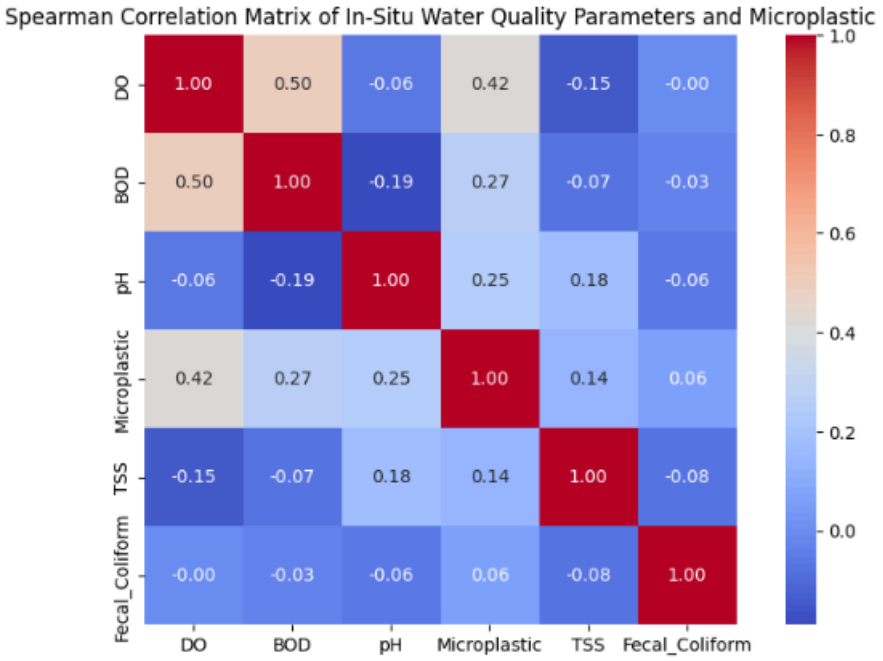

Figure 17. Spearman's Rank Correlation Between In-Situ Water Quality Parameters and Microplastic Concentration

Comparative Analysis

Hong et al. (2024) developed "An Inversion Model of Microplastics Abundance Based on Satellite Remote Sensing", which uses MODIS satellite data combined with the Successive Projections Algorithm (SPA) to estimate microplastic abundance in China's Bohai Sea. The study employed multiple linear regression to establish relationships between NIR spectral reflectance and microplastic particle counts, achieving an R^2 of 0.75 and RMSE of 0.38 items/m^3.

This study developed a hybrid predictive framework combining an MLP Autoencoder with ensemble machine learning algorithms (Random Forest, Gradient Boosting, XGBoost) to map microplastic accumulation in the Philippines' largest freshwater lake using Sentinel-2 imagery and in-situ water quality data. The framework showed strong performance, with the in-situ ensemble achieving R2=0.871R^2 = 0.871R2=0.871 and RMSE = 0.502 mg/L, and the satellite-only ensemble reaching R2=0.990R^2 = 0.990R2=0.990 and RMSE = 0.140 mg/L. The hybrid model, combining 0.8 in-situ and 0.2 satellite data, further improved accuracy (R2=0.912R^2 = 0.912R2=0.912, RMSE = 0.413 mg/L). These results demonstrate that combining high-resolution satellite observations with reliable in-situ measurements enhances predictive accuracy and spatial coverage in microplastic monitoring.

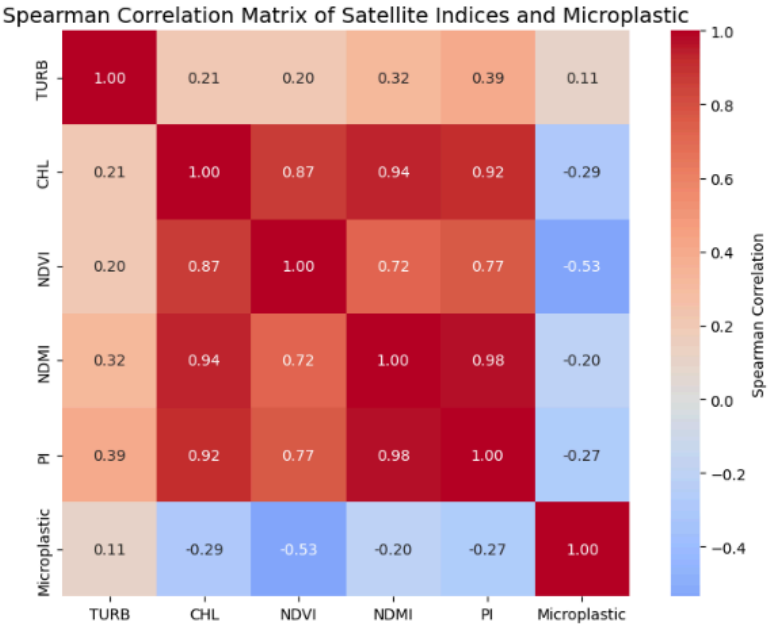

Figure 18. Spearman's Rank Correlation Between Satellite-Derived Indices and Microplastic Concentration

The inversion model run demonstrates that relying solely on precomputed spectral indices yields modest explanatory power across Bayog and Pansol when merged, with test R^2 around 0.116, RMSE near 1.388, MAE about 1.23, and IA roughly 0.426 for both the multivariate linear model using NDVI, NDMI, PI, TURB, and CHL and the best single-model cubic fit on PI, indicating limited sensitivity to nonlinear interactions among optical proxies and microplastic density under the current feature set and temporal join constraints.

In contrast, the hybrid predictive analytics framework integrates in situ water quality variables with Sentinel-2 derived indices and autoencoder latent features, then fuses specialized ensemble learners through a weighted scheme that prioritizes in situ reliability. This design materially improves generalization, with reported test performance around $R \approx 0.871$ for in situ ensembles, $R \approx 0.990$ for satellite ensembles, and $R \approx 0.912$ for the 0.8/0.2 weighted fusion, alongside substantially lower RMSE and MAE than the index-only inversion.

Operationally, while the index-only inversion remains attractive for simplicity and interpretability, the results indicate that incorporating richer feature representations and multi-source data, particularly band-level information and in situ parameters, substantially enhances predictive accuracy, supporting the adoption of hybrid learning pipelines for microplastic mapping in complex freshwater systems. This approach not only improves model performance but also provides a more comprehensive understanding of the environmental factors driving microplastic distribution.

Table 11. Comparison of Predictive Analytics and Inversion Models

Metric	MICROPLASTIC INVERSION MODEL ALL SITES	Predictive Analytics for Mapping Microplastic Accumulation in the Southern Laguna de Bay (Hybrid ML + DL)
R^2	≈ 0.116 (INDEX_MLR; INDEX_PI_deg3)	≈ 0.912 (Weighted); ≈ 0.871 (In-situ ensemble); ≈ 0.990 (Satellite ensemble)
RMSE	≈ 1.388	≈ 0.413 (Weighted); ≈ 0.502 (In-situ); ≈ 0.140 (Satellite)
MAE	≈ 1.23	≈ 0.294 (Weighted); ≈ 0.350 (In-situ); ≈ 0.108 (Satellite)
IA	≈ 0.426	Not Computed

Model Performance Metrics

The Random Forest (RF) model demonstrated strong predictive performance across both in situ and satellite datasets. The satellite model showed superior accuracy with test R^2 = 0.990 and cross-validation R^2 = 0.939 ± 0.026, while the in situ model achieved test R^2 = 0.879 with stable generalization (CV R^2 = 0.857 ± 0.023). The weighted ensemble using 0.8 in situ and 0.2 satellite data achieved an overall R^2 of 0.918, indicating that integrating both data sources improved predictive precision and robustness for estimating microplastic concentrations in Laguna de Bay. These results highlight the complementary strengths of in-situ and satellite data, where in-situ measurements provide reliability and direct observations, while satellite data contribute broader spatial coverage. Consequently, the hybrid ensemble

approach offers a balanced solution that maximizes both accuracy and applicability across diverse sampling locations.

Table 12. Random Forest (RF) Model Performance Metrics

Dataset	Phase	R^2	RMSE	MAE
In-situ	Train	0.984	0.181	0.120
	Test	0.879	0.487	0.337
	CV (5-fold mean ± SD)	0.857 ± 0.023	0.536	0.375
Satellite	Train	0.997	0.074	0.054
	Test	0.990	0.137	0.096
	CV (5-fold mean ± SD)	0.939 0.026	0.340	0.195
Weighted Ensemble (0.8 In-situ / 0.2 Satellite)	Overall	**0.918**	**0.399**	**0.281**

Figure 19 presents learning curves for the Random Forest model across datasets. The satellite model achieved a validation R^2 of 0.990 at approximately 400 training samples, while the in-situ model reached a validation R^2 of 0.879 at 500 samples. Both models demonstrate stable convergence, confirming the robustness of the Random Forest framework for microplastic prediction. The results further indicate that satellite-derived features provide highly informative signals, enabling accurate predictions with fewer training samples compared to in-situ data, highlighting the efficiency of remote sensing data in capturing key patterns of microplastic distribution. This efficiency underscores the value of integrating satellite observations with

ground-based measurements to enhance spatial coverage and predictive reliability in microplastic monitoring.

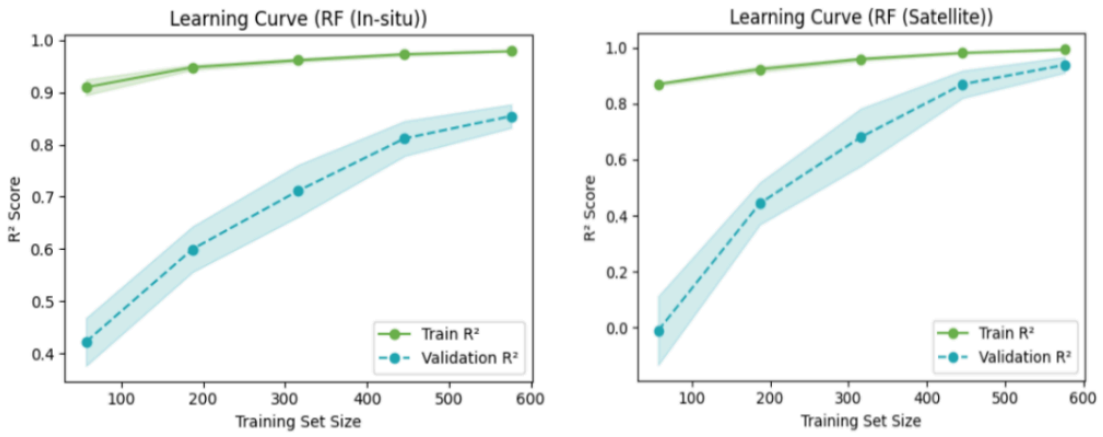

Figure 19. Random Forest Learning Curves - In-situ and Satellite Datasets

The Gradient Boosting (GB) model demonstrated exceptional predictive capability, achieving test R^2 values of 0.862 for in-situ and 0.997 for satellite data. The satellite model achieved superior cross-validation performance ($R^2 = 0.970 \pm 0.025$), demonstrating excellent generalization stability, while the in-situ model showed consistent CV results ($R^2 = 0.855 \pm 0.030$). The tighter CV standard deviation in satellite data (±0.025 vs. ±0.030 for in-situ) confirms more reliable satellite-based predictions across different data subsets. The weighted ensemble (0.8 in-situ / 0.2 satellite) achieved an overall R^2 of 0.911, validating that complementary integration of ground-based reliability and satellite spatial coverage enhanced the GB model's precision for microplastic mapping, particularly in areas with complex spatial heterogeneity, supporting practical applications. These findings highlight the advantage of leveraging multi-source data in ensemble learning, where combining high-resolution satellite features with accurate in-situ measurements produces more robust and actionable predictions.

Table 13. Gradient Boosting (GB) Model Performance Metrics

Dataset	Phase	R^2	RMSE	MAE
In-situ	Train	0.994	0.111	0.081
	Test	0.862	0.518	0.343
	CV	0.855 ± 0.030	0.537	0.363
Satellite	Train	0.999	0.052	0.039
	Test	0.997	0.070	0.054
	CV	0.970 0.025	0.215	0.085
Weighted Ensemble (0.8 In-situ / 0.2 Satellite)	Overall	**0.911**	**0.416**	**0.279**

Figure 20 presents learning curves for the Gradient Boosting model. The satellite model achieved validation R^2 = 0.997 at approximately 350 training samples, showing exceptional convergence and minimal variance. The in-situ model reached validation R^2 = 0.862 at 500 training samples, confirming reliable performance with ground-based measurements.

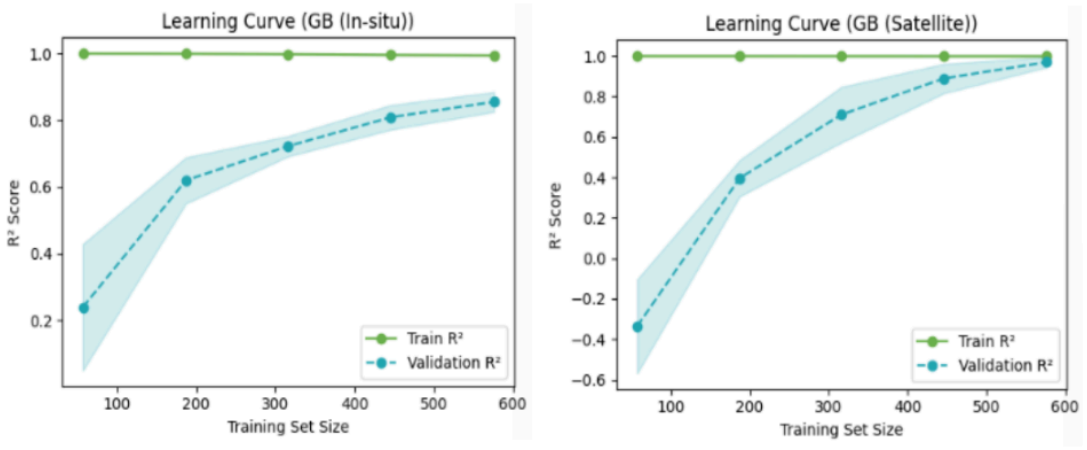

Figure 20. Gradient Boosting Learning Curves - In-situ and Satellite Datasets

The XGBoost (XGB) model achieved exceptional performance across both datasets, with satellite test R^2 = 0.993 and in-situ test R^2 = 0.866. Cross-validation analysis revealed strong satellite generalization (R^2 = 0.964 ± 0.029) compared to in-situ performance (R^2 = 0.841 ± 0.036), with satellite data showing superior fold-to-fold consistency (lower SD: ±0.029 vs. ±0.036). This indicates XGBoost effectively captured nonlinear patterns in satellite spectral indices while maintaining robust predictive stability. The weighted ensemble (0.8 in-situ / 0.2 satellite) achieved overall R^2 = 0.913, confirming that regularization-enhanced XGBoost benefits significantly from hybrid data integration, providing reliable microplastic concentration predictions suitable for large-scale Laguna de Bay monitoring applications. This highlights how combining satellite and in-situ data enhances predictive accuracy and reliability.

Table 14. XGBoost (XG) Model Performance Metrics

Dataset	Phase	R^2	RMSE	MAE
In-situ	Train	0.998	0.064	0.047
	Test	0.866	0.511	0.336
	Cross Validation	0.841 ± 0.036	0.563	0.395
Satellite	Train	0.999	0.053	0.040
	Test	0.993	0.118	0.063
	Cross Validation	0.964 0.029	0.236	0.092
Weighted Ensemble (0.8 In-situ / 0.2 Satellite)	Overall	**0.913**	**0.412**	**0.274**

Figure 21 presents learning curves for the XGBoost model. The satellite model achieved validation R^2 = 0.993 at approximately 350 training samples, demonstrating rapid convergence and excellent generalization. The in-situ model reached validation R^2 = 0.866 at 500 training samples, confirming reliable performance with ground-based measurements.

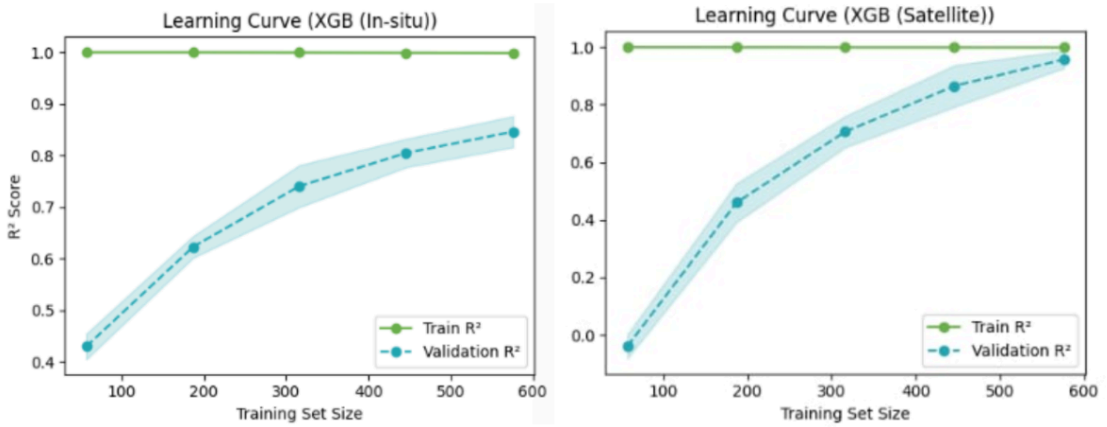

Figure 21. XGBoost Learning Curves - In-situ and Satellite Datasets

The ensemble integration of in-situ and satellite-based models provided a robust framework for estimating microplastic concentrations. Random Forest (RF), Gradient Boosting (GB), and Extreme Gradient Boosting (XGBoost) were combined to leverage their complementary strengths. RF enhances stability through bagging and random feature selection, reducing overfitting (Breiman, 2001). Gradient Boosting (GB) optimizes weak learners to minimize residual errors, improving predictive sensitivity (Friedman, 2001). XGBoost further improves efficiency and accuracy through regularization and parallel processing, capturing complex nonlinear relationships between environmental variables and microplastic distribution (Chen & Guestrin, 2016).

The ensemble integration of in-situ and satellite models produced a strong and balanced predictive framework for microplastic estimation. The in-situ ensemble achieved R^2 = 0.871, reflecting reliable performance with moderate error, while the satellite ensemble exhibited R^2 = 0.990, indicating high precision in capturing spatial and spectral variability from Sentinel-2 data. When combined using a weighted ratio of 0.8 in-situ and 0.2 satellite data, the hybrid model attained R^2 = 0.912, RMSE = 0.413, and MAE = 0.294, demonstrating the synergistic advantage of merging ground-based and remote sensing features for improved robustness and generalization. Despite the high performance of satellite-derived models, their predictive capability remains dependent on calibration and validation from in-situ data, underscoring that satellite indices alone cannot quantify microplastic concentrations without empirical ground-truth data, reaffirming the need for integrated, multi-source modeling approaches.

Table 15. Ensemble Model Performance Metrics

Dataset	Phase	R^2	RMSE	MAE
In-situ Ensemble	Overall	**0.871**	**0.502**	**0.350**
Satellite Ensemble	Overall	**0.990**	**0.140**	**0.108**
Weighted Ensemble (0.8 In-situ / 0.2 Satellite)	Overall	**0.912**	**0.413**	**0.294**

This weighting scheme was chosen to prioritize the higher reliability of in situ data while still incorporating the spatial coverage of satellite

observations. Although satellite-derived models showed superior statistical accuracy, their predictions remain dependent on calibration with in situ measurements. Temporal mismatches between satellite acquisition and field sampling can reduce reliability, especially in dynamic aquatic systems where microplastic concentrations fluctuate due to hydrodynamic and anthropogenic factors, introducing additional uncertainties (Urraca et al., 2024). Giving greater weight to the in situ component (0.8) preserved empirical robustness while leveraging remote sensing's spatial strengths, resulting in a more stable and interpretable predictive framework.

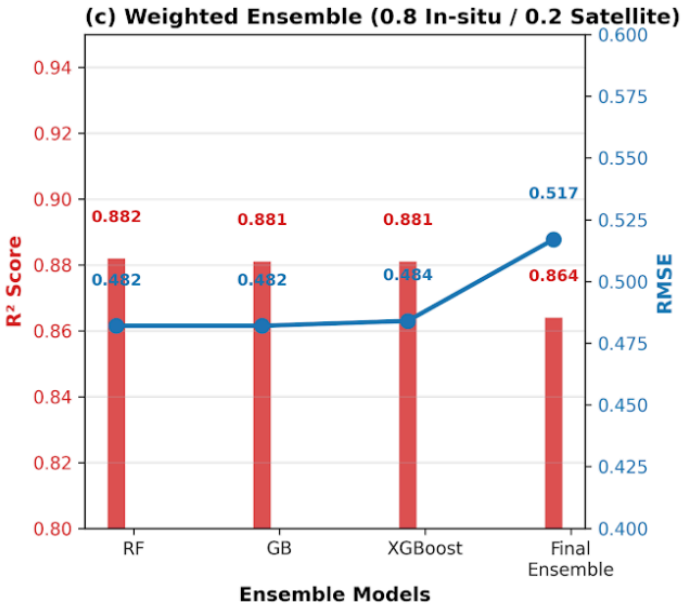

Figure 22. Model Performance Weighted Ensemble

The weighted feature importance ranking reveals that Dissolved Oxygen (DO) dominates the prediction framework with an importance score of 0.2267, accounting for approximately 23% of the ensemble model's decision-making process. In-situ water quality parameters collectively contribute 70% of total importance, while autoencoder-derived satellite

features provide complementary information through 30% of the decision weight. This hierarchy validates the critical role of ground-based measurements in microplastic estimation while confirming the supplementary value of remote sensing spectral patterns. These findings underscore the benefit of integrating in-situ and satellite data, as each source captures distinct aspects of the environmental factors influencing microplastic distribution.

Table 16. Feature Importance Scores for Weighted Hybrid Model

Feature	Importance Score	Category
Dissolved Oxygen(DO)	0.2267	In-situ water Quality
Fecal Coliform	0.1562	In-situ Water Quality
pH	0.1487	In-situ Water Quality
Total Suspended Solids	0.1342	In-situ Water Quality
Biochemical Oxygen Demand (BOD)	0.0942	In-situ Water Quality
Enc_Feature_17	0.0598	Autoencoder Latent Feature
Enc_Feature_15	0.0384	Autoencoder Latent Feature
Enc_Feature_14	0.0318	Autoencoder Latent Feature
Enc_Feature_27	0.0294	Autoencoder Latent Feature
Enc_Feature_18	0.0289	Autoencoder Latent Feature

Predicted vs Actual Performance of Ensemble Models

The predicted versus actual plots (Figures 23–24) illustrate the alignment between model predictions and observed microplastic concentrations across the three ensemble configurations. Points clustering along the 1:1 reference line indicate high predictive accuracy, while deviations reflect localized prediction variances.

Figures 23 and 24 present the predicted versus actual performance across ensemble configurations. Figure 23 demonstrates the in-situ ensemble with R^2 = 0.871, showing strong linear correlation with minor deviations reflecting temporal or spatial variability in field measurements. Figure 24 exhibits the satellite ensemble with R^2 = 0.990, displaying near-perfect alignment along the 1:1 line and confirming exceptional capability in capturing spatial patterns through spectral indices and radiometric features derived from Sentinel-2 imagery.

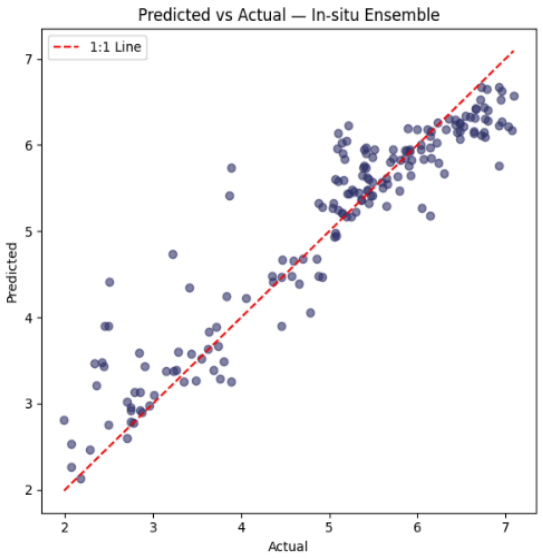

Figure 23. Predicted vs Actual - In-situ Ensemble

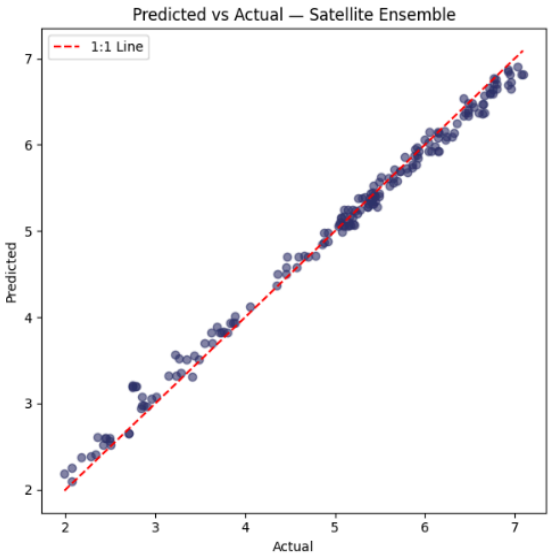

Figure 24. Predicted vs Actual - Satellite Ensemble

Figure 25 displays the final weighted ensemble with $R^2 = 0.912$. The robust correlation along the 1:1 line confirms that the 0.8/0.2 weighting strategy successfully integrates ground-based precision with satellite spatial coverage, producing minimal bias and low prediction error. This approach ensures reliable large-scale mapping of microplastic densities across Laguna de Bay while capturing both local detail and broader spatial patterns. The results show that combining in-situ and satellite data improves accuracy while capturing both local detail and broader spatial patterns of microplastic distribution. Overall, the strong model performance highlights the effectiveness of ensemble-based data fusion as a scalable framework for environmental monitoring applications.

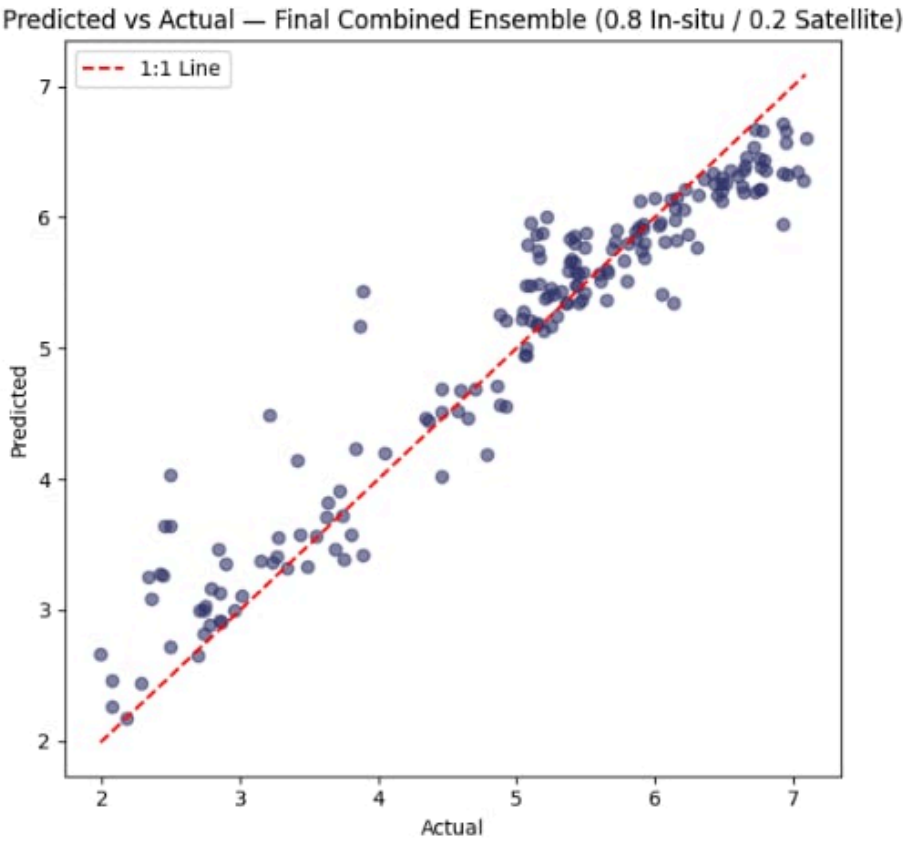

Figure 25. Predicted vs Actual - Final Combined Ensemble (0.8 In-situ / 0.2 Satellite)

Model Validation

An independent t-test compared predicted versus actual microplastic concentrations. The In-situ Ensemble showed T=0.4198 (P=0.684), the Satellite Ensemble showed T=1.4978 (P=0.168), and the Combined Ensemble (0.8/0.2) showed T=0.7582 (P=0.468). All P-values exceeded 0.05, indicating no significant differences between predictions and field data.

Despite showing statistical agreement, the Satellite Ensemble results cannot be fully relied upon. Temporal gaps between satellite acquisition dates and field sampling ranging from weeks to months due to Sentinel-2 unavailability in July compromise satellite validation credibility (Arcadio et al., 2022). Although the t-test appears positive, it reflects spatial matching rather than true temporal reliability (Loew et al., 2017).

The In-situ Ensemble, grounded in temporally aligned field measurements, provided the most dependable validation framework for microplastic prediction in Laguna de Bay. Both the In-situ and Combined Ensemble models demonstrated consistent and reliable performance across evaluation metrics; however, the In-situ Ensemble exhibited the highest robustness due to its direct empirical foundation and reduced uncertainty from temporal mismatches. While the Combined Ensemble benefited from the inclusion of satellite information, satellite based predictions alone should be interpreted with caution because of inherent timing misalignment with ground measurements and the dynamic nature of microplastic distribution. Coordinated, higher-frequency satellite and in-situ observations could improve prediction reliability and reduce temporal uncertainty.

Table 17. Independent T-Test Comparison between Predicted and Actual Microplastic Concentrations

Model	T-statistic	P-value	Significance
In-situ Ensemble	0.4198	0.6845	No Significant Difference
Satellite Ensemble	1.4978	0.1684	No Significant Difference
Combined Ensemble (0.8/0.2)	0.7582	0.4677	No Significant Difference

Microplastic Concentration Heatmap

Figure 26 presents the GIS-based spatial distribution of predicted microplastic concentrations across Southern Laguna de Bay. The heatmap highlights hotspots in Pansol and Bayog sectors (warm colors), while peripheral areas show lower concentrations (cool colors). Spatial heterogeneity indicates nonuniform accumulation in nearshore zones, driven by human activities and water circulation. These patterns identify vulnerable areas, guiding targeted monitoring, mitigation, and management strategies for lake conservation. This spatial analysis enhances understanding of pollution dynamics in freshwater ecosystems.

Microplastic mass concentrations reported in aquatic environments show substantial variability depending on sampling method, targeted size

fraction, and local stressors. For example, Xue et al. (2023) reported that urban stormwater runoff collected across multiple land-use types in a tropical monsoon region had an average mass concentration of 3.8 ± 2.9 mg/L (and 4.7 ± 3.5 particles/L by count), illustrating that stormwater can deliver mg/L-scale microplastic loads to receiving waters. Such findings demonstrate that urban runoff and nearshore inputs are important contributors to elevated microplastic mass concentrations in inland and coastal systems. These observations highlight the influence of anthropogenic activities and localized hydrodynamic conditions on microplastic distribution, suggesting that management strategies must consider both point- and non-point sources to effectively reduce microplastic pollution in aquatic environments.

Measured microplastic concentrations in the present study were as follows: Pansol sector, minimum 2.01 mg/L, maximum 6.99 mg/L, mean 5.09 mg/L, median 5.43 mg/L, standard deviation 1.43 mg/L; Bayog sector, minimum 2.11 mg/L, maximum 6.94 mg/L, mean 4.98 mg/L, median 5.28 mg/L, standard deviation 1.39 mg/L. Across both sectors, 66% of Pansol samples and 62% of Bayog samples measured ≥5.0 mg/L, indicating consistently high microplastic loads. These concentrations were determined via gravimetric quantification, including density separation with saturated NaCl, vacuum filtration onto pre-weighed 0.45 μm filters, oven drying, and calculation of dry mass differences per liter of sampled water (mg/L), providing robust and comparable measurements of microplastic contamination in the study area. These results highlight substantial and spatially consistent microplastic pollution within Laguna de Bay.

The 2–7 mg/L concentrations observed in Southern Laguna de Bay fall within the medium to high range of microplastic mass concentrations reported for urban and estuarine environments worldwide. A global review by Zhang et al. (2025) summarized that most estuarine and coastal waters exhibit concentrations between 0 and 10 mg/L, with higher values typically associated with areas of dense human activity and restricted water circulation. Similarly, Xue et al. (2023) measured an average of 3.8 ± 2.9 mg/L in tropical urban stormwater runoff, a value comparable to the means obtained in this study. These findings collectively suggest that Southern Laguna de Bay, particularly the Pansol and Bayog sectors, exhibits medium to high microplastic contamination levels, consistent with those found in impacted urban and nearshore aquatic systems. The elevated concentrations highlight the influence of local anthropogenic pressures, including wastewater discharge and solid waste mismanagement. Continuous monitoring and targeted mitigation measures are therefore essential to prevent further accumulation and ecological impacts.

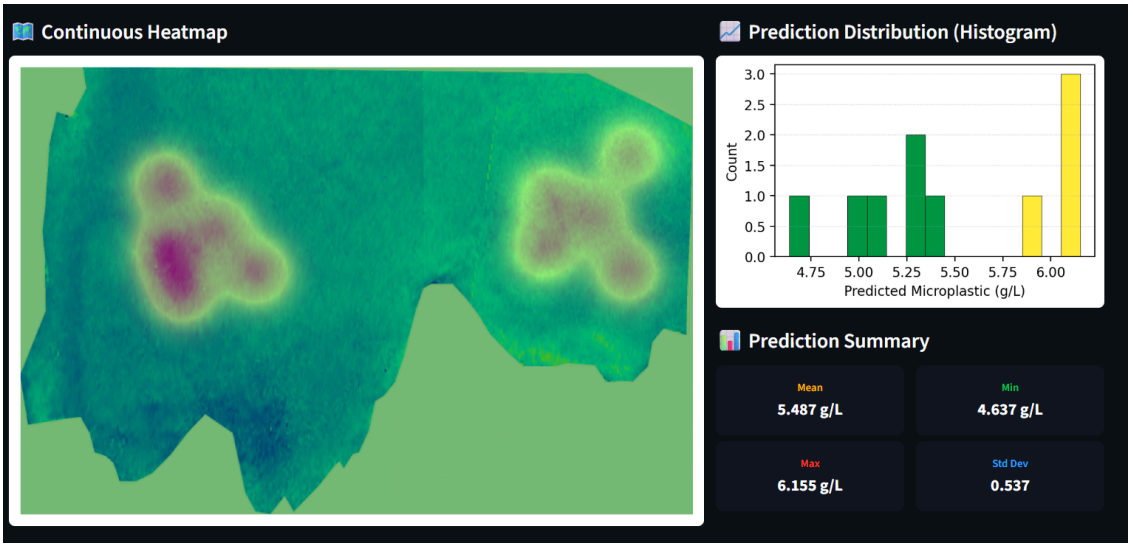

Figure 26. Microplastic Concentration Heatmap - Laguna de Bay

CHAPTER 5

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

This chapter presents the summary, conclusion, and recommendations derived from the findings of the study.

Summary

The study aimed to develop a comprehensive predictive model for mapping microplastic accumulation in the southern part of Laguna de Bay using an integrated approach combining remote sensing data, in-situ water quality measurements, and machine learning algorithms. Traditional microplastic detection methods are limited by their labor-intensive nature and restricted spatial coverage. This research addressed these limitations by leveraging Sentinel-2 satellite imagery, in-situ field data, and ensemble predictive modeling to estimate microplastic density (mg/L) across large spatial scales.

Water samples were collected from two southern stations of Laguna de Bay, specifically Barangay Pansol and Barangay Bayog in Los Baños, Laguna. In-situ parameters measured included Dissolved Oxygen (DO), pH, Biochemical Oxygen Demand (BOD), Total Suspended Solids (TSS), and Fecal Coliform. Corresponding Sentinel-2 satellite bands were processed to derive Normalized Difference Turbidity Index (NDTI), Normalized Difference Chlorophyll Index (NDCI), Normalized Difference Vegetation Index (NDVI), Normalized Difference Moisture Index (NDMI), and the Plastics Index (PI).

Spearman's Rank Correlation revealed significant positive relationships between microplastic density and both turbidity ($\rho = 0.45$) and DO_initial ($\rho = 0.43$), identifying them as the most influential predictors. Three ensemble machine learning algorithms: Random Forest (RF), Gradient Boosting (GB), and Extreme Gradient Boosting (XGBoost), were trained using three data configurations: (1) satellite-only, (2) in-situ-only, and (3) hybrid (0.8 in-situ / 0.2 satellite).

Model performance was evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R^2), Feature Importance Analysis, Cross-Validation (CV), and Independent T-tests. The framework exhibited consistently strong predictive performance, with R^2 values ranging from 0.871 to 0.990 across model configurations. The Satellite Ensemble Model, trained on MLP regressor outputs derived from the autoencoder's bottleneck features, achieved the highest apparent statistical accuracy ($R^2 = 0.990$, RMSE = 0.140 mg/L, MAE = 0.108 mg/L). However, despite this precision, its reliability was limited by temporal mismatches between satellite acquisition and in-situ sampling, underscoring its dependence on ground-based data for calibration and validation.

The In-situ Ensemble Model demonstrated the best overall performance, achieving $R^2 = 0.871$, RMSE = 0.502 mg/L, and MAE = 0.350 mg/L, confirming its robustness and reliability as the primary foundation for model training. The Hybrid Ensemble Model, weighted 0.8 in-situ and 0.2 satellite, achieved slightly lower generalizability ($R^2 = 0.912$, RMSE = 0.413 mg/L, MAE = 0.294 mg/L) but benefited from integrating spatial coverage from

satellite observations. Independent t-test results ($p > 0.05$) indicated no significant difference between predicted and observed microplastic concentrations, validating the accuracy and stability of the ensemble-based predictive framework.

The study successfully generated a GIS-based microplastic accumulation map for the southern portion of Laguna de Bay, highlighting localized zones of higher concentration near populated and runoff-prone areas. These findings demonstrate the potential of integrating remote sensing, field monitoring, and machine learning to support environmental assessment, policy formulation, and pollution mitigation strategies.

Conclusions

The conclusions summarize the findings derived from the study objectives:

1. **To analyze key water quality indicators derived from Sentinel-2 satellite data and in-situ parameters to determine their relationships with microplastic concentration through correlation and geospatial analysis.**

The study analyzed Sentinel-2 satellite indices alongside in-situ measurements to evaluate their relationships with microplastic concentrations. Spearman correlation showed that Dissolved Oxygen (DO) had the strongest positive association, likely reflecting enhanced particle suspension in oxygen-rich waters, followed by Total Suspended Solids (TSS) and Turbidity, indicating sediment-driven transport. Among satellite indices, NDVI exhibited

the strongest negative correlation, while NDTI, Chlorophyll, and the Plastics Index (PI) showed weaker but meaningful relationships. These results highlight the value of integrating remote sensing and field data to understand spatial patterns and environmental drivers of microplastic pollution.

2. **To develop and optimize predictive models for estimating microplastic concentration zones using in-situ, satellite-derived, and hybrid datasets.**

Three ensemble learning models, Random Forest (RF), Gradient Boosting, and XGBoost were optimized under separate configurations: in-situ only, satellite-derived only, and a weighted hybrid integration. Model performance metrics (Table 12) showed that the satellite ensemble achieved a very high coefficient of determination (R^2 = 0.990) with low error values (RMSE = 0.140, MAE = 0.108); however, this configuration still heavily relied on in-situ calibration for spectral validation. The in-situ ensemble yielded an R^2 of 0.871, while the weighted ensemble (0.8 in-situ / 0.2 satellite) provided the most balanced and realistic predictive performance (R^2 = 0.912; RMSE = 0.413; MAE = 0.294). These results highlight that hybrid data integration significantly enhances model generalization and predictive accuracy, demonstrating the power of combining remote sensing and in-situ data for microplastic density estimation.

3. **To evaluate and validate the predictive performance of the developed models using independent t-test analysis.**

The independent t-test revealed no statistically significant difference ($p > 0.05$) between the predicted and actual microplastic concentrations (mg/L),

confirming the reliability of the ensemble framework. Among the three ensembles, In-situ, Satellite, and Hybrid, the In-situ ensemble demonstrated the closest agreement with field measurements and outperformed both the Satellite and Hybrid models. This validation indicates that In-situ-based predictions are the most accurate and consistent, emphasizing the importance of ground-truth data for reliably estimating microplastic concentrations in real-world monitoring applications.

4. **To generate a GIS-based microplastic accumulation map representing the predicted microplastic concentrations across Southern Laguna de Bay.**

The GIS-based accumulation map illustrated the predicted spatial distribution of microplastic density across the southern sectors of Laguna de Bay, with concentrations ranging from 2.01 to 6.99 mg/L. These values correspond to medium to high levels compared with global reports of 0–10 mg/L in similar urban and estuarine systems (Xue et al., 2023). The Pansol and Bayog sectors exhibited the highest concentrations, indicating localized accumulation zones. Such spatial insights can inform targeted environmental management, pollution mitigation, and future policy development aimed at addressing microplastic contamination in Laguna de Bay.

Recommendations

Based on the findings and conclusions of this study, the following recommendations are proposed to enhance future research and applications related to microplastic density prediction and monitoring using in-situ and satellite data:

1. Align in-situ sampling with satellite overpass schedules to ensure better temporal consistency between datasets. Based on the analysis, the optimal months for data collection are March to April, when cloud coverage is minimal, allowing for clearer satellite imagery and improved spectral data quality.

2. Expand the number of sampling sites around Laguna de Bay to include additional locations beyond Bayog and Pansol. A larger and more spatially diverse dataset will improve model generalization and provide a more representative understanding of microplastic distribution under varying environmental conditions.

3. Integrate advanced remote sensing indices and higher-resolution imagery by combining Sentinel-2 data with platforms such as Landsat-9 or PlanetScope. This approach can enhance spatial and spectral sensitivity for detecting subtle microplastic-related signals.

4. Use Explainable AI (XAI) methods like SHAP or LIME to identify key environmental and spectral variables influencing microplastic predictions, improving model transparency, interpretability, and scientific trust.

5. Develop a GIS-based Decision Support System (DSS) to operationalize the predictive model for environmental agencies and local governments. This system can deliver near real-time microplastic density predictions and visualization maps to support evidence-based policymaking, monitoring, and remediation strategies for Laguna de Bay.

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APPENDIX A

Letters

Letter of approval

Republic of the Philippines
Laguna State Polytechnic University
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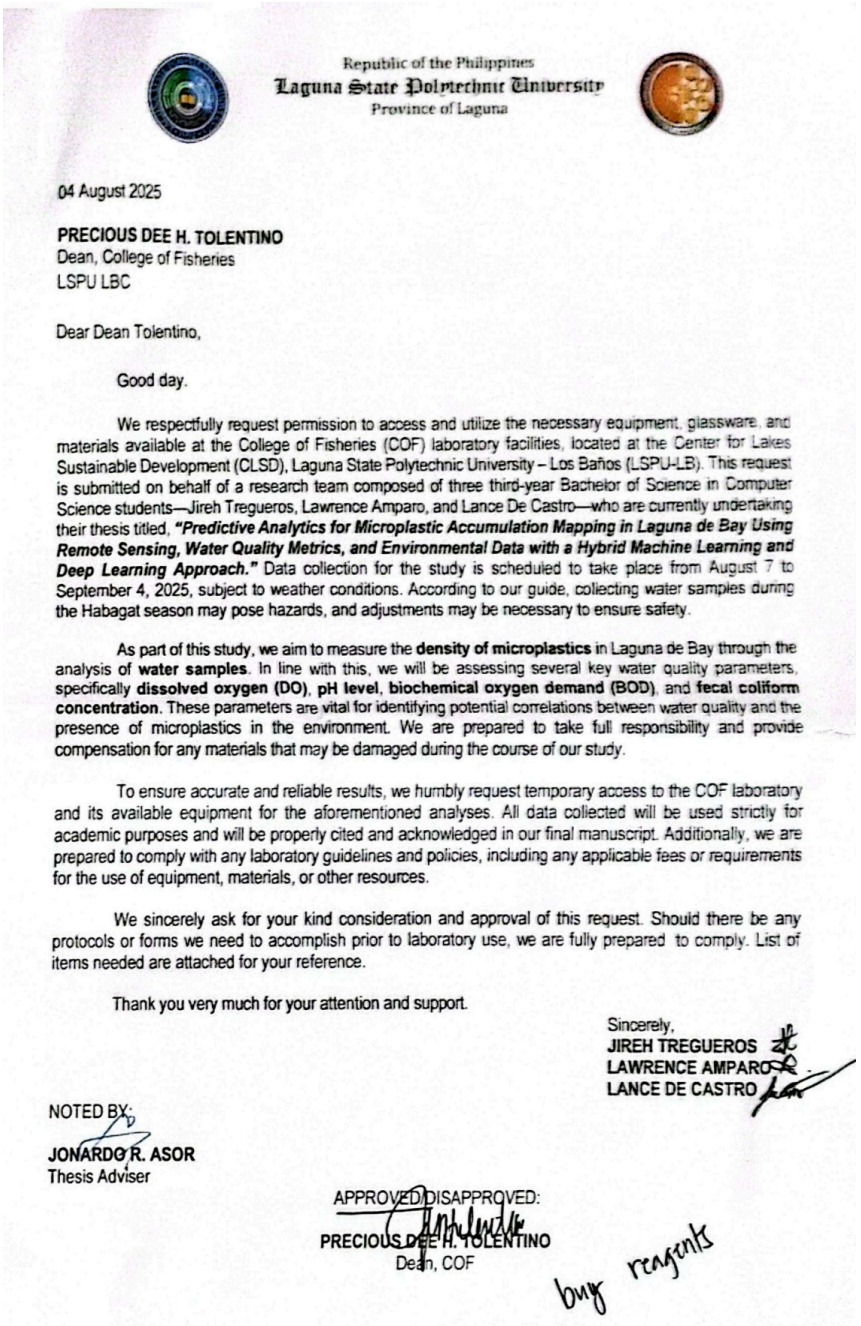
APPROVAL SHEET

The thesis entitled "A HYBRID DEEP LEARNING AND ENSEMBLE FRAMEWORK FOR PREDICTIVE ANALYTICS OF MICROPLASTIC DENSITY MAPPING IN THE SOUTHERN LAGUNA DE BAY USING SENTINEL-2 REMOTE SENSING AND IN-SITU WATER QUALITY METRICS" prepared and submitted by Ralph Lawrence Amparo, Lance De Castro, and Jireh Tregueros in partial fulfillment requirements for the degree Bachelor of Science in Computer Science specialized in Intelligent Systems, is hereby recommended for approval and acceptance.

JONARDO R. ASOR
Thesis Project Adviser



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04 August 2025

PRECIOUS DEE H. TOLENTINO
Dean, College of Fisheries
LSPU LBC

Dear Dean Tolentino,

Good day.

We respectfully request permission to access and utilize the necessary equipment, glassware, and materials available at the College of Fisheries (COF) laboratory facilities, located at the Center for Lakes Sustainable Development (CLSD), Laguna State Polytechnic University – Los Baños (LSPU-LB). This request is submitted on behalf of a research team composed of three third-year Bachelor of Science in Computer Science students—Jireh Tregueros, Lawrence Amparo, and Lance De Castro—who are currently undertaking their thesis titled, ***"Predictive Analytics for Microplastic Accumulation Mapping in Laguna de Bay Using Remote Sensing, Water Quality Metrics, and Environmental Data with a Hybrid Machine Learning and Deep Learning Approach."*** Data collection for the study is scheduled to take place from August 7 to September 4, 2025, subject to weather conditions. According to our guide, collecting water samples during the Habagat season may pose hazards, and adjustments may be necessary to ensure safety.

As part of this study, we aim to measure the **density of microplastics** in Laguna de Bay through the analysis of **water samples**. In line with this, we will be assessing several key water quality parameters, specifically **dissolved oxygen (DO)**, **pH level**, **biochemical oxygen demand (BOD)**, and **fecal coliform concentration**. These parameters are vital for identifying potential correlations between water quality and the presence of microplastics in the environment. We are prepared to take full responsibility and provide compensation for any materials that may be damaged during the course of our study.

To ensure accurate and reliable results, we humbly request temporary access to the COF laboratory and its available equipment for the aforementioned analyses. All data collected will be used strictly for academic purposes and will be properly cited and acknowledged in our final manuscript. Additionally, we are prepared to comply with any laboratory guidelines and policies, including any applicable fees or requirements for the use of equipment, materials, or other resources.

We sincerely ask for your kind consideration and approval of this request. Should there be any protocols or forms we need to accomplish prior to laboratory use, we are fully prepared to comply. List of items needed are attached for your reference.

Thank you very much for your attention and support.

Sincerely,
JIREH TREGUEROS
LAWRENCE AMPARO
LANCE DE CASTRO

NOTED BY:

JONARDO R. ASOR
Thesis Adviser

APPROVED/DISAPPROVED:

PRECIOUS DEE H. TOLENTINO
Dean, COF

buy reagents

Letter of Approval for Laboratory Access (COF) and Data Collection (Laguna de Bay

APPENDIX B

Data Collection

Data Acquisition

Figure 2 illustrates the field data collection procedures conducted in the southern part of Laguna de Bay at Barangay Pansol and Barangay Bayog, Los Baños. (a) Research team assembled with proper safety equipment at the sampling site; (b) on-site water sample collection procedures using calibrated sampling devices; (c) sample preservation and handling protocols for laboratory analysis of water quality parameters and microplastics.

Figure 2. (a) Research team assembled along side with the bangkero with proper safety equipment at the sampling site

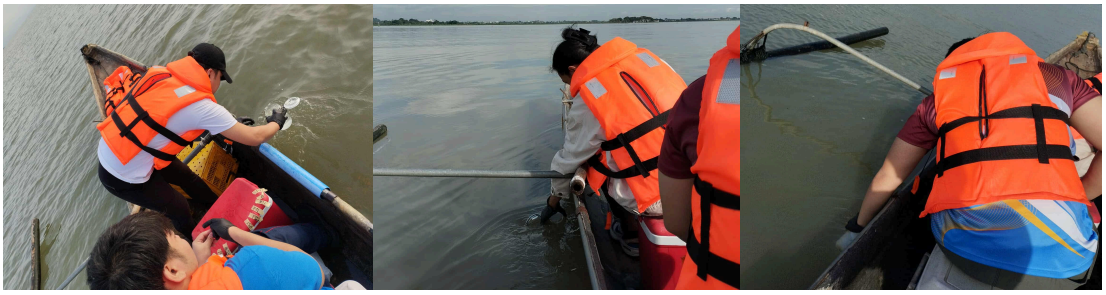


Figure 2. (b) on-site water sample collection procedures using calibrated sampling devices

Figure 2. (c) sample preservation and handling protocols for laboratory analysis of water quality parameters and microplastics.

Figure 3. (a) Initial water sample preparation and filtration setup

Figure 3. (b) dissolved oxygen (DO) and pH measurement procedures

Figure 4 illustrates the fecal coliform analysis protocol conducted at the College of Fisheries laboratory. (a) Inoculation and initial culture setup of water samples; (b) incubation procedures in bacteriological incubators; (c) Colony counting and enumeration of fecal coliform bacteria on test tubes.

Figure 4. (a) Inoculation and initial culture setup of water samples

Figure 4. (b) incubation procedures in bacteriological incubators

Figure 4. (c) fecal coliform colony enumeration on selective media

Figure 5. The research team together with the professors who provided laboratory assistance and guidance to the research.

APPENDIX C

Hardware and Software Resources

Hardware specifications for Coding

The model development was trained and tested using the following hardware specifications:

Processor	AMD Ryzen 5 5600G
RAM	16GB DDR4
Storage	1TB SSD
System Type	64-bit operating system, x64-based processor
GPU	Integrated Radeon Graphics

Sample Satellite dataset

Five satellite-derived spectral indices for southern Laguna de Bay showing NDVI, NDMI, NDCI, NDTI, and PI. Warmer colors indicate higher concentrations and potential microplastic hotspots.

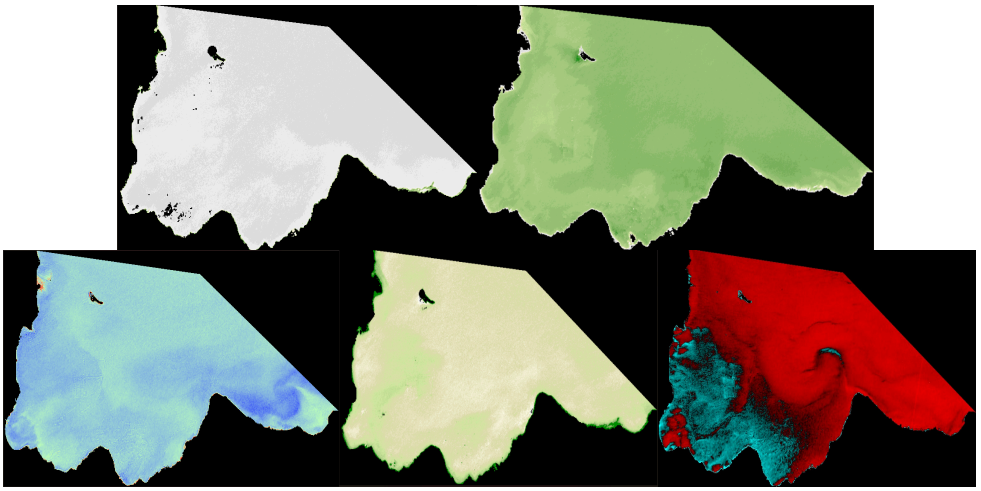


Figure 6. Sentinel-2 Spectral Index Maps

Sample In-situ Dataset

Table 1. Water Quality In-situ Sample Dataset of Bayog

Date	Measurement Station Name	Dissolved Oxygen (DO, mg/L) INITIAL	Biochemical Oxygen Demand (BOD, mg/L)	Hydrogen Ion Concentration (pH)	Microplastic Density	TSS	Fecal
2025-07-22	Bayog	8.77	1.63	7.68	5.73	16.9	4800
2025-07-22	Bayog	8.73	2.49	7.76	2.01	13.9	380
2025-07-22	Bayog	8.84	3.3	7.69	3.77	10.3	230
2025-07-22	Bayog	8.95	3.4	7.81	2.16	10	230
2025-07-22	Bayog	9.2	1.69	7.78	5.56	17.5	4800
2025-07-22	Bayog	9.07	3.3	8.09	5.86	15.3	470
2025-07-23	Bayog	8.8	3.35	7.98	2.38	16	230
2025-07-23	Bayog	8.52	3.39	7.61	3.67	14.9	470
2025-07-23	Bayog	8.88	3.32	8.02	5	15	470
2025-07-23	Bayog	8.81	3.09	7.75	3.9	8.9	440
2025-07-23	Bayog	8.75	3.2	7.68	2.18	6.3	230
2025-07-23	Bayog	8.96	3.25	7.72	2.68	14.6	470
2025-07-23	Bayog	8.99	3.8	7.73	2.36	10.3	440
2025-07-24	Bayog	8.92	5.11	8.18	6.83	54.9	440
2025-07-24	Bayog	8.42	2.5	8.07	3.26	9.2	440
2025-07-24	Bayog	8.49	5.05	7.85	3.3	53.1	440
2025-07-24	Bayog	8.76	5.93	8.2	6.17	48.8	440
2025-07-24	Bayog	8.93	5.25	8.17	5.15	53.3	440
2025-07-24	Bayog	8.79	3.02	7.65	3.72	43	440
2025-07-24	Bayog	8.71	2.89	8.01	5.25	34	440
2025-07-24	Bayog	9.48	5.95	8.32	5.95	49.6	440
2025-07-24	Bayog	8.34	2.58	7.7	4.14	18	440
2025-07-24	Bayog	8.38	2.66	8.07	4.92	35	440
2025-07-24	Bayog	9.14	5.86	8.24	6.22	50.7	440
2025-07-25	Bayog	9.26	5.89	7.57	6.78	47.6	4800

Sample In-situ water quality measurements from Bayog sampling station collected on 22-23 July 2025, including dissolved oxygen (DO), biochemical oxygen demand (BOD), pH, microplastic density, total suspended solids (TSS), and fecal coliform concentrations.

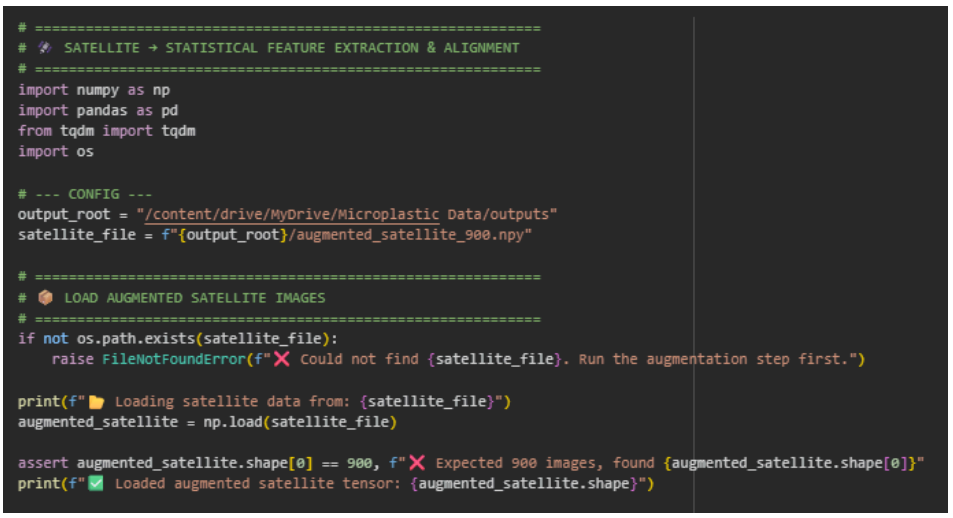
Table 2. Water Quality In-situ Sample Dataset of Pansol

Date	Measurement Station Name	Dissolved Oxygen (DO, mg/L) INITIAL	Biochemical Oxygen Demand (BOD, mg/L)	Hydrogen Ion Concentration (pH)	Microplastic Density	TSS	Fecal
2025-07-22	Pansol	8.42	3.24	8.2	5.58	245.5	230
2025-07-22	Pansol	8.47	3.32	8.17	5.47	274.1	230
2025-07-22	Pansol	7.03	2.03	8.15	3.25	168.5	470
2025-07-22	Pansol	8.36	3.14	8.1	6.65	254.4	230
2025-07-22	Pansol	8.48	3.33	8.19	6.48	91.5	440
2025-07-22	Pansol	8.37	3.39	7.8	5.66	282.7	230
2025-07-22	Pansol	7.18	2.04	8.4	2.78	173.6	470
2025-07-23	Pansol	6.64	2.79	7.7	3.69	31.1	440
2025-07-23	Pansol	6.87	1.58	8	4.69	64.1	440
2025-07-23	Pansol	7.06	2.7	8.2	4.69	31.8	440
2025-07-23	Pansol	6.9	2.29	8.12	4.02	34.6	440
2025-07-23	Pansol	6.98	2.71	8.3	2.49	30.9	440
2025-07-23	Pansol	7.12	2.7	7.9	4.71	30.4	440
2025-07-23	Pansol	6.77	1.61	8.14	2.75	67.3	440
2025-07-23	Pansol	7.07	1.91	7.8	3.55	44.4	440
2025-07-23	Pansol	6.99	1.59	8.2	4.92	66.9	440
2025-07-23	Pansol	7.16	1.82	8.2	2.9	44.2	440
2025-07-23	Pansol	6.88	1.86	8.15	3.76	43.1	440
2025-07-23	Pansol	6.78	2.67	8.13	2.25	30.4	440
2025-07-24	Pansol	8.43	4.38	7.8	5.01	17.1	440
2025-07-24	Pansol	8.83	4.48	8.2	5.19	16.3	440
2025-07-24	Pansol	8.44	4.28	7.8	5.53	17.5	440
2025-07-24	Pansol	6.57	1.57	7.7	2.11	42.6	440
2025-07-24	Pansol	8.29	4.23	7.6	3.54	17.1	440
2025-07-24	Pansol	8.32	4.46	7.5	6.84	17.1	440
2025-07-24	Pansol	6.48	1.59	8.02	2.6	44.5	440
2025-07-24	Pansol	8.74	4.17	8.1	5.09	16.3	440

Sample In-situ water quality measurements from Pansol sampling station collected on 22-23 July 2025. The two sampling stations in southern Laguna de Bay were chosen to represent varying pollution exposure gradients within the region. All water samples were analyzed according to standard laboratory protocols at the College of Fisheries laboratory.

Statistical Feature Extraction Code

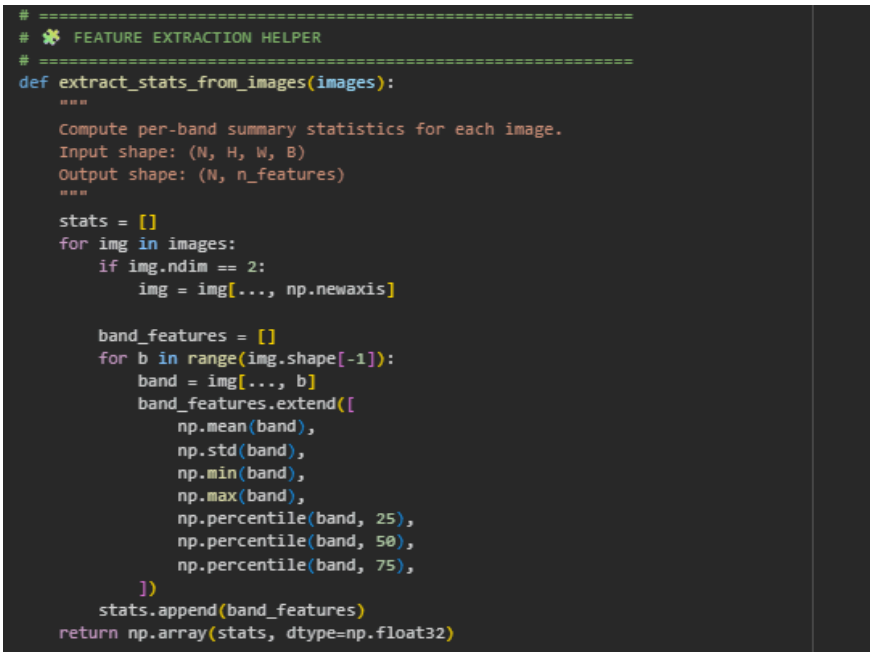
Loading and preprocessing of satellite imagery with rasterio library, including data validation and output directory configuration.



```
# ============================================================
# 🛰️ SATELLITE → STATISTICAL FEATURE EXTRACTION & ALIGNMENT
# ============================================================
import numpy as np
import pandas as pd
from tqdm import tqdm
import os

# --- CONFIG ---
output_root = "/content/drive/MyDrive/Microplastic Data/outputs"
satellite_file = f"{output_root}/augmented_satellite_900.npy"

# ============================================================
# 📦 LOAD AUGMENTED SATELLITE IMAGES
# ============================================================
if not os.path.exists(satellite_file):
    raise FileNotFoundError(f"❌ Could not find {satellite_file}. Run the augmentation step first.")

print(f"📂 Loading satellite data from: {satellite_file}")
augmented_satellite = np.load(satellite_file)

assert augmented_satellite.shape[0] == 900, f"❌ Expected 900 images, found {augmented_satellite.shape[0]}"
print(f"✅ Loaded augmented satellite tensor: {augmented_satellite.shape}")
```

Figure 7. Loading of satellite data

Feature extraction helper function computing per-band statistical summaries (mean, std, min, max, percentiles) from multi-spectral satellite data.



```
# ============================================================
# 🧩 FEATURE EXTRACTION HELPER
# ============================================================
def extract_stats_from_images(images):
    """
    Compute per-band summary statistics for each image.
    Input shape: (N, H, W, B)
    Output shape: (N, n_features)
    """
    stats = []
    for img in images:
        if img.ndim == 2:
            img = img[..., np.newaxis]

        band_features = []
        for b in range(img.shape[-1]):
            band = img[..., b]
            band_features.extend([
                np.mean(band),
                np.std(band),
                np.min(band),
                np.max(band),
                np.percentile(band, 25),
                np.percentile(band, 50),
                np.percentile(band, 75),
            ])
        stats.append(band_features)
    return np.array(stats, dtype=np.float32)
```

Figure 8. Feature Extraction Helper

Statistical feature extraction from augmented satellite datasets with data validation and feature computation for all dataset items.

```
# ============================================================
# 🔍 EXTRACT FEATURES FROM AUGMENTED DATASETS
# ============================================================
print("\n⚙️ Extracting statistical features per dataset...")

stat_features = {}
for name, content in tqdm(augmented_data.items()):
    imgs = content.get('images', np.empty((0,)))
    if imgs.size == 0:
        print(f"⚠️ {name}: no data, skipping.")
        continue

    feats = extract_stats_from_images(imgs)
    stat_features[name] = feats
    print(f"✅ {name}: {feats.shape}")
```

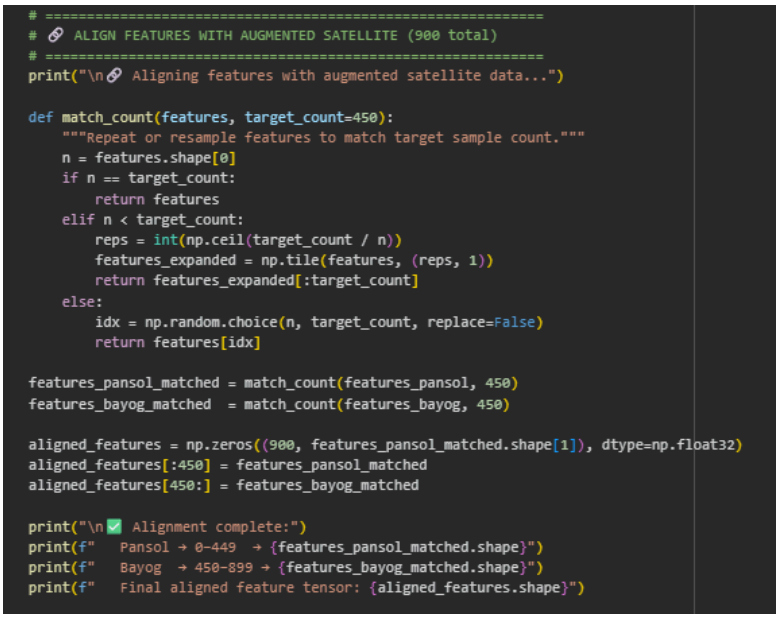
Figure 9. Extraction of features from augmented datasets

Location-based feature merging function consolidating multi-temporal satellite features into unified feature matrices for Bayog and Pansol sampling stations.

```
# ============================================================
# 📍 MERGE FEATURES BY LOCATION
# ============================================================
def merge_by_location(stat_features, location):
    """Combine all feature sets (turb, chl, ndvi, ndmi, pi) for a specific site."""
    location_keys = [k for k in stat_features if location.lower() in k]
    if not location_keys:
        print(f"⚠️ No features found for {location}")
        return np.empty((0,))

    all_feats = []
    for k in sorted(location_keys):
        all_feats.append(stat_features[k])
    merged = np.concatenate(all_feats, axis=1)
    print(f"🌎 {location.title()} merged → {merged.shape}")
    return merged

features_pansol = merge_by_location(stat_features, "pansol")
features_bayog  = merge_by_location(stat_features, "bayog")
```

Figure 10. Merging Features by Location

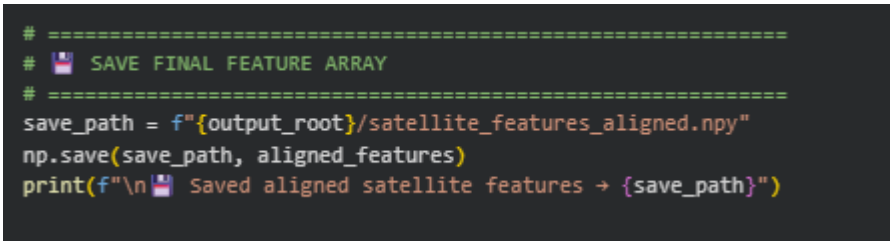
Feature alignment and standardization across Bayog and Pansol stations using tile/resample strategies to match target sample counts (450 per location).



```
# ============================================================
# 🔗 ALIGN FEATURES WITH AUGMENTED SATELLITE (900 total)
# ============================================================
print("\n🔗 Aligning features with augmented satellite data...")

def match_count(features, target_count=450):
    """Repeat or resample features to match target sample count."""
    n = features.shape[0]
    if n == target_count:
        return features
    elif n < target_count:
        reps = int(np.ceil(target_count / n))
        features_expanded = np.tile(features, (reps, 1))
        return features_expanded[:target_count]
    else:
        idx = np.random.choice(n, target_count, replace=False)
        return features[idx]

features_pansol_matched = match_count(features_pansol, 450)
features_bayog_matched  = match_count(features_bayog, 450)

aligned_features = np.zeros((900, features_pansol_matched.shape[1]), dtype=np.float32)
aligned_features[:450] = features_pansol_matched
aligned_features[450:] = features_bayog_matched

print("\n✅ Alignment complete:")
print(f"   Pansol → 0–449  → {features_pansol_matched.shape}")
print(f"   Bayog  → 450–899 → {features_bayog_matched.shape}")
print(f"   Final aligned feature tensor: {aligned_features.shape}")
```

Figure 11. Aligning Features with Augmented Satellite Data

Storage and serialization of final aligned satellite feature arrays to binary NumPy format for machine learning model input.

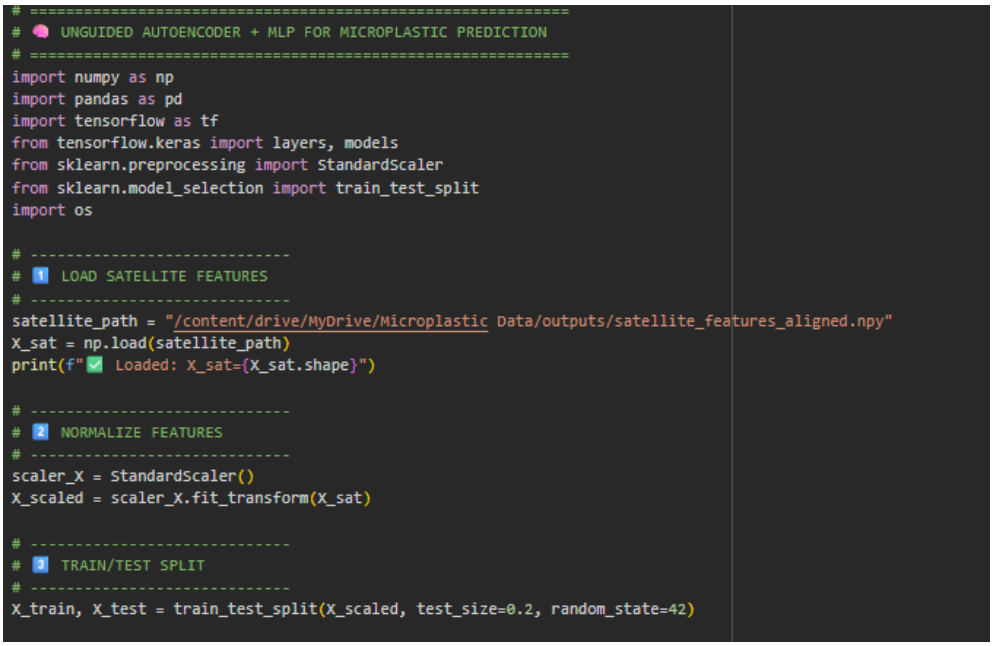


```
# ============================================================
# 💾 SAVE FINAL FEATURE ARRAY
# ============================================================
save_path = f"{output_root}/satellite_features_aligned.npy"
np.save(save_path, aligned_features)
print(f"\n💾 Saved aligned satellite features → {save_path}")
```

Figure 12. Final Feature Array Storage

Unguided Autoencoder + MLP For Microplastic Prediction

Machine learning pipeline showing satellite feature loading, normalization, train-test splitting, and unsupervised autoencoder architecture design with encoder-decoder layers.



```
# ============================================================
# 🧠 UNGUIDED AUTOENCODER + MLP FOR MICROPLASTIC PREDICTION
# ============================================================
import numpy as np
import pandas as pd
import tensorflow as tf
from tensorflow.keras import layers, models
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
import os

# ----------------------------
# 1️⃣ LOAD SATELLITE FEATURES
# ----------------------------
satellite_path = "/content/drive/MyDrive/Microplastic Data/outputs/satellite_features_aligned.npy"
X_sat = np.load(satellite_path)
print(f"✅ Loaded: X_sat={X_sat.shape}")

# ----------------------------
# 2️⃣ NORMALIZE FEATURES
# ----------------------------
scaler_X = StandardScaler()
X_scaled = scaler_X.fit_transform(X_sat)

# ----------------------------
# 3️⃣ TRAIN/TEST SPLIT
# ----------------------------
X_train, X_test = train_test_split(X_scaled, test_size=0.2, random_state=42)
```

Figure 13. Unguided Autoencoder + MLP for Microplastic Prediction

Autoencoder model training and data scaling preparation for MLP classifier, including model compilation, training history logging, and scaler serialization.

```
# ----------------------------
# 4 DEFINE UNGUIDED AUTOENCODER
# ----------------------------
input_dim = X_train.shape[1]
inputs = layers.Input(shape=(input_dim,))
encoded = layers.Dense(128, activation='relu')(inputs)
encoded = layers.Dense(64, activation='relu')(encoded)
bottleneck = layers.Dense(32, activation='relu', name="bottleneck")(encoded)

decoded = layers.Dense(64, activation='relu')(bottleneck)
decoded = layers.Dense(128, activation='relu')(decoded)
decoded = layers.Dense(input_dim, activation='linear', name="reconstruction")(decoded)

autoencoder = models.Model(inputs=inputs, outputs=decoded)
autoencoder.compile(optimizer='adam', loss='mse')
autoencoder.summary()

# ----------------------------
# 5 TRAIN AUTOENCODER
# ----------------------------
history_ae = autoencoder.fit(
    X_train, X_train,
    validation_data=(X_test, X_test),
    epochs=100,
    batch_size=32,
    verbose=1
)

# ----------------------------
# 6 SAVE AUTOENCODER AND SCALED DATA
# ----------------------------
save_dir = "/content/drive/MyDrive/Microplastic Data/outputs"
os.makedirs(save_dir, exist_ok=True)
autoencoder.save(os.path.join(save_dir, "autoencoder_satellite_unguideds.h5"))
np.save(os.path.join(save_dir, "X_sat_scaled.npy"), X_scaled)
```

Figure 14. Autoencoder Training and Scaled Data Storage

MLP neural network architecture design with multiple dense layers, model compilation, and training with validation monitoring for microplastic density regression.

```python
# ----------------------------
# 7️⃣ EXTRACT BOTTLENECK FEATURES
# ----------------------------
encoder = models.Model(inputs=autoencoder.input, outputs=autoencoder.get_layer("bottleneck").output)
encoded_features = encoder.predict(X_scaled)
np.save(os.path.join(save_dir, "encoded_features_satellite_unguideds.npy"), encoded_features)
print(f"\n✅ Encoded satellite features saved: {encoded_features.shape}")

# ----------------------------
# 8️⃣ LOAD MICROPLASTIC TARGETS
# ----------------------------
insitu_path = "/content/drive/MyDrive/Microplastic Data/outputs/augmented_insitu(1).csv"
df = pd.read_csv(insitu_path)
y = df["Microplastic"].values

# ----------------------------
# 9️⃣ SPLIT BOTTLENECK FEATURES FOR MLP
# ----------------------------
X_train_b, X_test_b, y_train, y_test = train_test_split(
    encoded_features, y, test_size=0.2, random_state=42
)

# ----------------------------
# 1️⃣0️⃣ STANDARDIZE BOTTLENECK FEATURES
# ----------------------------
scaler_b = StandardScaler()
X_train_b = scaler_b.fit_transform(X_train_b)
X_test_b = scaler_b.transform(X_test_b)

# ----------------------------
# 1️⃣1️⃣ DEFINE MLP REGRESSOR
# ----------------------------
input_dim_b = X_train_b.shape[1]
inputs_b = layers.Input(shape=(input_dim_b,))
x = layers.Dense(64, activation='relu')(inputs_b)
x = layers.Dense(32, activation='relu')(x)
outputs_b = layers.Dense(1, activation='linear')(x)

mlp_model = models.Model(inputs=inputs_b, outputs=outputs_b)
mlp_model.compile(optimizer='adam', loss='mae')
mlp_model.summary()
```

Figure 15. MLP Regressor Model Development and Training

```python
# ----------------------------
# 1️⃣2️⃣ TRAIN MLP
# ----------------------------
history_mlp = mlp_model.fit(
    X_train_b, y_train,
    validation_data=(X_test_b, y_test),
    epochs=100,
    batch_size=16,
    verbose=1
)

# ----------------------------
# 1️⃣3️⃣ PREDICT MICROPLASTIC USING MLP
# ----------------------------
y_pred = mlp_model.predict(X_test_b).flatten()

# ----------------------------
# 1️⃣4️⃣ SAVE EVERYTHING FOR T-TEST OR LATER USE
# ----------------------------
# Bottleneck features
np.save(os.path.join(save_dir, "X_bottleneck.npy"), encoded_features)
np.save(os.path.join(save_dir, "X_bottleneck_train.npy"), X_train_b)
np.save(os.path.join(save_dir, "X_bottleneck_test.npy"), X_test_b)

# Microplastic targets
np.save(os.path.join(save_dir, "y_train.npy"), y_train)
np.save(os.path.join(save_dir, "y_test.npy"), y_test)

# Predictions
np.save(os.path.join(save_dir, "y_pred_mlp.npy"), y_pred)

# Save MLP model
mlp_model.save(os.path.join(save_dir, "mlp_on_bottleneck.h5"))

print("\n✅ Autoencoder, bottleneck features, MLP model, and predictions saved successfully.")
```

Figure 16. MLP Model Prediction and Results Storage

Final model predictions, feature extraction, training history logging, and comprehensive results storage including model weights and prediction outputs.

APPENDIX D

Sample Input / Output

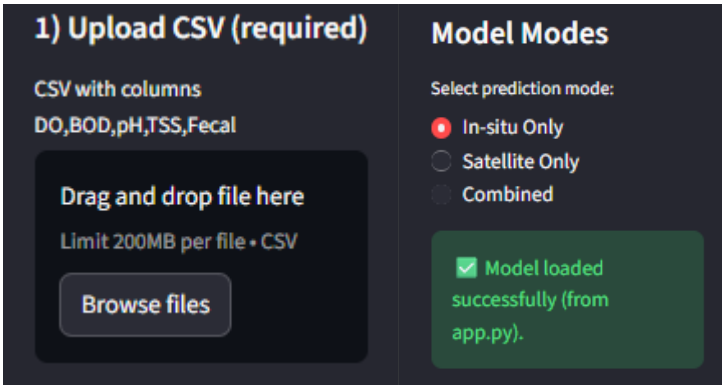


Figure 17. Model Interface for Microplastic Prediction

The figure presents the graphical user interface (GUI) of the developed predictive model. It shows the selection of model modes: In-situ Only, Satellite Only, or Combined along with the file upload section for input CSV data containing water quality parameters (DO, BOD, pH, TSS, and Fecal Coliform). The interface confirms successful model loading for prediction tasks.

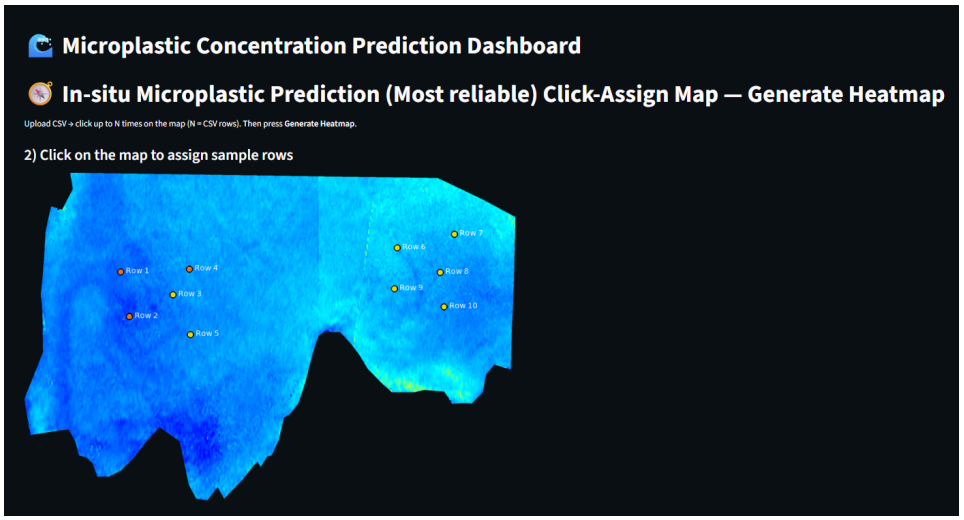


Figure 18. Microplastic Concentration Prediction Dashboard

The figure above displays the prediction dashboard output generated by the system. The map visualizes the spatial distribution of predicted microplastic concentrations across the southern portion of Laguna de Bay, with color gradients representing varying density levels. Users can

interactively assign sampling points and generate heatmaps for geospatial analysis.

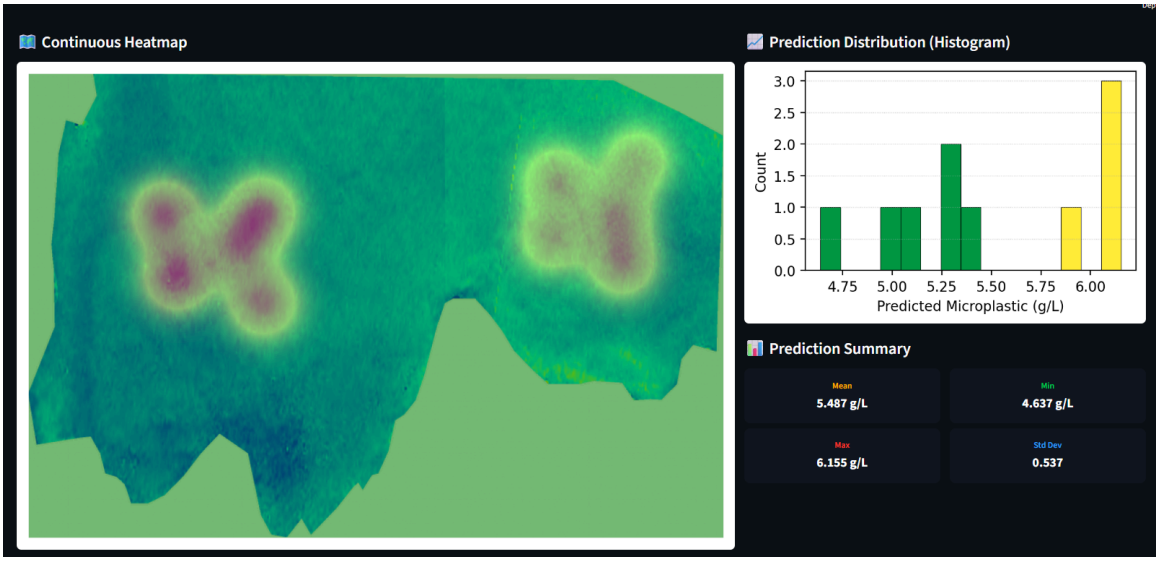


Figure 19. Continuous Heatmap and Prediction Distribution Dashboard

The figure illustrates the continuous heatmap visualization of predicted microplastic concentrations across the southern sector of Laguna de Bay. Areas with higher intensity indicate zones of elevated microplastic density. The accompanying histogram shows the distribution of predicted concentration values (mg/L), while the summary section presents key prediction statistics, including mean and range values derived from the model output.

4) Predictions & Coordinates

	DO	BOD	pH	TSS	Fecal	Predicted Microplastic (g/L)
0	8.42	3.24	8.2	245.5	230	6.0819
1	8.47	3.32	8.17	274.1	230	6.1549
2	7.03	2.03	8.15	168.5	470	5.3199
3	8.36	3.14	8.1	254.4	230	6.0875
4	8.48	3.33	8.19	91.5	440	5.9012
5	8.77	1.63	7.68	16.9	4800	5.3275
6	8.73	2.49	7.76	13.9	380	4.637
7	8.84	3.3	7.69	10.3	230	5.0449
8	8.95	3.4	7.81	10	230	4.9613
9	9.2	1.69	7.78	17.5	4800	5.352

Figure 20. Predicted Microplastic Concentration and Coordinate Table

The figure presents the tabular output generated by the model, displaying predicted microplastic concentration values alongside their corresponding spatial coordinates and associated in-situ parameters. The table enables quantitative assessment and spatial referencing of each prediction point within the study area, supporting data validation and geospatial mapping.

APPENDIX E

Turnitin and Grammarly

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A HYBRID DEEP LEARNING AND ENSEMBLE FRAMEWORK FOR PREDICTIVE ANALYTICS OF MICROPLASTIC DENSITY MAPPING IN THE SOUTHERN LAGUNA DE BAY USING SENTINEL-2 REMOTE SENSING AND IN-SITU WATER QUALITY METRIC
by Jireh Tregueros

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(Chapter 1 & 2) A HYBRID DEEP LEARNING AND ENSEMBLE FRAMEWORK FOR PREDICTIVE ANALYTICS OF MICROPLASTIC DENSITY MAPPING IN THE SOUTHERN LAGUNA DE BAY USING SENTINEL-2 REMOTE SENSING AND IN-SITU WATER QUALITY METRICS

by Jabez Estanislao

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by Jabez Estanislao

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APPENDIX F

Publishable Paper (IEEE format)

PREDICTIVE ANALYTICS FOR MAPPING MICROPLASTIC ACCUMULATION IN THE SOUTHERN LAGUNA DE BAY USING REMOTE SENSING, WATER QUALITY METRICS: HYBRID MACHINE LEARNING AND DEEP LEARNING APPROACH

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Article Info

Article history:

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Deep Learning
Remote Sensing
Machine Learning
Water Quality

ABSTRACT

This study developed a predictive model to estimate microplastic density (mg/L) in the southern part of Laguna de Bay using an integrated machine learning and deep learning framework. Data from two sampling sites, Bayog and Pansol, included in-situ measurements of water quality parameters (Dissolved Oxygen, pH, Biochemical Oxygen Demand, Total Suspended Solids, and Fecal Coliform) and five Sentinel-2 spectral indices: NDTI, NDCI, NDVI, NDMI, and PI. An MLP Autoencoder was applied for deep feature extraction, combined with ensemble learning algorithms (Random Forest, Gradient Boosting, and XGBoost). The hybrid model achieved strong predictive performance ($R^2 = 0.912$). Although satellite data effectively captured spatial and spectral variability, it could not independently quantify microplastic concentrations without calibration using in-situ measurements, highlighting the necessity of ground-truth data for reliable prediction. Statistical validation through an independent t-test ($p > 0.05$) showed no significant difference between predicted and observed concentrations, confirming the robustness of the framework. Data collection is recommended during March–April when cloud cover is minimal. Overall, the study demonstrates the potential of deep learning and multispectral remote sensing, calibrated with in-situ data, for accurate monitoring of microplastic pollution in Laguna de Bay.

Corresponding Author:

1. INTRODUCTION

Microplastic pollution, defined as particles smaller than 5 mm, is pervasive in aquatic ecosystems and poses ecological and human health risks due to bioaccumulation and contaminant transport. In the Philippines, Laguna de Bay exemplifies this crisis, with significant microplastic contamination primarily from domestic and industrial sources. Pollution from runoff, aquaculture, and industrial discharge threatens fisheries, aquaculture, and public health through contaminated seafood.

Remote sensing enables scalable monitoring of microplastics using satellites, UAVs, and radar, but direct detection is limited by low resolution, optical confusion, and atmospheric interference. Indirect proxies such as chlorophyll-a, turbidity, and floating algae indices can identify potential hotspots, yet freshwater systems remain underrepresented compared to marine environments.

Representation learning with autoencoders compresses high-dimensional environmental features into compact latent codes, capturing non-linear patterns from heterogeneous data while addressing the scarcity of labeled samples. In this study, an MLP-based autoencoder was applied to Sentinel-2 imagery over Southern Laguna de Bay to extract 16-dimensional latent features. These were combined with in-situ water-quality measurements to form a 24-dimensional hybrid dataset, which was used as input to Random Forest, Gradient Boosting, and XGBoost models for microplastic prediction. For operational deployment, only the trained encoder is retained, with inputs standardized to zero mean and unit variance.

Microplastic detection has advanced from manual stereomicroscopy to spectroscopic and semi-automated approaches. The proposed hybrid framework integrates unsupervised feature extraction and ensemble learning with remote sensing and water-quality data to enable scalable monitoring, hotspot identification, and informed mitigation strategies in freshwater ecosystems.

The objective of this study is to develop a predictive model that maps microplastic accumulation in Southern Laguna de Bay to support targeted environmental intervention and management.

RESEARCH OBJECTIVES

This study aims to develop a comprehensive predictive model for mapping microplastic accumulation in the southern part of Laguna de Bay using remote sensing and water quality metrics through an integrated machine learning and deep learning approach.

I. Analyze key water quality indicators derived from Sentinel-2 satellite data (Normalized Difference Turbidity Index [NDTI], Normalized Difference Chlorophyll Index [NDCI], Normalized Difference Vegetation Index [NDVI], Normalized Difference Moisture Index [NDMI], and Plastics Index [PI]) and in situ measurements (Dissolved Oxygen, pH, Biochemical Oxygen Demand, Total Suspended Solids, and Fecal Coliform) using Spearman's rank correlation to evaluate their relevance to microplastic presence through multivariate analysis and geospatial mapping.

II. Develop and implement predictive models capable of estimating microplastic concentration zones using three configurations: satellite-derived indices only, in situ parameters only, and an integrated hybrid dataset. The modeling framework employs an MPN Autoencoder for feature extraction and dimensionality reduction, combined with Random Forest (RF), Gradient Boosting, and XGBoost algorithms for predictive classification.

III. Evaluate and validate model performance by comparing predicted microplastic concentrations with actual sampling data using performance metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R^2), Feature Importance Analysis, Cross-Validation (CV), and Independent T-test.

IV. Generate a GIS-based microplastic accumulation map representing the predicted microplastic concentration across Bayog and Pansol to provide actionable insights for policymakers and environmental managers in support of targeted monitoring, mitigation strategies, and regulatory planning.

2. METHODOLOGY

MODEL DEVELOPMENT

The model development process followed a structured pipeline, as illustrated in Figure 1. First, data preprocessing harmonized satellite-derived spectral indices (NDVI, NDMI, NDCI, NDTI, and PI) with in-situ water quality measurements through normalization and georeferencing. Georeferencing aligned satellite imagery with real-world geographic coordinates, ensuring accurate spatial correspondence between remote sensing data and field measurements. An MLP Autoencoder compressed the satellite features into a compact representation, which were then combined with in-situ parameters using weighted integration (0.8 for in-situ, 0.2 for satellite). The higher weight for in-situ data reflected its lower measurement uncertainty and higher reliability compared to satellite-derived features. Next, the model training phase employed ensemble learning algorithms (Random Forest, Gradient Boosting, XGBoost) to identify relationships between environmental variables and microplastic concentrations. Model performance was evaluated using a held-out dataset through independent t-test analysis alongside MAE, RMSE, R^2, and cross-validation (CV) metrics. Once validated, the model was applied to the southern part of Laguna de Bay to generate GIS-based microplastic predictions. The final stage involved identifying potential microplastic accumulation hotspots through spatial mapping, leveraging the combined strength of

remote sensing and field-based observations. This end-to-end approach prioritized methodological transparency while addressing data limitations in freshwater monitoring.

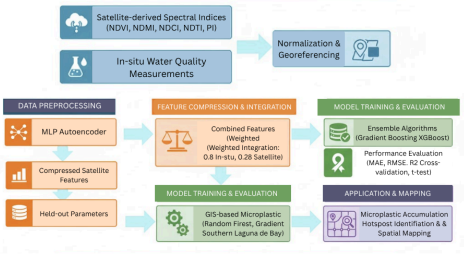

Figure 1. Model Development

DATA COLLECTION

Sampling Station Selection of Insitu Water Quality Parameters

Two sampling stations were selected in the southern part of Laguna de Bay, an area identified as being particularly vulnerable to pollution due to high levels of anthropogenic activity. According to the Laguna Lake Development Authority (LLDA, 2021), the lake receives substantial pollutant input from domestic wastewater, industrial discharge, and agricultural runoff, with the southern bay region being one of the primary receiving areas due to its proximity to urban settlements and tributary inflows. Arcadio et al. (2022) also noted that population density and aquaculture operations in the southern and western parts of the lake contribute significantly to environmental degradation and water quality decline. These conditions make the southern bay an appropriate site for monitoring microplastic accumulation.

The two sampling stations were selected to reflect these high-pressure zones and to represent varying pollution exposure gradients within the region. GPS coordinates were recorded at each site to ensure spatial precision, enabling accurate pairing of in-situ data with satellite-derived observations. This alignment supports the integration of datasets in the model development process and ensures consistency in both temporal and spatial dimensions during analysis.

Satellite Data

Satellite data are acquired from the Sentinel-2 mission, which provides high-resolution multispectral imagery suitable for monitoring large water bodies. The satellite-derived parameters include spectral indices such as Normalized Difference Vegetation Index (NDVI), Normalized Difference Moisture Index (NDMI), Normalized Difference Chlorophyll Index (NDCI), Normalized Difference Turbidity Index (NDTI), and Plastics Index (PI), which serve as indicators of water quality and potential microplastic presence. These remotely sensed data allow for spatially extensive analysis, capturing broader environmental patterns that complement the localized in-situ measurements. By integrating satellite observations with ground-based data, a comprehensive dataset is created to support predictive modeling of microplastic distribution across aquatic systems.

DATA PREPROCESSING

Data preprocessing was performed to prepare both satellite-derived spectral indices and in-situ water quality data for model training and prediction. Sentinel-2 imagery was processed to compute five primary spectral indices Normalized Difference Vegetation Index (NDVI), Normalized Difference Moisture Index (NDMI), Normalized Difference Chlorophyll Index (NDCI), Normalized Difference Turbidity Index (NDTI), and Plastics Index (PI). From these indices, per-band statistical features such as mean, standard deviation, variance, and percentile values were extracted from relevant Sentinel-2 bands (B03, B04, B05, and B08). The extracted features were subsequently processed through a dense autoencoder to generate compressed latent representations suitable for integration with in-situ measurements, which included Dissolved Oxygen (DO), pH, Biochemical Oxygen Demand (BOD), Chemical Oxygen Demand (COD), and Fecal Coliform.

To ensure temporal and spatial consistency between datasets, Sentinel-2 imagery was temporally aligned with in-situ sampling events conducted on the same or nearest available dates. Georeferencing was applied using ground control points (GCPs) to assign accurate geographic coordinates to each pixel, ensuring proper spatial correspondence between remote sensing observations and sampling locations. After alignment, all features from both datasets were standardized using **Z-score normalization** through the StandardScaler method, transforming variables to have a mean of zero and a standard deviation of one. This process minimized scale bias among features, ensuring balanced model learning during correlation and predictive analyses.

Spearman's rank correlation analysis was employed to identify the relationships

between spectral indices, in-situ parameters, and measured microplastic concentrations, as it effectively handles non-parametric and non-linear environmental data. To enhance model robustness and generalization, data augmentation techniques were applied. Satellite data were augmented through controlled spatial transformations, including rotations, flips, and slight rescaling, while in-situ measurements underwent physics-guided noise injection to simulate realistic variability in sampling and environmental conditions. The resulting standardized and augmented dataset provided a reliable and integrated foundation for ensemble-based and deep learning model development.

MODEL SELECTION

MLP Autoencoder Feature Extraction Architecture

The Multilayer Perceptron (MLP) Autoencoder was designed to extract compact latent representations from satellite-derived spectral and statistical features optimized for microplastic prediction. Satellite inputs including normalized Sentinel-2 indices, reflectance values, and texture-based features were standardized using *StandardScaler* before being passed through an encoder–decoder network (64 → 32 → bottleneck → 64). The 32-neuron bottleneck layer compressed the satellite data while retaining essential spectral and spatial information.

A supervised guidance branch was connected to the bottleneck to predict normalized microplastic concentration, enabling the Autoencoder to learn representations aligned with ground observations. The model was trained using the *Adam* optimizer for 200 epochs with a composite Mean Squared Error (MSE) loss (0.3 reconstruction, 0.7 guidance). This dual-objective design allowed the network to produce latent features that effectively bridge satellite-derived data and observed microplastic levels.

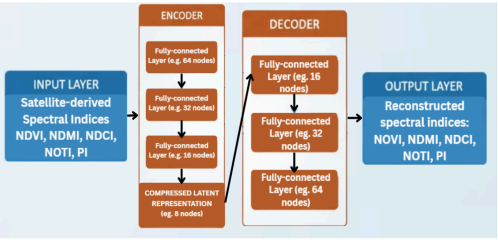

Figure 2. Figure 8. Architecture of the Dense Autoencoder for Spectral Feature Compression

Weighting Strategy for Ensemble Integration

To address temporal gaps caused by cloud-obstructed satellite imagery particularly during July sampling a weighting adjustment was applied between datasets prior to model training. The ensemble framework prioritized in-situ data (weight = 0.8) over satellite data (weight = 0.2) to balance spatial coverage and temporal accuracy. This approach ensured robust model training despite incomplete satellite observations, maintaining consistency across predictive experiments.

Integration of Ensemble Learning Models for Predicting Microplastic Accumulation

Three ensemble algorithms Random Forest (RF), Gradient Boosting (GB), and Extreme Gradient Boosting (XGBoost) were developed to predict microplastic concentrations using both in-situ and satellite-encoded datasets. The models integrated key physicochemical parameters (DO, BOD, pH, TSS, Fecal Coliform) and Autoencoder-derived features. Performance was evaluated through R^2, RMSE, MAE, and cross-validation metrics, with feature importance analysis conducted using XGBoost to identify the most influential predictors.

Figure 3. Ensemble Regression Framework (RF, GB, and XGBoost)

Hybrid Ensemble Integration Framework

The final predictive model combined the outputs of in-situ and satellite-based ensembles through a weighted averaging mechanism (0.8 in-situ, 0.2 satellite). This hybrid system leveraged the interpretability and accuracy of ensemble regressors with the dimensionality reduction power of the MLP Autoencoder. The resulting hybrid framework effectively mapped spatial microplastic accumulation across Southern Laguna de Bay, generating a GIS-based surface that highlighted key accumulation hotspots for environmental monitoring and policymaking.

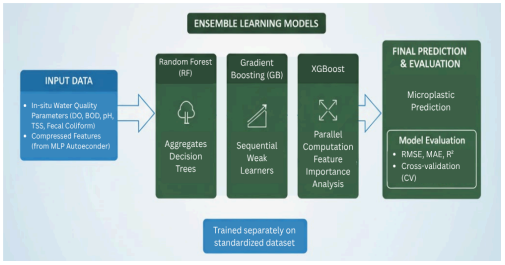

Figure 4. Hybrid Weighted Ensemble (MLP Autoencoder + RF/GB/XGBoost)

MODEL EVALUATION

The hybrid framework integrating the MLP Autoencoder with ensemble regressors Random Forest (RF), Gradient Boosting (GB), and XGBoost (XGB) was evaluated using standard regression performance metrics, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), the coefficient of determination (R^2), feature importance analysis, and *k*-fold cross-validation (CV). An independent *t*-test was further applied to assess statistical consistency between predicted and observed microplastic concentrations on unseen data.

MSE assessed the reconstruction performance of the Autoencoder by measuring the average squared difference between the input and reconstructed satellite features, with lower values indicating efficient latent feature learning. RMSE and MAE evaluated ensemble prediction accuracy, where RMSE emphasized larger deviations and MAE provided a robust average error magnitude. The R^2 score quantified the proportion of variance in microplastic concentration explained by the model, serving as an indicator of predictive strength.

K-fold cross-validation was employed to ensure model generalization and minimize overfitting by repeatedly training and validating across multiple data subsets. Feature importance analysis, derived from the XGBoost model, identified the most influential predictors among satellite-derived indices (NDVI, NDMI, NDCI, NDTI, and PI) and in-situ water quality variables, enhancing interpretability and guiding environmental prioritization. Finally, the independent *t*-test verified that predicted and measured concentrations were statistically comparable ($p > 0.05$), confirming the model's reliability for spatial prediction of microplastic accumulation.

3. RESULTS AND DISCUSSION

Model Performance Metrics

Table 1. Ensemble Model Performance Metrics

Dataset	Phase	R^2	RMSE
In-situ Ensemble	Overall	**0.871**	**0.502**
Satellite Ensemble	Overall	**0.990**	**0.140**
Weighted Ensemble (0.8 In-situ / 0.2 Satellite)	Overall	**0.912**	**0.413**

The ensemble integration of in-situ and satellite models produced a robust and complementary predictive framework for microplastic estimation. The in-situ ensemble achieved an R^2 of 0.871 with an RMSE of 0.502, indicating strong predictive capability and consistent performance under field conditions. In contrast, the satellite ensemble demonstrated exceptionally high accuracy ($R^2 = 0.990$, RMSE = 0.140), reflecting its strength in capturing spatial and spectral patterns from Sentinel-2 data.

However, despite its apparent statistical accuracy, the satellite-derived estimates are less reliable due to temporal mismatches between satellite image acquisition and in-situ sampling dates. In dynamic aquatic environments, microplastic concentrations can fluctuate rapidly due to currents, tides, and local inputs. Consequently, when satellite observations and field sampling are not perfectly synchronized, reflectance-based predictions may not accurately represent real-time surface conditions. This spatio-temporal mismatch can introduce discrepancies in validation outcomes and compromise the reliability of satellite-derived estimates, as similarly highlighted by Urraca et al. (2024) in their assessment of solar radiation measurements. Therefore, while the satellite model demonstrates strong statistical performance, its practical reliability remains limited compared to in-situ observations.

When combined using a weighted ratio of 0.8 in-situ and 0.2 satellite data, the weighted ensemble achieved an R^2 of 0.912 and an RMSE of 0.413, effectively balancing the ground-based precision of in-situ data with the spatial coverage of remote sensing inputs. This

integration underscores the synergistic advantage of merging both data sources, yielding a more stable and generalized predictive framework that enhances accuracy while mitigating uncertainties associated with temporal misalignment in satellite observations.

Model Validation

An independent t-test was conducted to assess the statistical significance of the differences between the predicted and actual microplastic concentration values for each ensemble model. This analysis aimed to verify whether the prediction performance of the In-situ, Satellite, and Combined Ensemble models significantly deviated from the actual field measurements, thereby validating the reliability and consistency of the models in estimating microplastic density.

Table 2. Independent T-Test Comparison between Predicted and Actual Microplastic Concentrations

Model	T-statistic	P-value	Significance
In-situ Ensemble	0.4198	0.684474	No Significant Difference
Satellite Ensemble	1.4978	0.168423	No Significant Difference
Combined Ensemble (0.8/0.2)	0.7582	0.467734	No Significant Difference

As shown in Table 2, the results of the independent t-test revealed no significant difference between the predicted and actual microplastic concentrations across all ensemble models. The In-situ Ensemble Model obtained a T-statistic of 0.4450 with a P-value of 0.661346, the Satellite Ensemble Model yielded a T-statistic of 0.9631 with a P-value of 0.347615, and the Combined Ensemble Model (0.8 In-situ / 0.2 Satellite) recorded a T-statistic of 0.6369 with a P-value of 0.531804.

Since all P-values are greater than the 0.05 significance threshold, the null hypothesis (H_0) cannot be rejected, indicating that there is no statistically significant difference between the predicted and actual values. This suggests that the ensemble models, particularly the combined model, are capable of generating predictions that closely align with the observed data, demonstrating stability and minimal bias in estimation. The results validate the robustness of the developed models and confirm their accuracy in predicting microplastic concentrations across varying environmental conditions.

4. CONCLUSIONS

The conclusions summarize the findings derived from the study objectives:

1. **To analyze key water quality indicators derived from Sentinel-2 satellite data and in-situ parameters to determine their relationships with microplastic concentration through correlation and geospatial analysis.**

Spearman's correlation analysis revealed that Dissolved Oxygen (DO) exhibited the strongest positive correlation with microplastic density, followed by Turbidity and Total Suspended Solids (TSS). Among the satellite-derived indices, NDTI, NDCI, and the Plastics Index (PI) showed meaningful relationships with observed concentrations, confirming that both physicochemical and spectral indicators effectively capture spatial variations in microplastic abundance. These findings validate the integration of remote sensing with field-based measurements as a robust approach for assessing microplastic pollution dynamics in aquatic systems.

2. **To develop and optimize predictive models for estimating microplastic concentration zones using in-situ, satellite-derived, and hybrid datasets.**

Three ensemble learning models, Random Forest (RF), Gradient Boosting, and XGBoost were optimized under separate configurations: in-situ only, satellite-derived only, and a weighted hybrid integration. Model performance metrics (Table 1) showed that the satellite ensemble achieved a very high coefficient of determination ($R^2 = 0.990$) with low error values (RMSE = 0.140, MAE = 0.108); however, this configuration still heavily relied on in-situ calibration for spectral validation. The in-situ ensemble yielded an R^2 of 0.871, while the weighted ensemble (0.8 in-situ / 0.2 satellite) provided the most balanced and realistic predictive performance ($R^2 = 0.912$; RMSE = 0.413; MAE = 0.294). These results highlight that hybrid data integration significantly enhances model generalization and predictive accuracy, demonstrating the power of combining remote sensing and in-situ data for microplastic density estimation.

3. **To evaluate and validate the predictive performance of the developed models using independent t-test analysis.**

The independent t-test revealed no statistically

significant difference ($p > 0.05$) between the predicted and actual microplastic concentrations (mg/L), confirming the reliability of the ensemble framework. This statistical validation indicates that the developed hybrid ensemble model can generate predictions that are not only accurate but also consistent with observed field measurements, supporting its application in real-world monitoring scenarios.

4. **To generate a GIS-based microplastic accumulation map representing the predicted microplastic concentrations across Southern Laguna de Bay.**

The GIS-based accumulation map visualized the predicted spatial distribution of microplastic density across the southern sectors of Laguna de Bay. Concentrations ranged from 2.01 to 6.99 mg/L, corresponding to medium to high levels when compared with global literature values ranging from 0–10 mg/L in similar urban and estuarine systems (Xue et al., 2023). The Pansol and Bayog sectors exhibited the highest concentrations, suggesting the presence of localized accumulation zones likely influenced by hydrodynamic behavior and human activity. These spatial outputs provide actionable insights for targeted environmental management, pollution mitigation, and future policy formulation addressing microplastic contamination in Laguna de Bay.

Recommendations

Based on the findings and conclusions of this study, the following recommendations are proposed to enhance future research and applications related to microplastic density prediction and monitoring using in-situ and satellite data:

1. Align in-situ sampling with satellite overpass schedules to ensure better temporal consistency between datasets. Based on the analysis, the optimal months for data collection are March to April, when cloud coverage is minimal, allowing for clearer satellite imagery and improved spectral data quality.
2. Expand the number of sampling sites around Laguna de Bay to include additional locations beyond Bayog and Pansol. A larger and more spatially diverse dataset will improve model generalization and provide a more representative understanding of microplastic distribution under varying environmental conditions.
3. Integrate advanced remote sensing indices and higher-resolution imagery by combining Sentinel-2 data with platforms such as Landsat-9 or PlanetScope. This approach can enhance spatial and spectral sensitivity for detecting subtle microplastic-related signals.
4. Incorporate Explainable Artificial Intelligence (XAI) techniques to improve the interpretability and transparency of predictive models. XAI methods such as SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-Agnostic Explanations) can identify which environmental or spectral variables most influence microplastic predictions, supporting trust, validation, and scientific understanding of model outputs.
5. Develop a GIS-based Decision Support System (DSS) to operationalize the predictive model for environmental agencies and local governments. This system can deliver near real-time microplastic density predictions and visualization maps to support evidence-based policymaking, monitoring, and remediation strategies for Laguna de Bay.

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APPENDIX G

Curriculum Vitae

Ralph Lawrence V. Amparo

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Los Baños Laguna Philippines

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TECHNICAL SKILLS

Programming Languages & Frameworks:

- Python
- HTML
- CSS
- Javascript

Web Development:

- React

Tools & Software:

- **Design & Editing:**
  - Canva
- **Microsoft Applications:**
- Google Workspace

Certification

CS50P - Harvard Online Course program (Python)

CS50X - Harvard Online Course program

TOP-CIT Certification (LEVEL 2)

ABOUT ME

Dedicated Computer Science student with strong skills in virtual assistance and data analysis. I have a solid background in cybersecurity and a growing expertise in web development. My goal is to combine technical knowledge and analytical skills to create secure, data-driven digital solutions.

EDUCATION

- **LAGUNA STATE POLYTECHNIC UNIVERSITY - LOS BANOS CAMPUS**
  Bachelor of Science in Computer Science Major in Intelligent System
  2022-2026
- **Los Banos Senior High School - HUMSS**
  2020-2022
- **Los Banos National High School - Batong Malake**
  Science, Technology, Engineering Program
  2016-2020

PROJECTS

Automatic Plant Watering System Using Arduino Uno

- A watering system that reveals the emotion of the plant based on the soil moisture, temperature, and humidity.

LSPU Reservation System

- Reservation for Lacson Gym

WORK EXPERIENCE

Freelance Researcher & Virtual Assistant
2024 – Present

• Conducts independent and collaborative research projects focused on data analysis, intelligent systems, and web technologies.

• Provides virtual assistance and data management services to clients, handling document organization, reports, and automation tasks.

- **Applies Python, Excel, and web-based tools to streamline workflows and extract insights from data.**

DE CASTRO, LANCE GEEFFREY S.

Contact

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TECHNICAL SKILLS

Programming Languages & Frameworks:

- HTML
- CSS
- JavaScript
- React.js
- PHP (Laravel)
- Python

Web Development:

- Full-Stack Development
- Web Design & Development

Tools & Software:

- **Design & Editing:** Canva, Photoshop, Figma
- **Microsoft Applications:** Excel, Word, PowerPoint

SOFT SKILLS

- Communication Skills
- Problem-Solving
- Time Management
- Flexibility
- Adaptability

Certification

TESDA Online Program

- SMART Technopreneurship 101 (2021)
- SMART Android Mobile Apps Development for Beginners (2021)

ABOUT ME

Dedicated and motivated Computer Science student with proven analytical and creative problem-solving abilities. Strong technical foundation combined with excellent interpersonal and communication skills, enabling effective collaboration across diverse teams and projects. Eager to contribute technical skills and innovative thinking to help organizations achieve their goals.

EDUCATION

- **LAGUNA STATE POLYTECHNIC UNIVERSITY - LOS BANOS CAMPUS**
  Bachelor of Science in Computer Science Major in Intelligent System
  2022-2026
- **TRACE COLLEGE**
  Bachelor of Science in Computer Engineering
  2021
- **CARD - MRI DEVELOPMENT INSTITUTE INC**
  ACCOUNTANCY AND BUSINESS MANAGEMENT
  2019-2021

PROJECTS

Smart Parking System(IOT)

- IoT-based parking management solution with real-time sensor tracking and automated space allocation.

Tomato Disease Detection(Machine Learning)

- ML model for automated detection and classification of plant diseases using image recognition.

Encryption & Decryption System

- Secure encryption/decryption system demonstrating cybersecurity principles and data protection algorithms.

OTHER PROJECTS

- Interactive 3D Web Models
- AI Chatbot
- SUMO BOT (IOT)
- TrashCash Detection(Machine Learning)

WORK EXPERIENCE

Website Developer
Sept 2023 - Present

- Developed full-stack healthcare website (https://goclinic.ph/)

International Freelancer
Jan 2025 - Present

- Leading website redesign and enhancement project for international client.

Jireh Magsino Tregueros

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TECHNICAL SKILLS

Programming Languages & Frameworks:

- Python
- HTML
- CSS
- Javascript

Web Development:

- React.js
- Node.js
- Tailwind CSS

Tools & Software:

- **Design & Editing:**
  - Canva
  - Figma
  - Adobe Photoshop
- **Microsoft Applications:**
  - Google Workspace
  - Microsoft Word
  - Excel

Certification

CS50X - Harvard Online Course program

ABOUT ME

I am a passionate and detail-oriented computer science practitioner with a strong interest in artificial intelligence, data analytics, and intelligent systems development. My work focuses on applying machine learning, deep learning, and automation to solve real-world problems in environmental monitoring, web technologies, and IoT applications. I enjoy building efficient, data-driven solutions and continuously exploring innovative ways to merge technology with meaningful impact.

EDUCATION

- **LAGUNA STATE POLYTECHNIC UNIVERSITY - LOS BANOS CAMPUS**
  Bachelor of Science in Computer Science Major in Intelligent System
  2022-2026
- **Tuntungin-Putho Integrated National High School**
  Senior High School - General Academic Strand (GAS)
  2020-2022
- **Tuntungin-Putho Integrated National High School**
  2016-2020
- **Paciano Rizal Elementary School**
  2010-2016

PROJECTS

Noise-Sensitive Classroom Monitor Bot
AI + IoT Project – 2025

- Designed an Arduino-based noise detection system integrated with a machine learning model trained using Edge Impulse and scikit-learn.

ASTRA AI: Intelligent Multi-Modal Chat Assistant (React + Node.js)
Web Development Project – 2025

- Built a modern React.js chatbot interface with voice input/output, theme switching, emoji/GIF support, and Markdown rendering.

WORK EXPERIENCE

Freelance Developer and Data Science Researcher
Remote — May 2020 – Present

- Designed intelligent systems and data-driven models for environmental and educational applications.
- Conducted exploratory data analysis, feature engineering, and model evaluation using Python and machine learning frameworks.
- Developed automation and web-based systems (e.g., RFID attendance, rent monitoring, and AI interfaces) to improve productivity and efficiency.
- Enhanced expertise in data processing, automation, AI system design, and full-stack web development.